Pontificia Universidad Católica del Perú
Facultad de Ciencias e Ingeniería
Departamento Académico de Ingeniería

1INF02 APRENDIZAJE DE MÁQUINA

Otros Esquemas de Aprendizaje

Prof. Rodrigo Uribe

Organización de las Clases

(10 min) Preguntas y consultas.

(30 min) Respuesta *Vibe Quest.*

(65 min) Otros esquemas de aprendizaje.



(10 min) Break

(35 min) Tendencias 2026.



Respuestas Vibe Quest

- Considerando los diversos métodos presentados para aprender representaciones de datos (como los autoencoders lineales que se relacionan con PCA, los autoencoders no lineales que aprenden variedades, y el objetivo de los embeddings de capturar relaciones semánticas como en el ejemplo de "rey" - "hombre" + "mujer" = "reina"): ¿Cuáles son las consideraciones arquitectónicas (por ejemplo, tipos de capas, funciones de activación, dimensionalidad del espacio latente) y las funciones de pérdida (más allá de la simple reconstrucción) más críticas al diseñar un sistema específicamente para generar embeddings que no solo compriman la información, sino que estructuren el espacio latente de manera que las distancias y direcciones reflejen relaciones semánticas útiles y transferibles a tareas posteriores?
- Considerando los avances en autoencoders variacionales (VAEs) y su capacidad para modelar distribuciones de datos complejas: ¿Cuáles son los desafíos y las oportunidades más significativos al intentar escalar los VAEs para generar datos de altísima dimensionalidad y fidelidad (por ejemplo, videos fotorrealistas o secuencias genómicas completas) manteniendo al mismo tiempo un espacio latente interpretable y controlable?
- La presentación discute la polisemanticidad y los esfuerzos por lograr características monosemánticas mediante autoencoders en modelos grandes. ¿De qué manera la capacidad emergente de "dirigir" o influenciar el comportamiento de modelos de lenguaje extensos, manipulando estas características internas, podría transformar las metodologías de auditoría, alineación y detección de sesgos en la inteligencia artificial, y qué nuevos riesgos éticos plantea esta capacidad de intervención detallada?
- Más allá de la similitud semántica básica que capturan los embeddings actuales en bases de datos vectoriales: ¿Cómo podrían evolucionar las técnicas de embedding y las arquitecturas de bases de datos vectoriales para representar y consultar eficientemente relaciones más complejas, como la causalidad, la jerarquía conceptual o la lógica argumentativa, directamente dentro del espacio vectorial?

¿Qué otro esquemas de aprendizajes conocen?



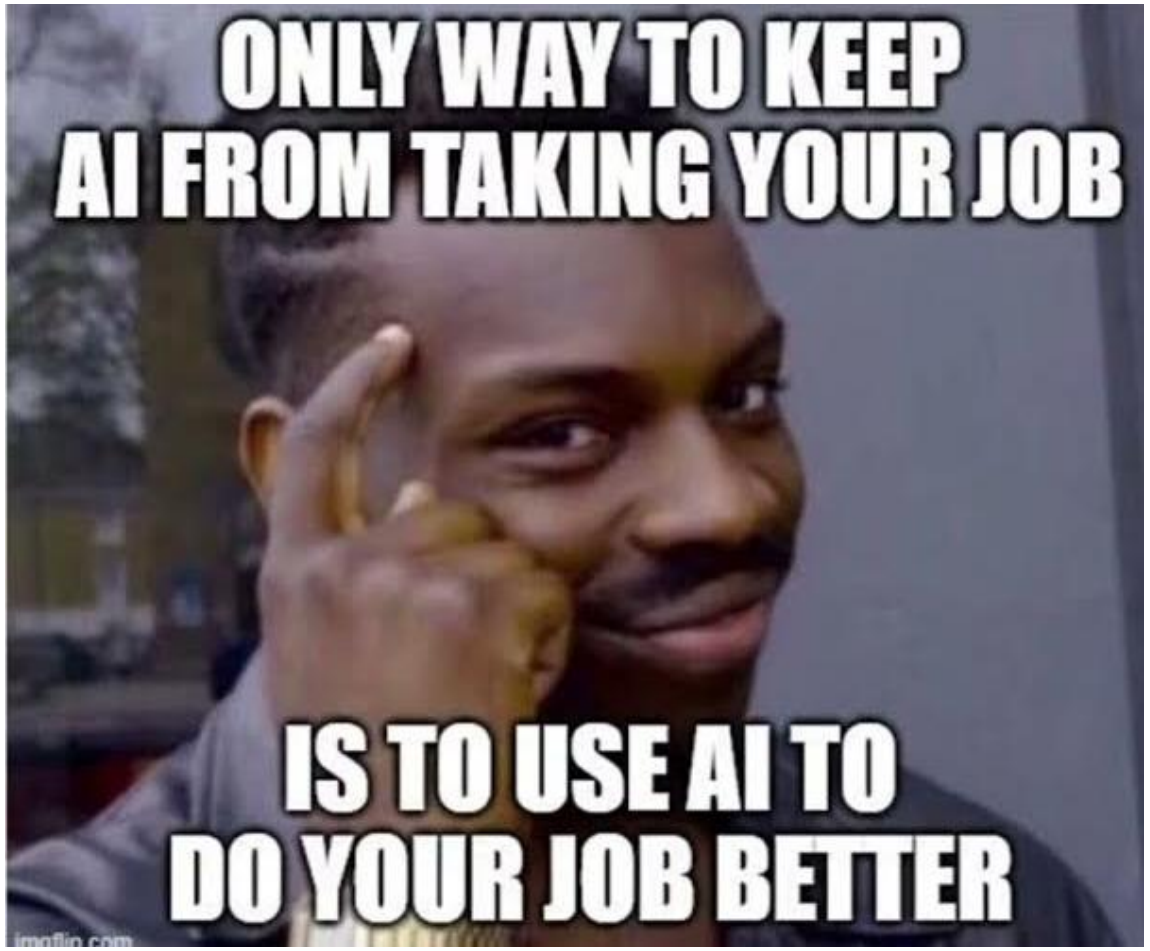
In-context Learning (Prompting Techniques)

Prompting Techniques

¿Qué es?

La “ingeniería de prompts” optimiza la comunicación con modelos de lenguaje para obtener mejores resultados

- Zero-shot y Few-shot (prompting sin o con pocos ejemplos)
- Chain-of-Thought (razonamiento paso a paso)
- Recuperación aumentada y generación de conocimiento
- Prompting reflexivo y multimodal
- Uso automático de herramientas y razonamiento



Zero-Shot Prompting

- Es una técnica en la que el modelo realiza una tarea sin necesidad de ejemplos previos.
- El prompt solo contiene instrucciones directas, sin demostraciones adicionales.
- Gracias a su entrenamiento a gran escala, los LLMs pueden generar respuestas precisas sin entrenamiento específico para cada tarea.

Prompt:

Classify the text into neutral, negative or positive.
Text: I think the vacation is okay.
Sentiment:

Output:

Neutral

Few-Shot Prompting

- Técnica que mejora el rendimiento del modelo al proporcionar ejemplos en el prompt.
- Funciona como un aprendizaje en contexto, donde las demostraciones guían la generación de respuestas.
- Permite que el modelo interprete mejor la tarea y genere respuestas más precisas.

Prompt:

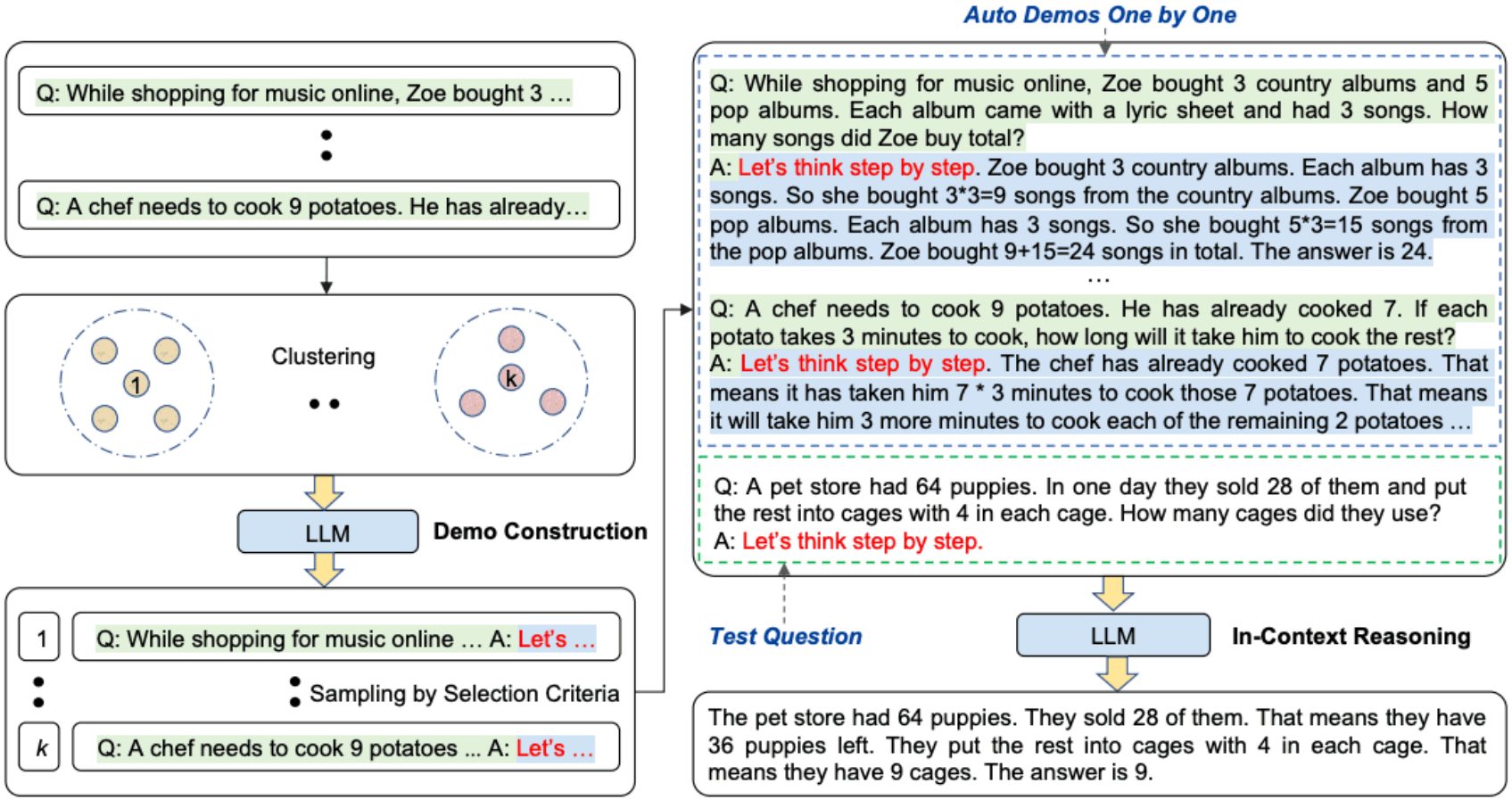
A "whatpu" is a small, furry animal native to Tanzania. An example of a sentence that uses the word whatpu is:
We were traveling in Africa and we saw these very cute whatpus.

To do a "farduddle" means to jump up and down really fast. An example of a sentence that uses the word farduddle is:

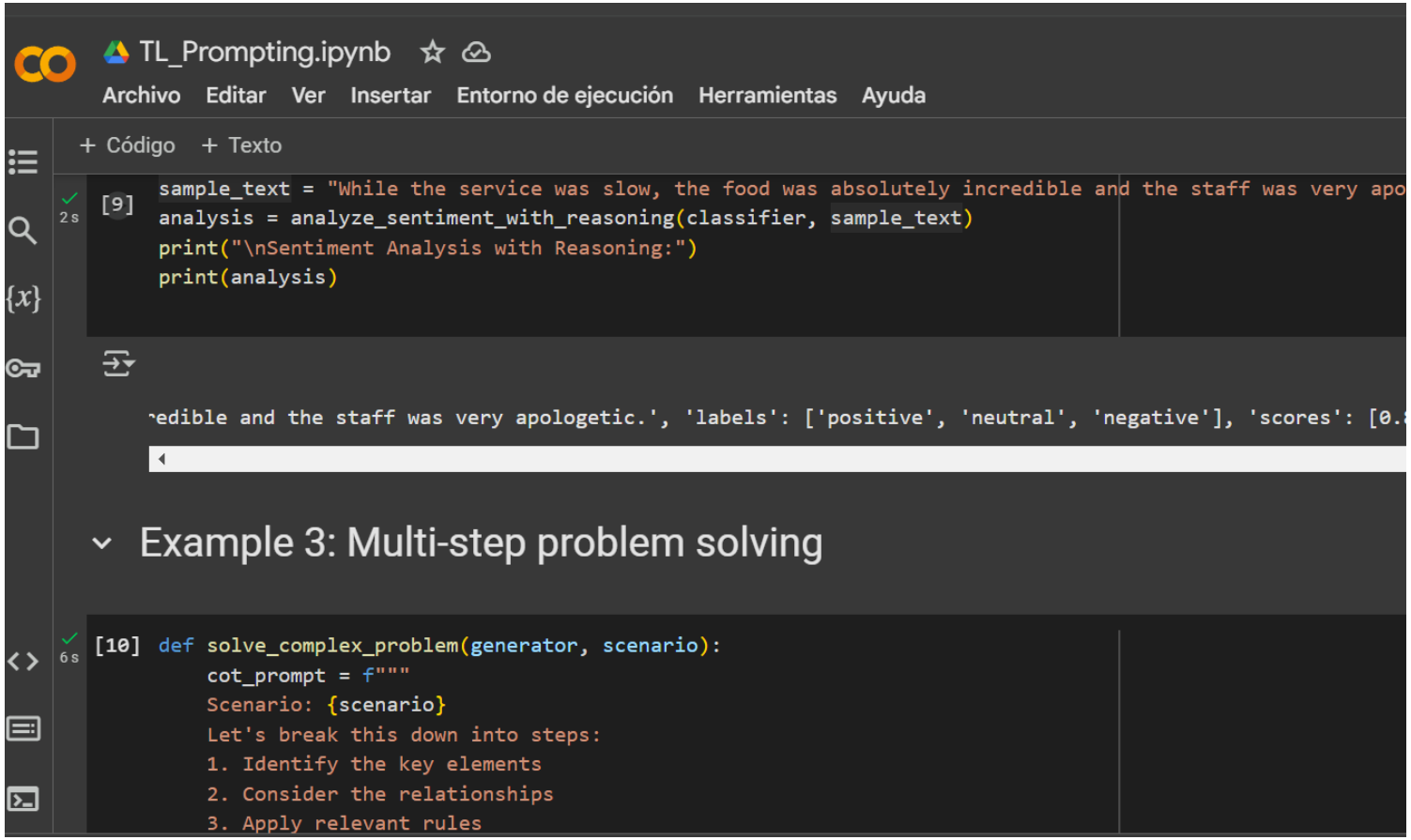
Output:

When we won the game, we all started to farduddle in celebration.

Chain-of-Thought



¡Ejemplo en Python!





Fine-tuning

Entrenamiento y Ajuste de Modelos

Entrenamiento

- **Entrenar el modelo**
  Mostrar al modelo una gran cantidad de datos para que pueda descubrir conexiones y aprender sobre la información.
- **Entrada:** Datos de entrenamiento
  Salida: Pesos del modelo base

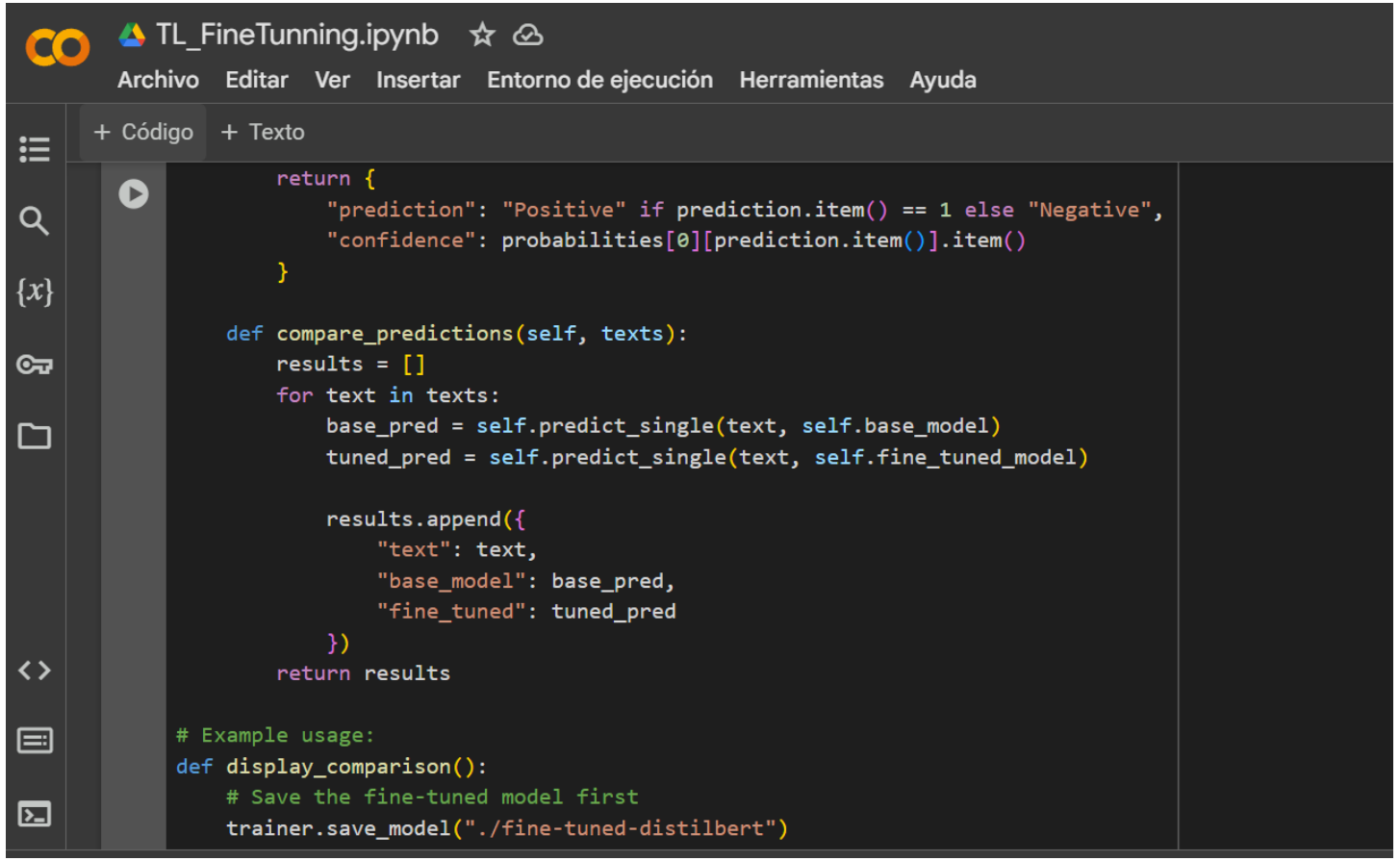
Ajuste fino supervisado

- **Enseñar al modelo cómo interactuar**
  Proporcionar al modelo un conjunto de guiones o patrones para las interacciones con las personas.
- **Entrada:** Pesos del modelo base
  Salida: Pesos del modelo ajustado

RLHF

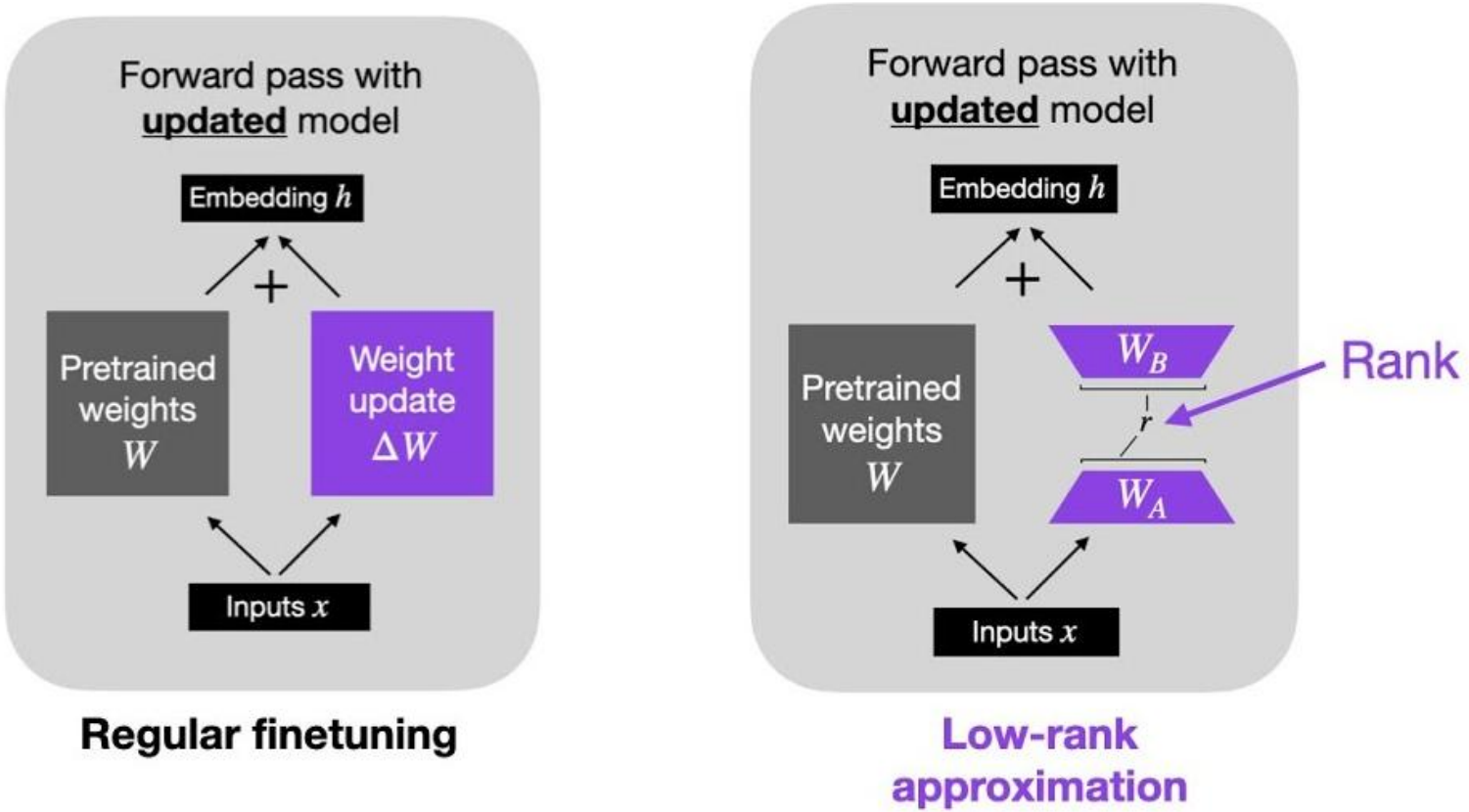
- **Enseñar al modelo 👍 & 👎**
  Dar al modelo un sentido de lo que es bueno y malo.
- **Entrada:** Pesos del modelo SFT
  Salida: Pesos del modelo ajustado

¡Ejemplo en Python!

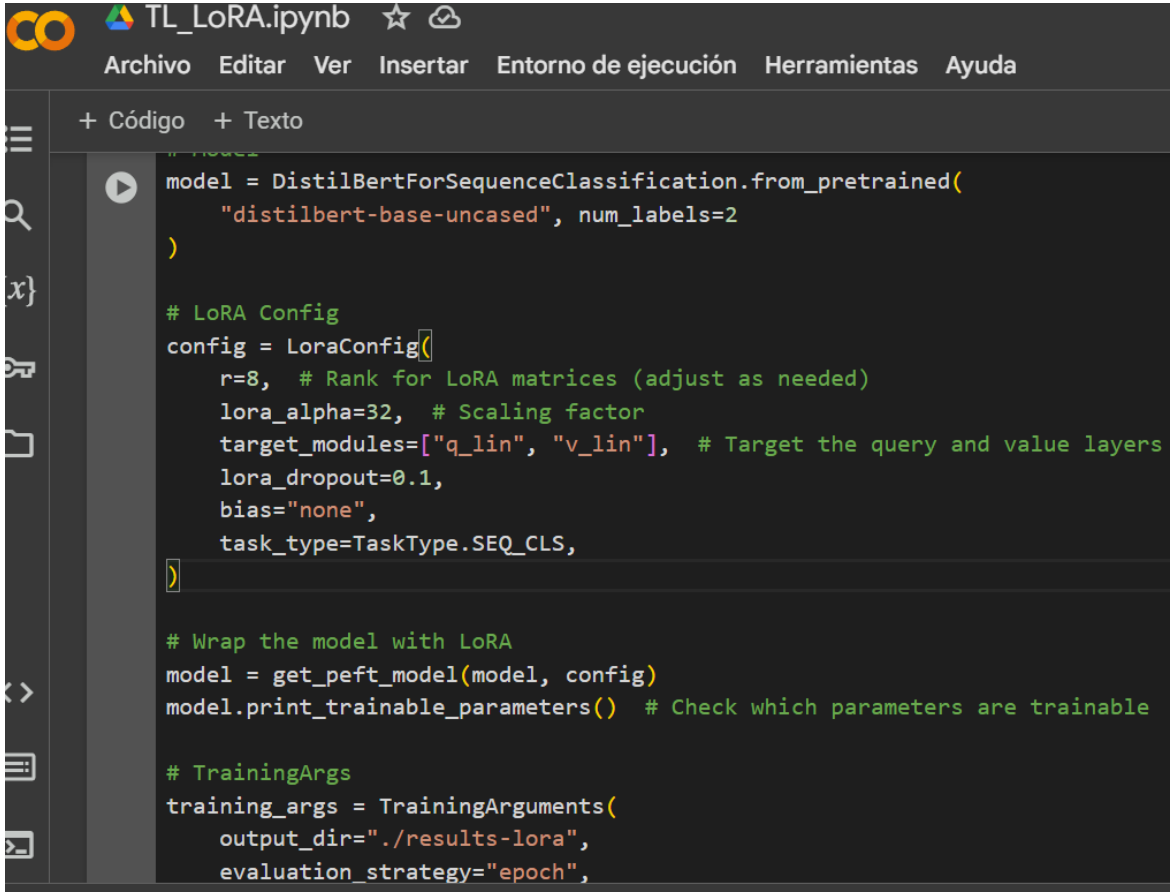


Low-Rank Adaptation (LoRA)

En lugar de actualizar todos los parámetros del modelo, LoRA **inyecta capas adicionales de bajo rango** que se ajustan durante el entrenamiento.



¡Ejemplo en Python!





Aprendizaje contrastivo

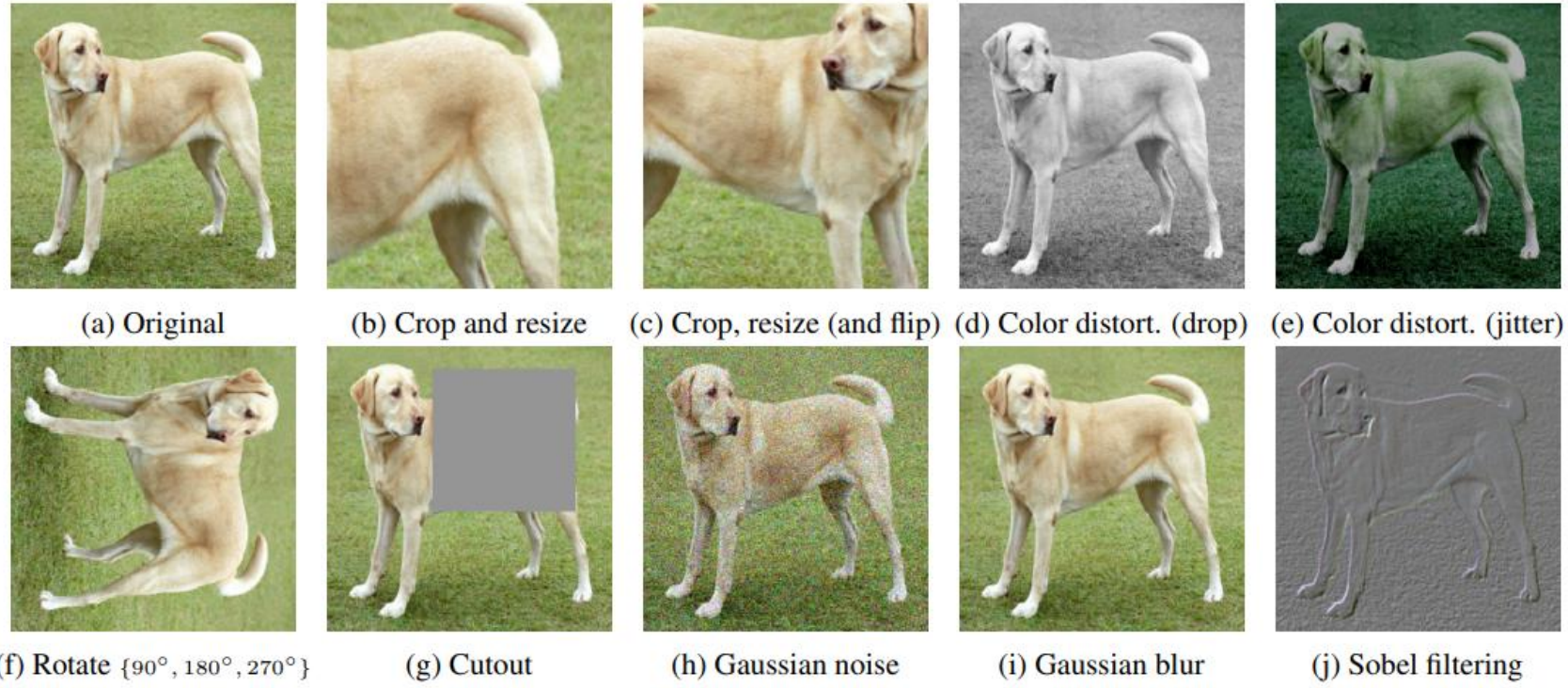


Aprendizaje contrastivo

- Un tema recurrente en este curso: una de las mayores ventajas del aprendizaje profundo es la capacidad de transferir conocimiento de una tarea de origen a una tarea de destino. Por ejemplo,
  - Entrene una red convolucional o ViT en ImageNet y ajústela en un conjunto de datos más pequeño
    - Quizás sea mejor ajustar solo la capa de clasificación final (a menudo llamada cabezal lineal o sonda lineal)
  - Perfeccionar un LLM en una tarea
  - Aprendizaje en contexto
- La ventaja de la transferencia es que la tarea de origen puede tener muchos más datos que la tarea de destino.
- Es ventajoso que la tarea de origen no esté supervisada (o sea autosupervisada) ya que es fácil recopilar muchos datos sin etiquetar.
  - Por ejemplo, el ajuste fino del LLM.
  - ¿Podemos hacer lo mismo con la visión?

Aprendizaje contrastivo

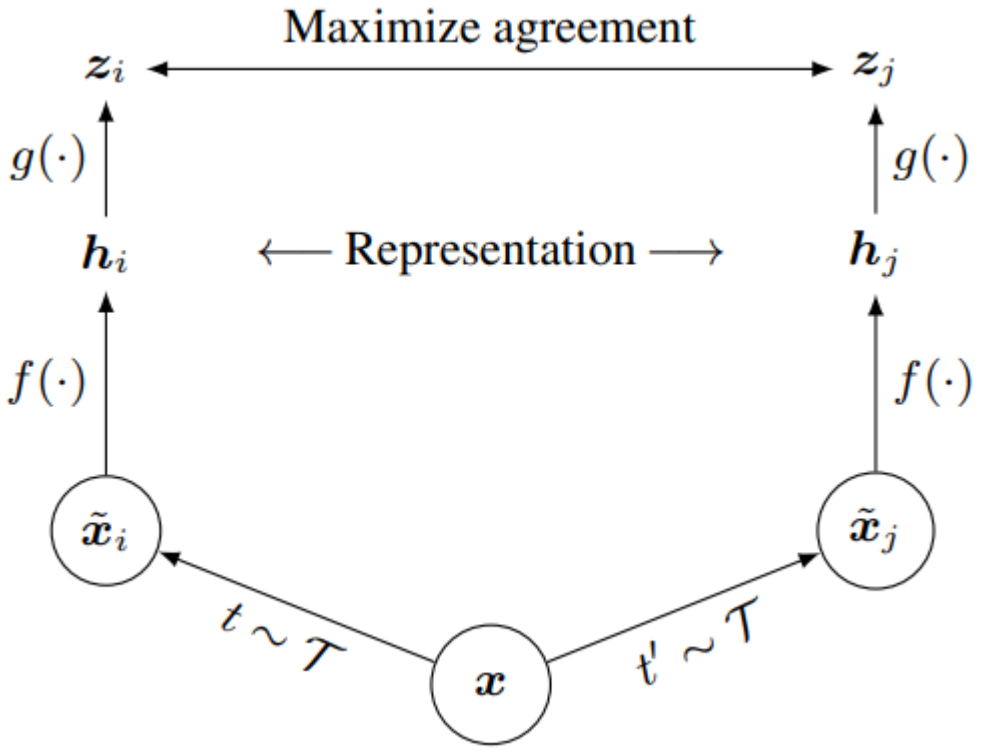
- ¿Cómo podemos entrenar buenas representaciones visuales de forma autosupervisada?
- Intuición: las características de una buenas para vision deben ser invariables a las transformaciones.
    - Las redes convolucionales poseen esto para las transformaciones de imágenes pequeñas, pero hay muchas más transformaciones interesantes.



Chen et al., 2020, “A simple framework for contrastive learning of visual representations”

Aprendizaje contrastivo

- El aprendizaje contrastivo simple de representaciones (SimCLR) (Chen et al., 2020) entrena una red para aprender representaciones que son invariantes a (algunas de) las transformaciones de la diapositiva anterior.
- La idea: muestrear dos transformaciones t y t′, ejecutar el codificador f de la red en ambas y penalizar su disimilitud.
- Pero debemos evitar la solución trivial donde todo se asigna a un vector constante. Las representaciones de imágenes no relacionadas deben ser diferentes.



Chen et al., 2020, “A simple framework for contrastive learning of visual representations”

Aprendizaje contrastivo

- Como sugiere el nombre, utilizan una pérdida contrastiva que hace que las imágenes relacionadas sean más similares que las imágenes no relacionadas.
- Muestrean un lote de 2N imágenes, es decir, 2 transformaciones independientes de la misma imagen base.
- Utilizan la entropía cruzada escalada por temperatura normalizada
- Objetivo (NT-Xent):

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(\boldsymbol{z}_i, \boldsymbol{z}_k)/\tau)}$$

Esto es básicamente pérdida de entropía cruzada para clasificar cuál de las 2N - 1 imágenes restantes es la relacionada.

- $\text{sim}(\boldsymbol{u}, \boldsymbol{v}) = \boldsymbol{u}^\top \boldsymbol{v} / \|\boldsymbol{u}\| \|\boldsymbol{v}\|$ es la similitud coseno.
- La función de similitud se usa en lugar de los logits de clasificación.
- τ es un parámetro de temperatura (mayor = predicciones más suaves).

Chen et al., 2020, “A simple framework for contrastive learning of visual representations”

Aprendizaje contrastivo

- La transferencia de aprendizaje desde un modelo SimCLR autosupervisado (entrenado en ImageNet no etiquetado) a varias tareas de visión funciona al menos tan bien como la transferencia desde un clasificador ImageNet supervisado.

	Food	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech-101	Flowers
Linear evaluation:												
SimCLR (ours)	**76.9**	**95.3**	80.2	48.4	**65.9**	60.0	61.2	**84.2**	**78.9**	89.2	**93.9**	**95.0**
Supervised	75.2	**95.7**	**81.2**	**56.4**	64.9	**68.8**	**63.8**	83.8	**78.7**	**92.3**	**94.1**	94.2
Fine-tuned:												
SimCLR (ours)	**89.4**	**98.6**	**89.0**	**78.2**	**68.1**	**92.1**	**87.0**	**86.6**	**77.8**	92.1	**94.1**	97.6
Supervised	88.7	98.3	**88.7**	**77.8**	67.0	91.4	**88.0**	86.5	**78.8**	**93.2**	**94.2**	**98.0**
Random init	88.3	96.0	81.9	**77.0**	53.7	91.3	84.8	69.4	64.1	82.7	72.5	92.5

Chen et al., 2020, “A simple framework for contrastive learning of visual representations”

Multimodalidad

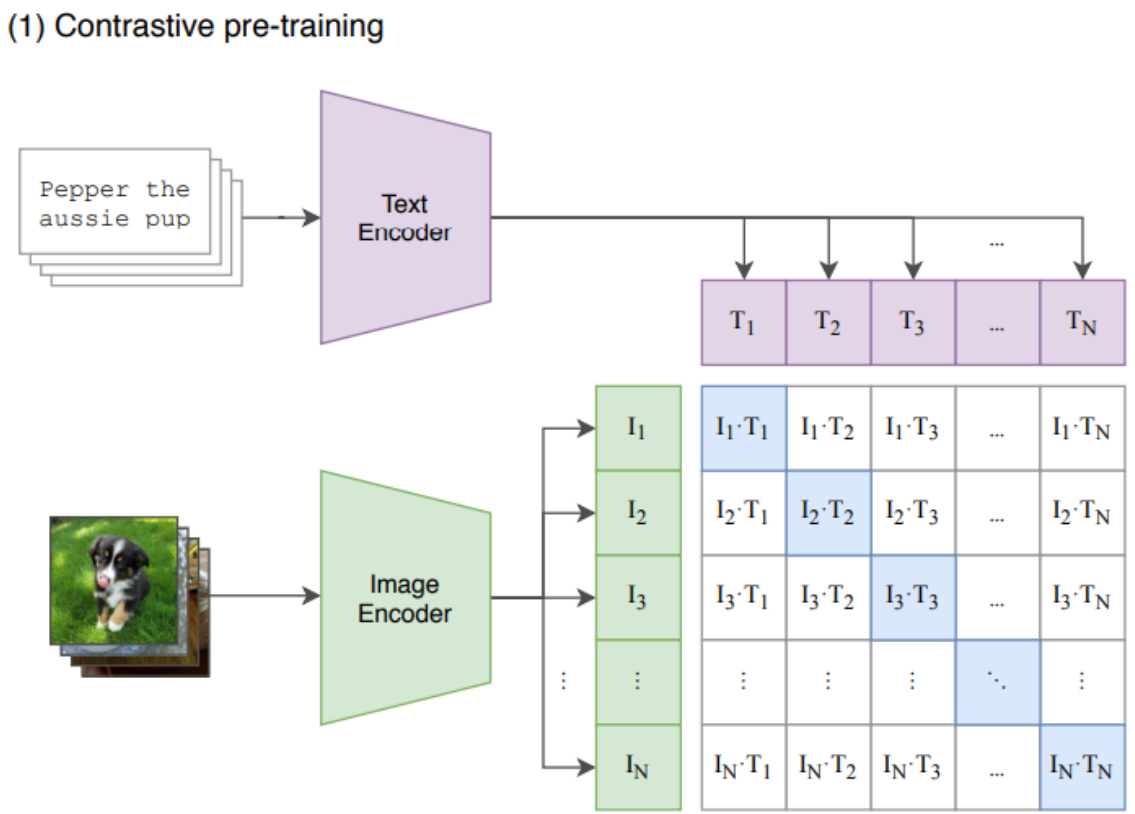
Multimodalidad

- ¿Se puede aprender a realizar una clasificación de imágenes a escala cero, es decir, sin un conjunto de datos etiquetado?
- Internet tiene muchísimas imágenes con texto. ¿Quizás podamos aprender de ellas?
- Preentrenamiento contrastivo de lenguaje e imagen (CLIP) (Radford et al., 2021)
  - Utiliza un codificador de visión y un codificador de texto, cada uno de los cuales produce una representación con valores vectoriales de una imagen o texto determinado.
  - Entrenado con un objetivo contrastivo simple.
    - Tomar un lote de B imágenes emparejadas con texto asociado
    - Para cada imagen, hacer que el modelo prediga cuál de los ejemplos de texto está emparejado con ella (usando la similitud de sus representaciones)

Radford et al., 2021, “Learning transferable visual models from natural language supervision”

Multimodalidad

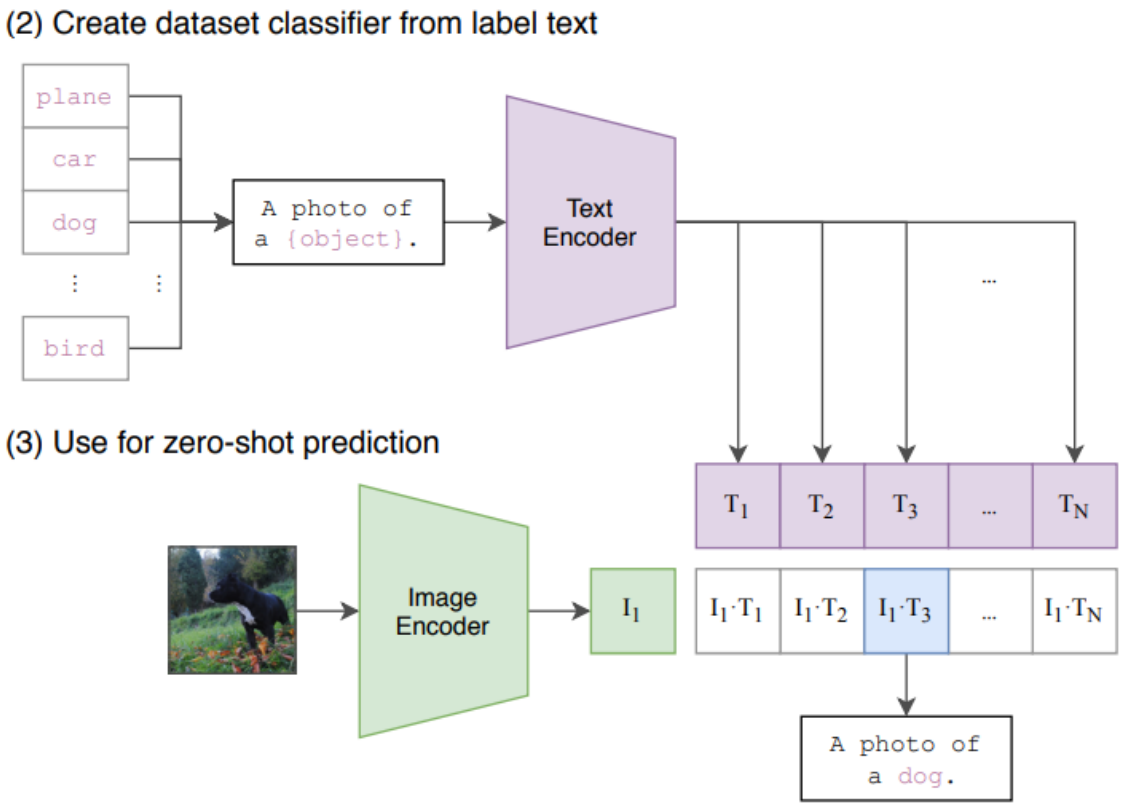
Entrenamiento previo CLIP:



Radford et al., 2021, “Learning transferable visual models from natural language supervision”

Multimodalidad

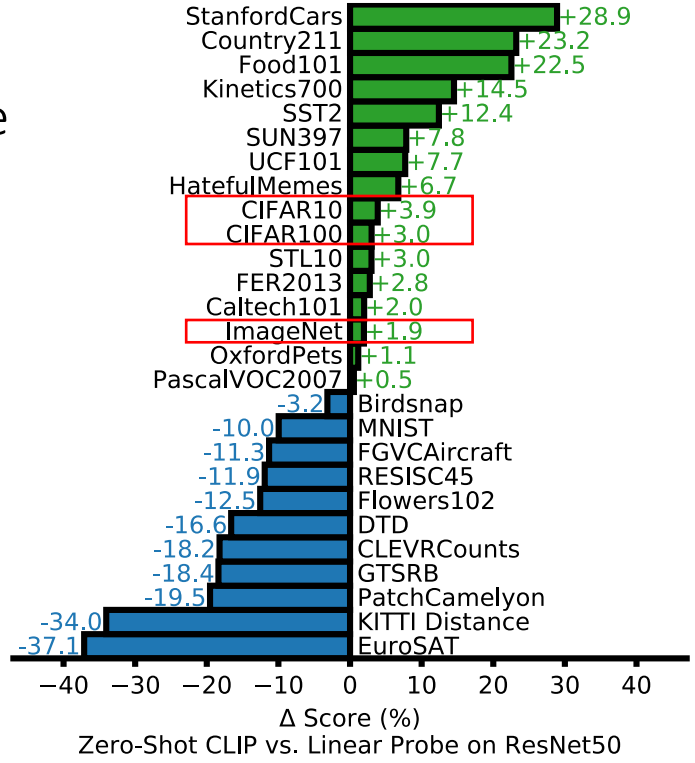
Para clasificar una imagen de ImageNet, simplemente evalúa la coincidencia con descripciones de la forma "Una foto de un _____".



Radford et al., 2021, “Learning transferable visual models from natural language supervision”

Multimodalidad

¡El rendimiento de zero shot de CLIP es competitivo (major que ResNet-50)!



Radford et al., 2021, “Learning transferable visual models from natural language supervision”



Modelos Generativos



Peruvian Inca astronaut on the moon



make it more peruvian



even more peruvian

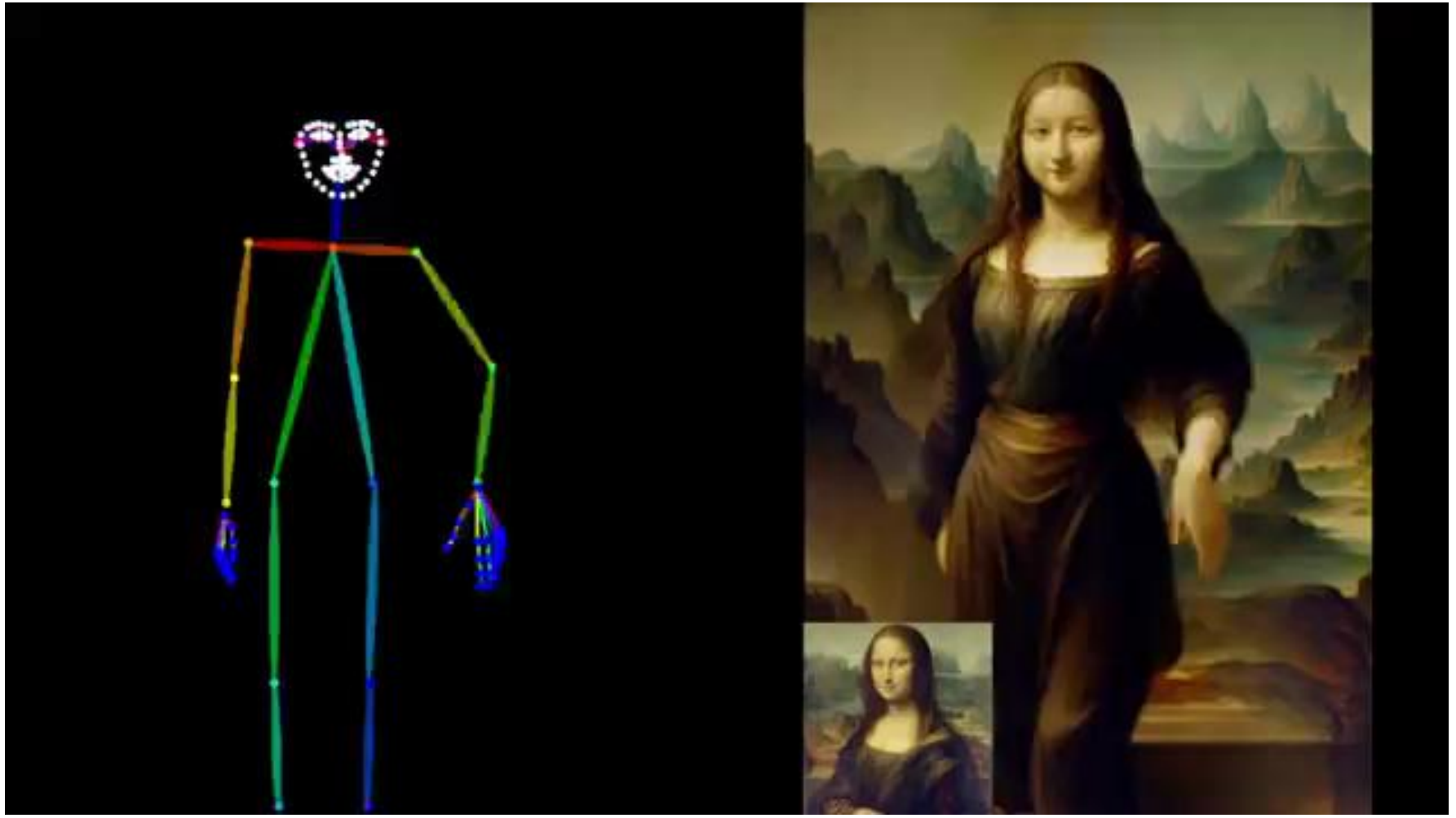


go extreamly more peruvian



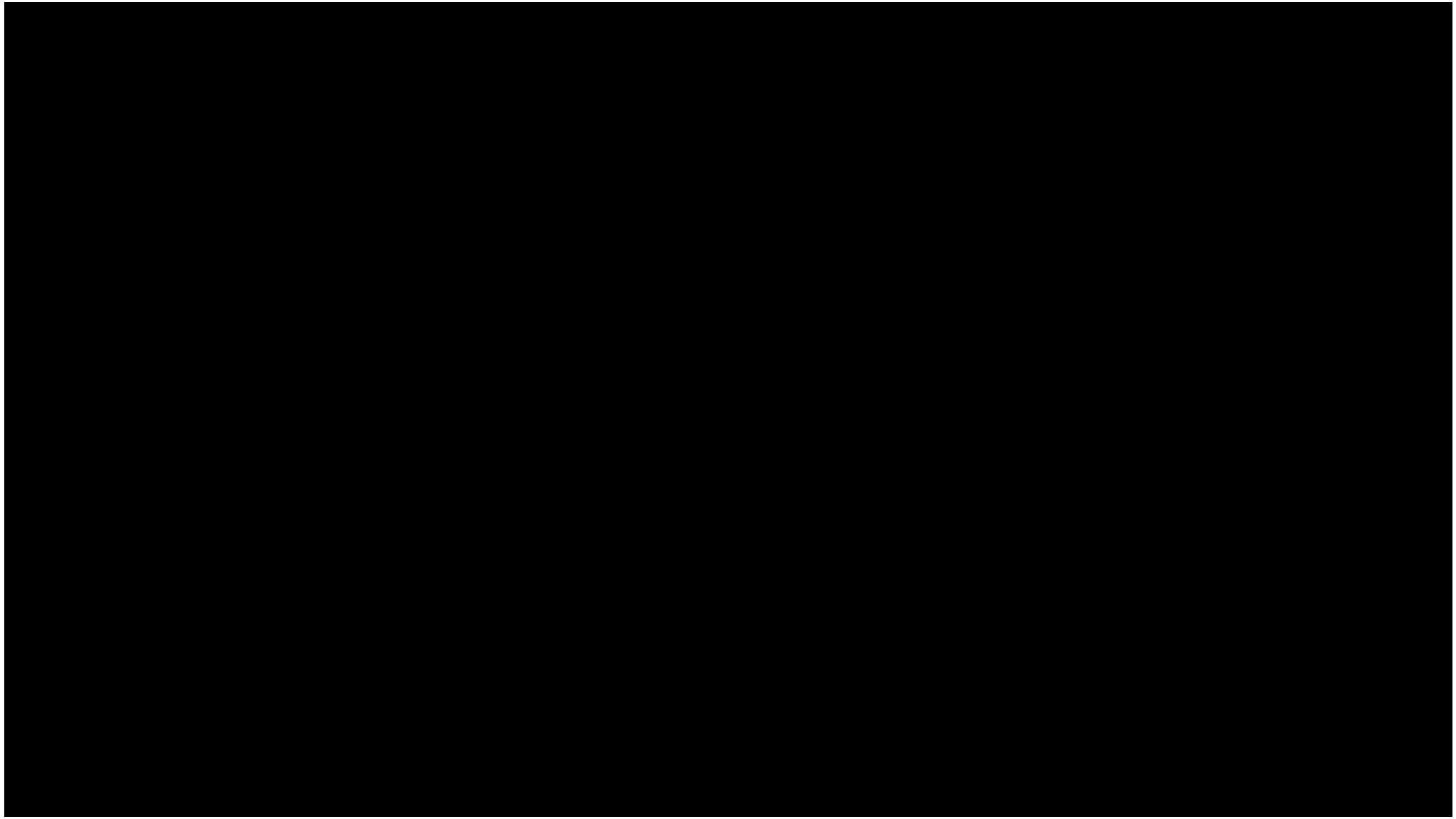
more peruvian

Habilidades de estos modelos desde el 2023



Desatando el Poder de los Modelos de Difusión

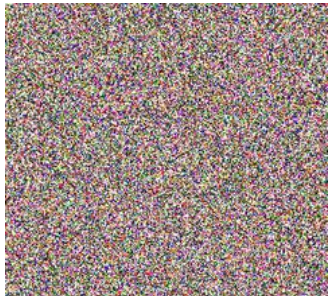




Los modelos generativos generan muestras a partir de la distribución de datos

Un modelo generativo convierte muestras de una distribución inicial (por ejemplo, gaussiana) en muestras de la distribución de datos:

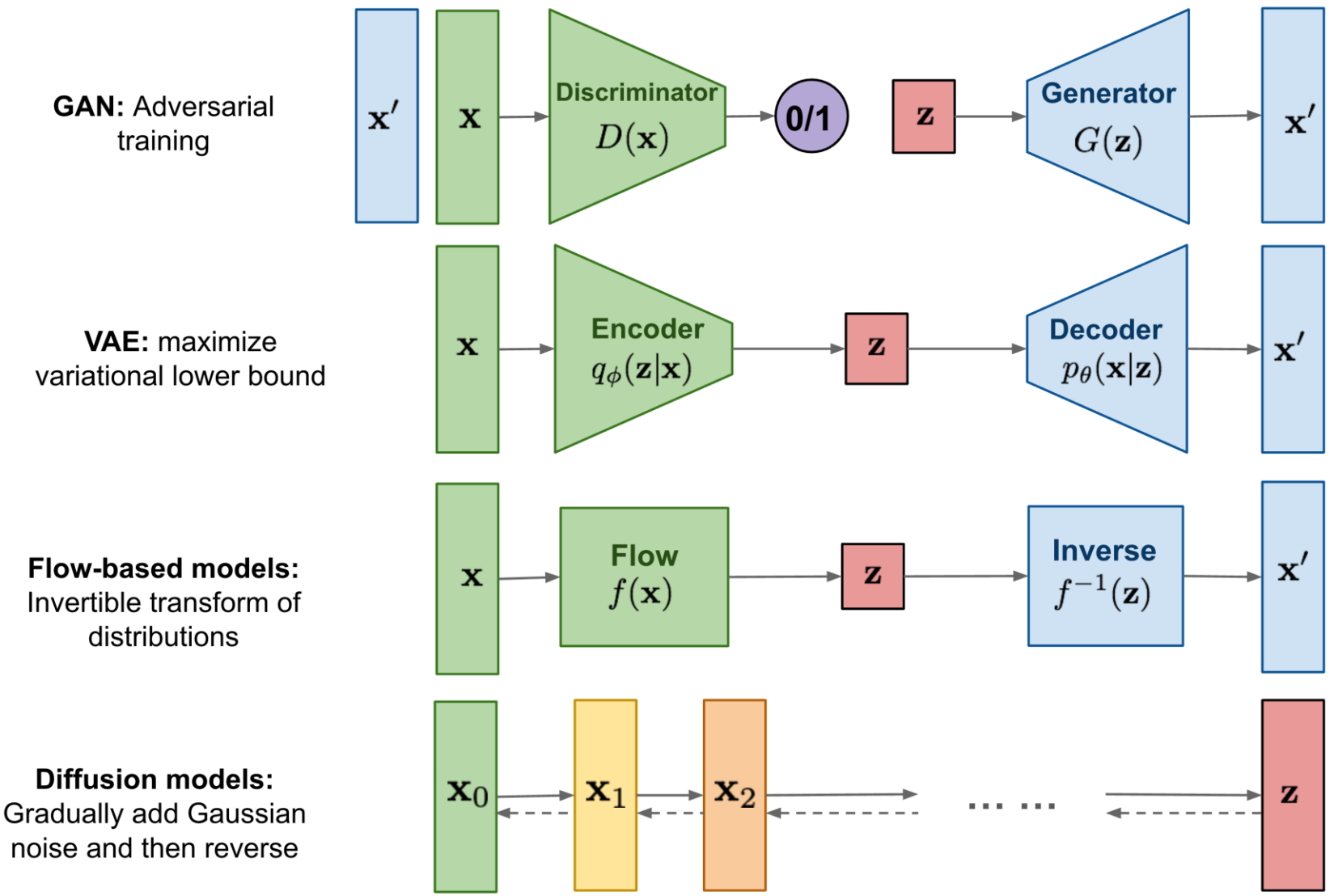
$$x \sim p_{\text{init}}$$



Generativo
Modelo

$$z \sim p_{\text{data}}$$




GAN: Adversarial training
$\mathbf{x}'$
$\mathbf{x}$
Discriminator
$D(\mathbf{x})$
0/1
$\mathbf{z}$
Generator
$G(\mathbf{z})$
$\mathbf{x}'$
VAE: maximize variational lower bound
$\mathbf{x}$
Encoder
$q_\phi(\mathbf{z}|\mathbf{x})$
$\mathbf{z}$
Decoder
$p_\theta(\mathbf{x}|\mathbf{z})$
$\mathbf{x}'$
Flow-based models: Invertible transform of distributions
$\mathbf{x}$
Flow
$f(\mathbf{x})$
$\mathbf{z}$
Inverse
$f^{-1}(\mathbf{z})$
$\mathbf{x}'$
Diffusion models: Gradually add Gaussian noise and then reverse
$\mathbf{x}_0$
$\mathbf{x}_1$
$\mathbf{x}_2$
... ...
$\mathbf{z}$



Modelos de difusión

Todo Comenzó…



v8 [cs.LG] 18 Nov 2015

Deep Unsupervised Learning using Nonequilibrium Thermodynamics

Jascha Sohl-Dickstein
Stanford University

Eric A. Weiss
University of California, Berkeley

Niru Maheswaranathan
Stanford University

Surya Ganguli
Stanford University

Abstract

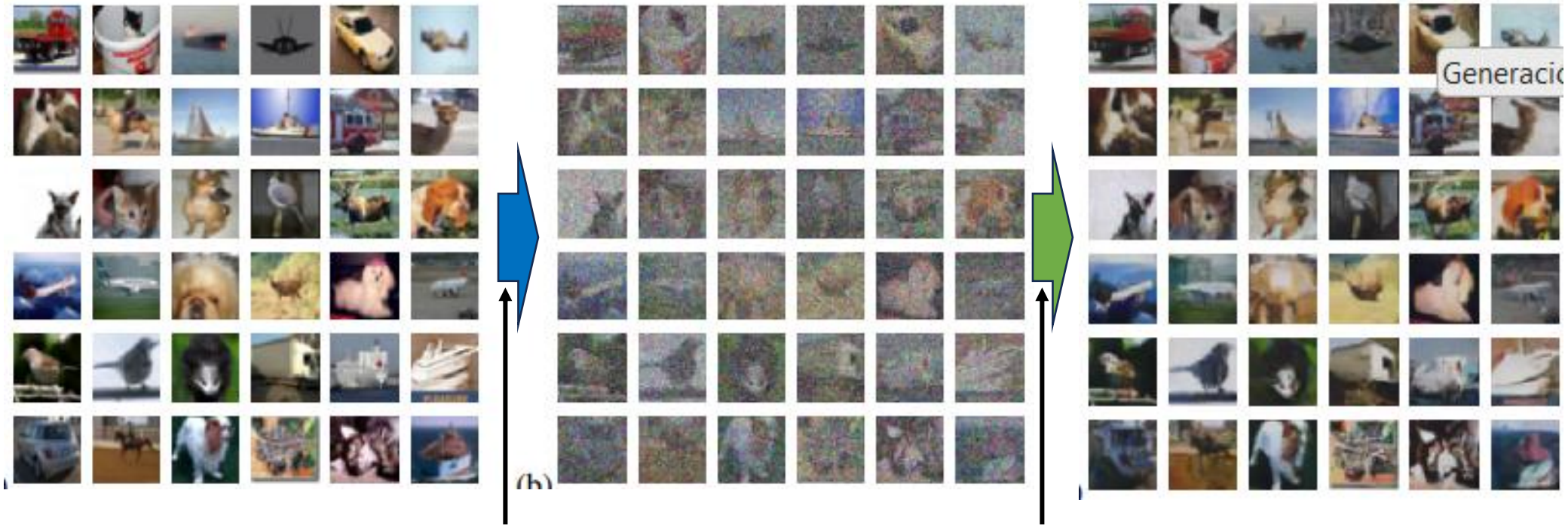
A central problem in machine learning involves modeling complex data-sets using highly flexible families of probability distributions in which learning, sampling, inference, and evaluation are still analytically or computationally tractable. Here, we develop an approach that simultaneously achieves both flexibility and tractability.

these models are unable t
datasets. On the other han
molded to fit structure in
can define models in term
$\phi(\mathbf{x})$ yielding the flexible
Z is a normalization cons
normalization constant is g
training, or drawing samples from such flexible models typ-



- **Publicado en 2015:** El concepto inicial de DDPM se presentó en 2015.
- **Basado en Conceptos Físicos:** Utilizaba una metodología inspirada en procesos de difusión física para la generación de datos.
- **No Alcanzó Resultados Prometedores**: El modelo inicial no logró resultados de vanguardia y fue superado en rendimiento por otros modelos generativos contemporáneos.

La idea original



AGREGAR RUIDO
(APRENDER)

QUITAR RUIDO
(APLICAR LO
APRENDIDO)

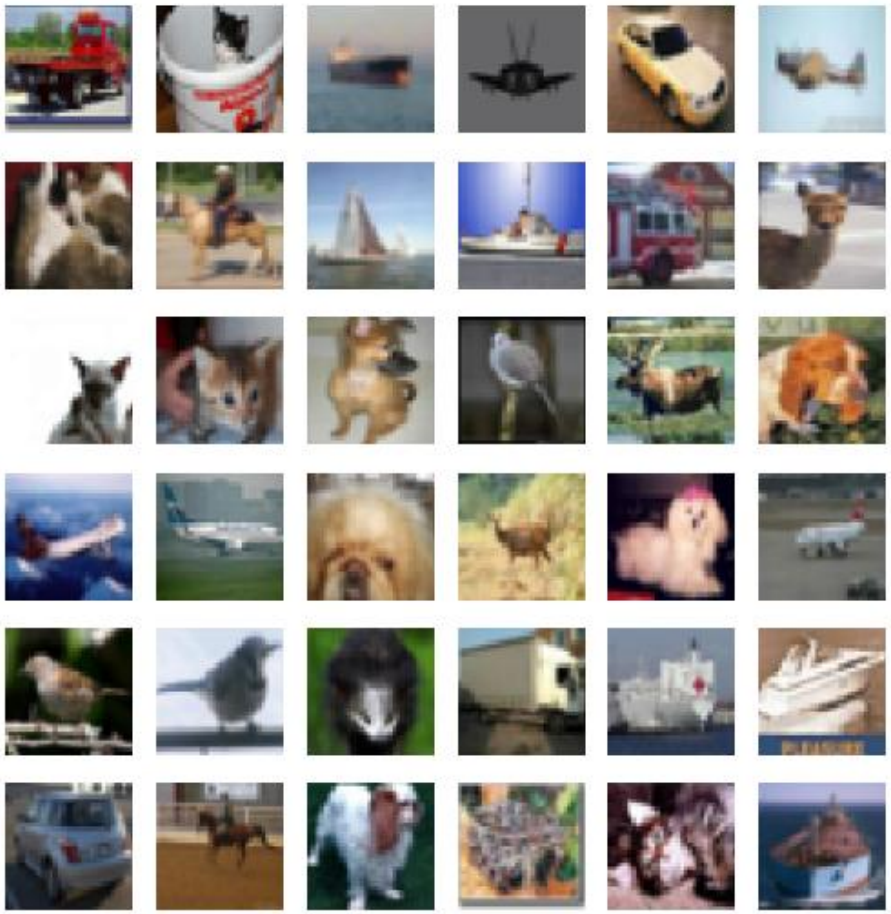
Extendiendo a perder toda señal



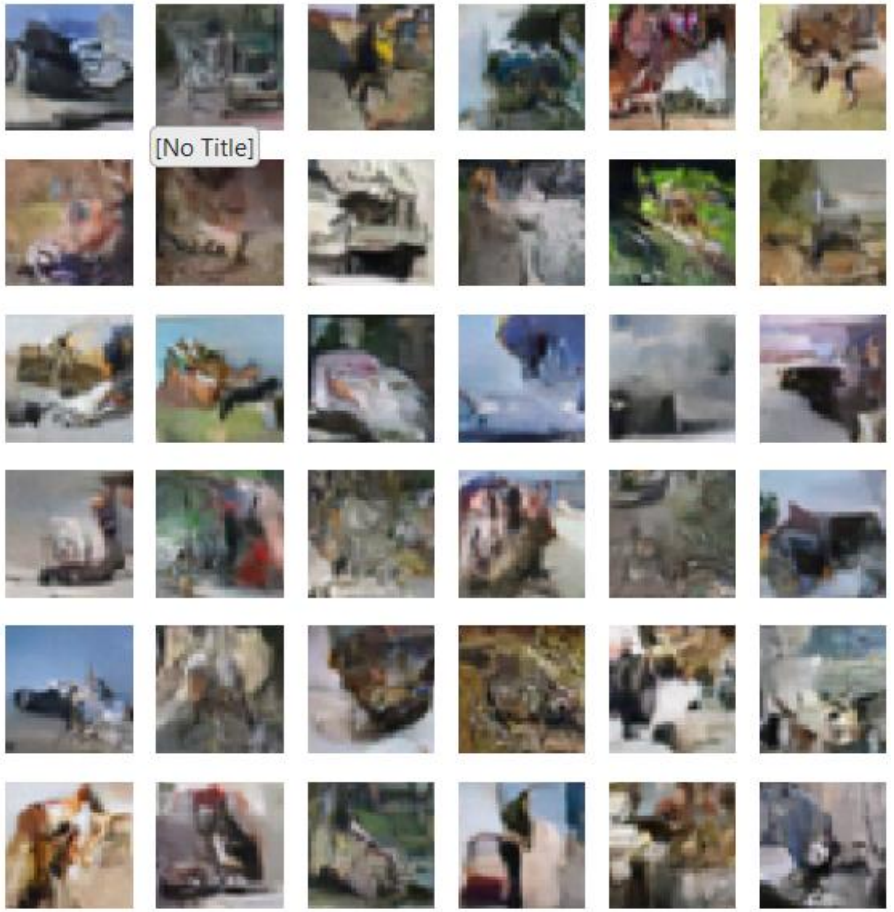
Y crear imágenes quitando el ruido



Generación de CIFAR10 en 2015



Dataset



Modelo de difusión

La Influencia de los Transformers



Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly

Primeros Modelos de DDPM (2015-2016):

- Inicio con modelos simples de DDPM, centrados en demostrar la viabilidad de la difusión de ruido en la generación de datos.

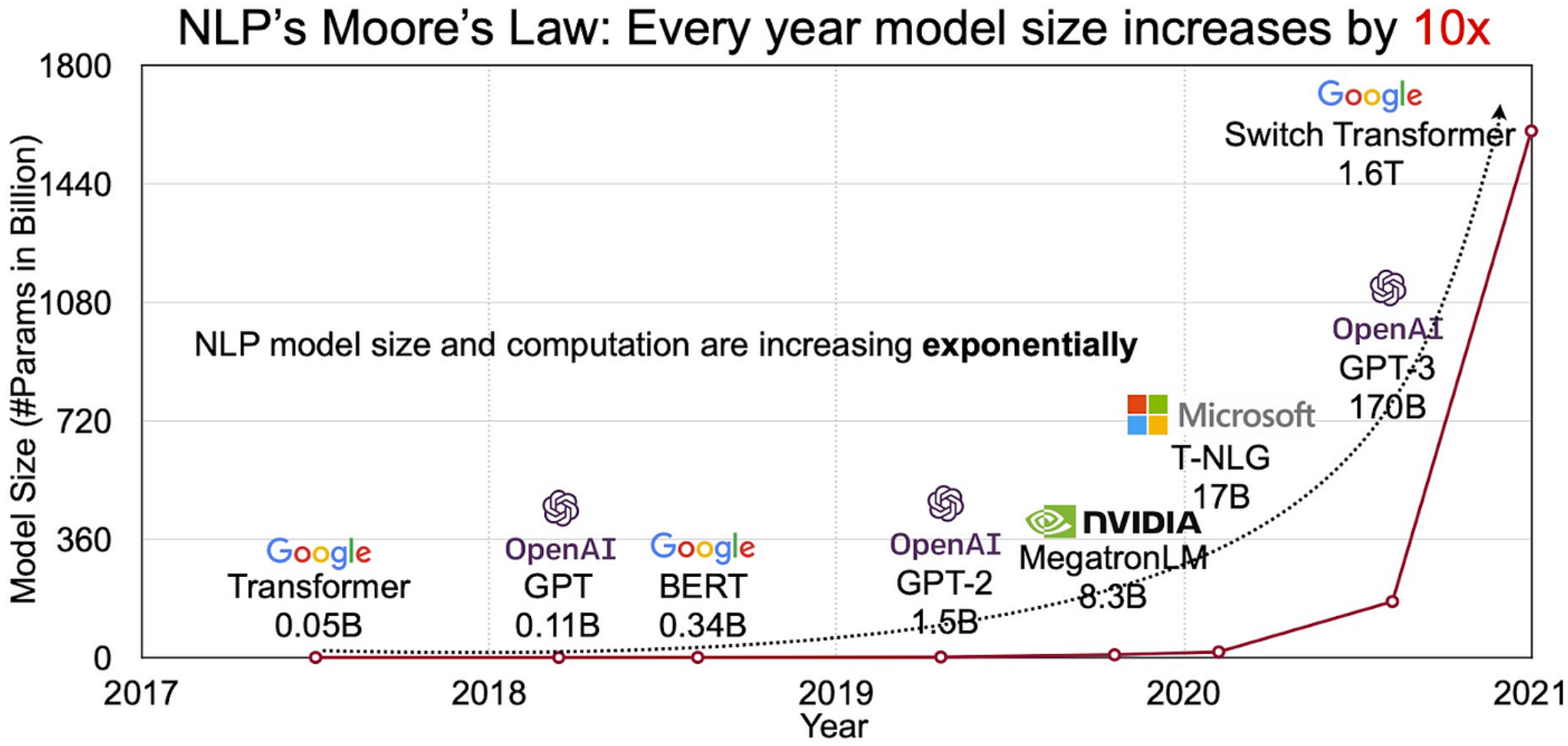
Aparición de los Transformers (2017):

- Lanzamiento de los Transformers, revolucionando el manejo de secuencias de datos y parámetros en aprendizaje profundo.

Inspiración y Cambio de Paradigma (2018-2019):

- El éxito de los Transformers inspira una nueva dirección en el diseño de modelos, incluyendo DDPMs, hacia estructuras más complejas y de mayor escala.
- Este período establece las bases para investigaciones futuras en DDPMs, marcando el inicio de una era de modelos más grandes y computacionalmente intensivos.

Aumento Exponencial de Parámetros



*más de esto después

Aumento de la escala para predecir el “Ruido”

9v2 [cs.LG] 16 Dec 2020

Denoising Diffusion Probabilistic Models

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Abstract

We present high quality image synthesis results using diffusion probabilistic models, a class of latent variable models inspired by considerations from nonequilibrium thermodynamics. Our best results are obtained by training on a weighted variational bound designed according to a novel connection between diffusion probabilistic models and denoising score matching with Langevin dynamics, and our models naturally admit a progressive lossy decompression scheme that can be interpreted as a generalization of autoregressive decoding. On the unconditional CIFAR10 dataset, we obtain an Inception score of 9.46 and a state-of-the-art FID score of 3.17. On 256x256 LSUN, we obtain sample quality similar to ProgressiveGAN. Our implementation is available at https://github.com/hojonathanho/diffusion.

1 Introduction

Deep generative models of all kinds have recently exhibited high quality samples in a wide variety of data modalities. Generative adversarial networks (GANs), autoregressive models, flows, and

- **Publicado en 2020:** "Denoising Diffusion Probabilistic Models" redefine los DDPMs.
- **Innovación:** Introduce un enfoque eficiente para revertir el proceso de difusión, usando una U-Net para invertir el ruido.
- **Mejoras Significativas:** Supera a otros modelos generativos en calidad y detalle, especialmente en la generación de imágenes.
- **Impulso** a la Investigación Futura: Establece un nuevo estándar.

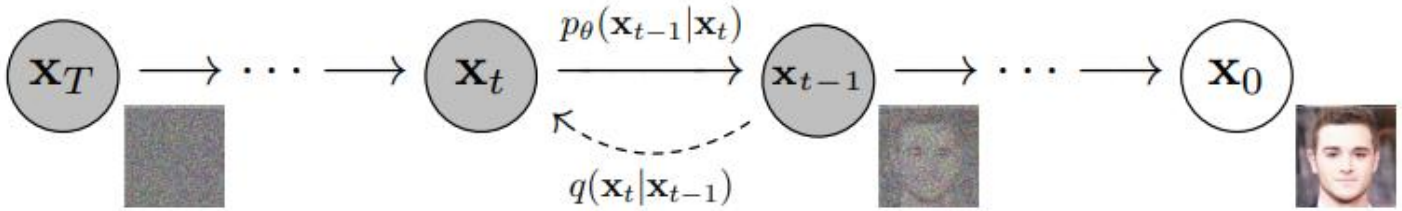
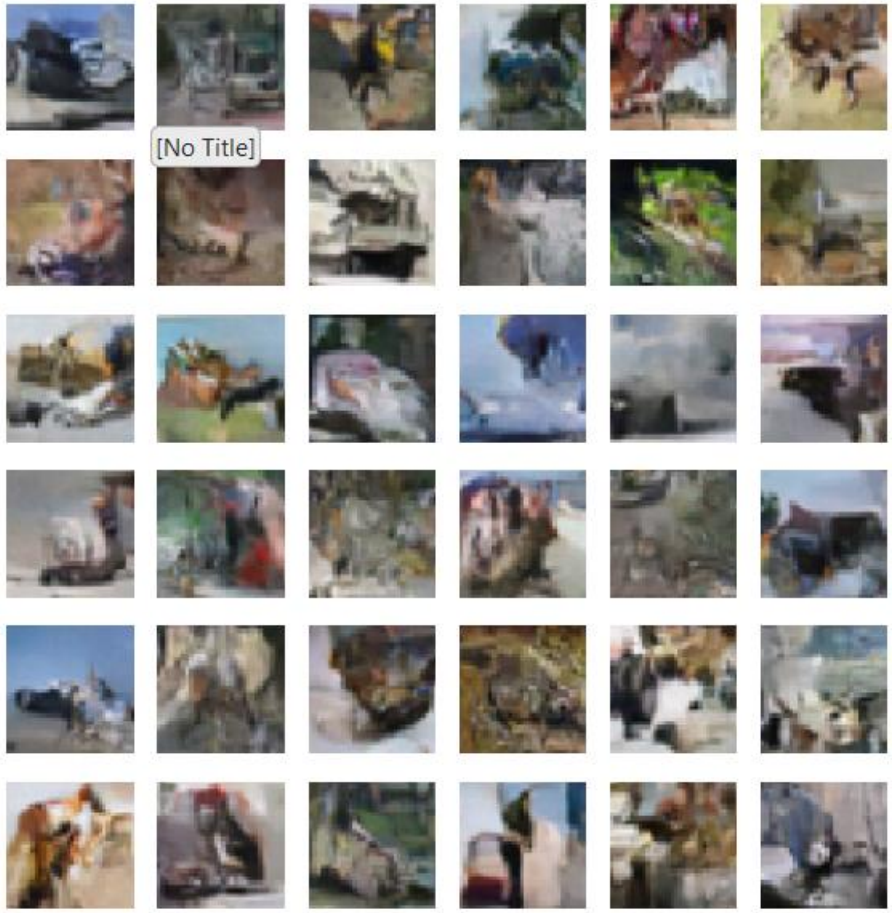


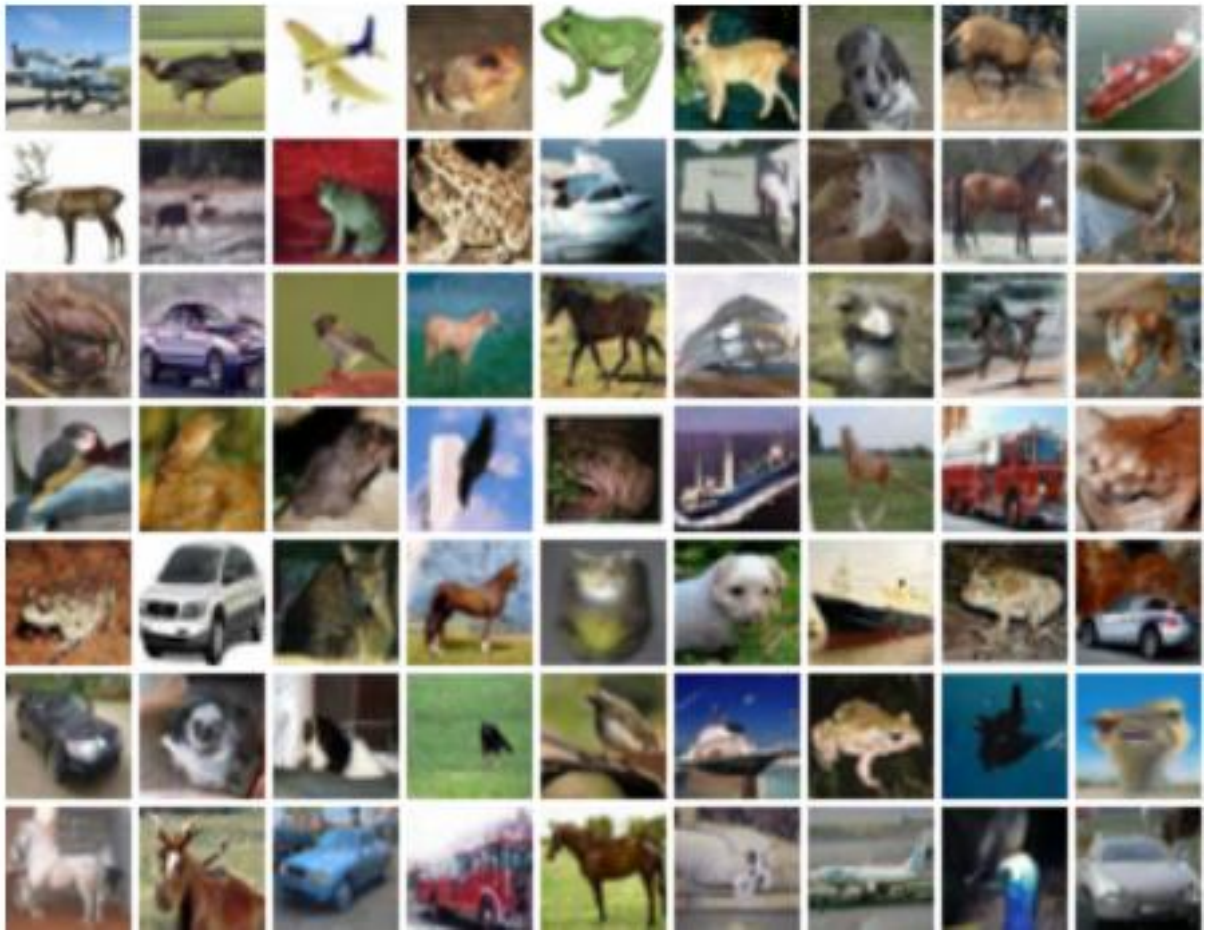
Figure 2: The directed graphical model considered in this work.



Generación de CIFAR10 en 2020



2015



2020

El Nuevo Chico en la Cuadra

Diffusion Models Beat GANs on Image Synthesis

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Abstract

We show that diffusion models can achieve image sample quality superior to the current state-of-the-art generative models. We achieve this on unconditional image synthesis by finding a better architecture through a series of ablations. For conditional image synthesis, we further improve sample quality with classifier guidance: a simple, compute-efficient method for trading off diversity for fidelity using gradients from a classifier. We achieve an FID of 2.97 on ImageNet 128×128, 4.59 on ImageNet 256×256, and 7.72 on ImageNet 512×512, and we match BigGAN-deep even with as few as 25 forward passes per sample, all while maintaining better coverage of the distribution. Finally, we find that classifier guidance combines well with upsampling diffusion models, further improving FID to 3.94 on ImageNet 256×256 and 3.85 on ImageNet 512×512.

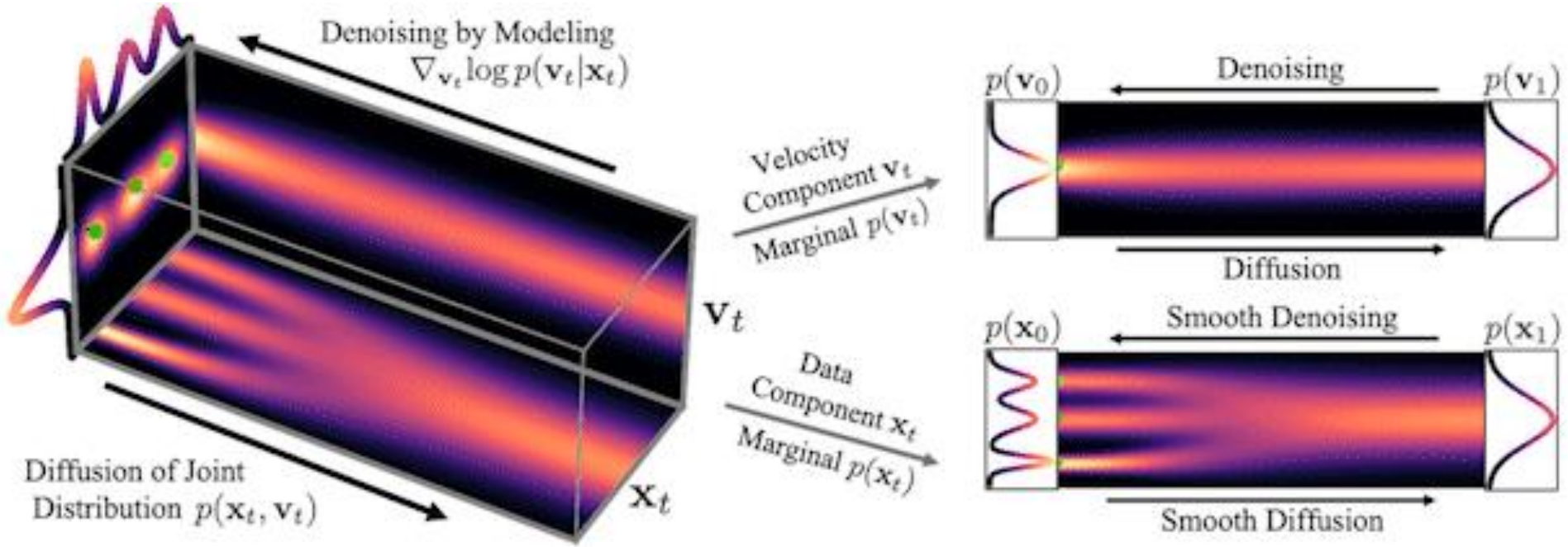
1 Introduction



*2021

DALL-E	**Jan 2021**	OpenAI
Midjourney V1	**Feb 2022**	Midjourney, Inc.
Midjourney V2	**Apr 2022**	
Midjourney V3	**Jul 2022**	
Stable Diffusion	**Aug 2022**	Stability AI
DALL-E 2	**Sep 2022**	OpenAI
Midjourney V4	**Nov 2022**	Midjourney, Inc.
Midjourney V5	**Mar 2023**	

¿Cómo funcionan estos modelos?



Forward
Diffusion Process



Image $\mathbf{x}_t$



Velocity $\mathbf{v}_t$

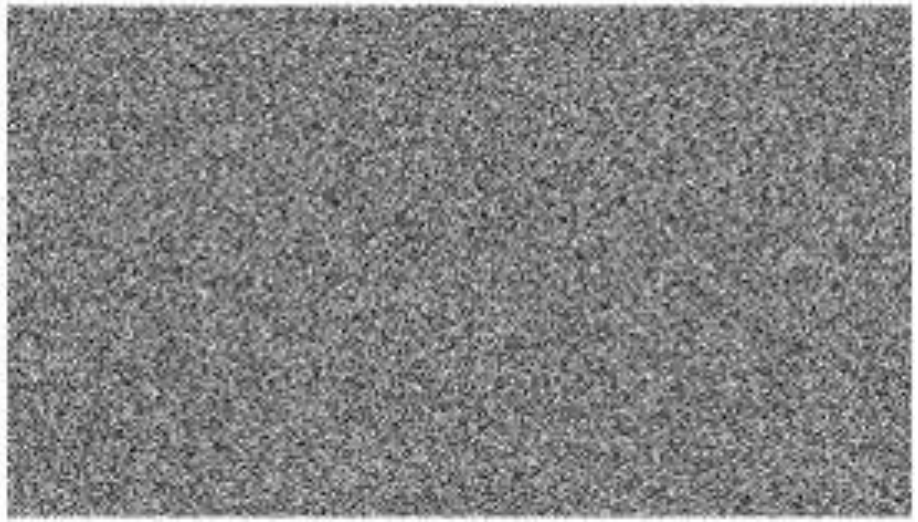
$t = 0.00$

Entender el “ruido” es lo principal



Add noise at training time

Denoise at synthesis time



Caso “La Tinka”



¿Ven el patrón?

23	5	17	9	34	13
46	5	45	19	29	26
36	27	28	26	29	3
11	37	7	46	42	15
49	19	35	9	17	18
41	38	46	9	40	31
46	2	17	48	15	40
4	24	37	36	14	2
42	15	22	9	7	18
6	27	32	16	24	47
28	9	16	1	46	40
8	50	24	33	10	15
29	31	8	47	12	10
31	17	28	9	21	5
15	20	48	37	27	23
14	9	2	25	8	6
47	50	20	13	43	17
43	47	26	35	15	12

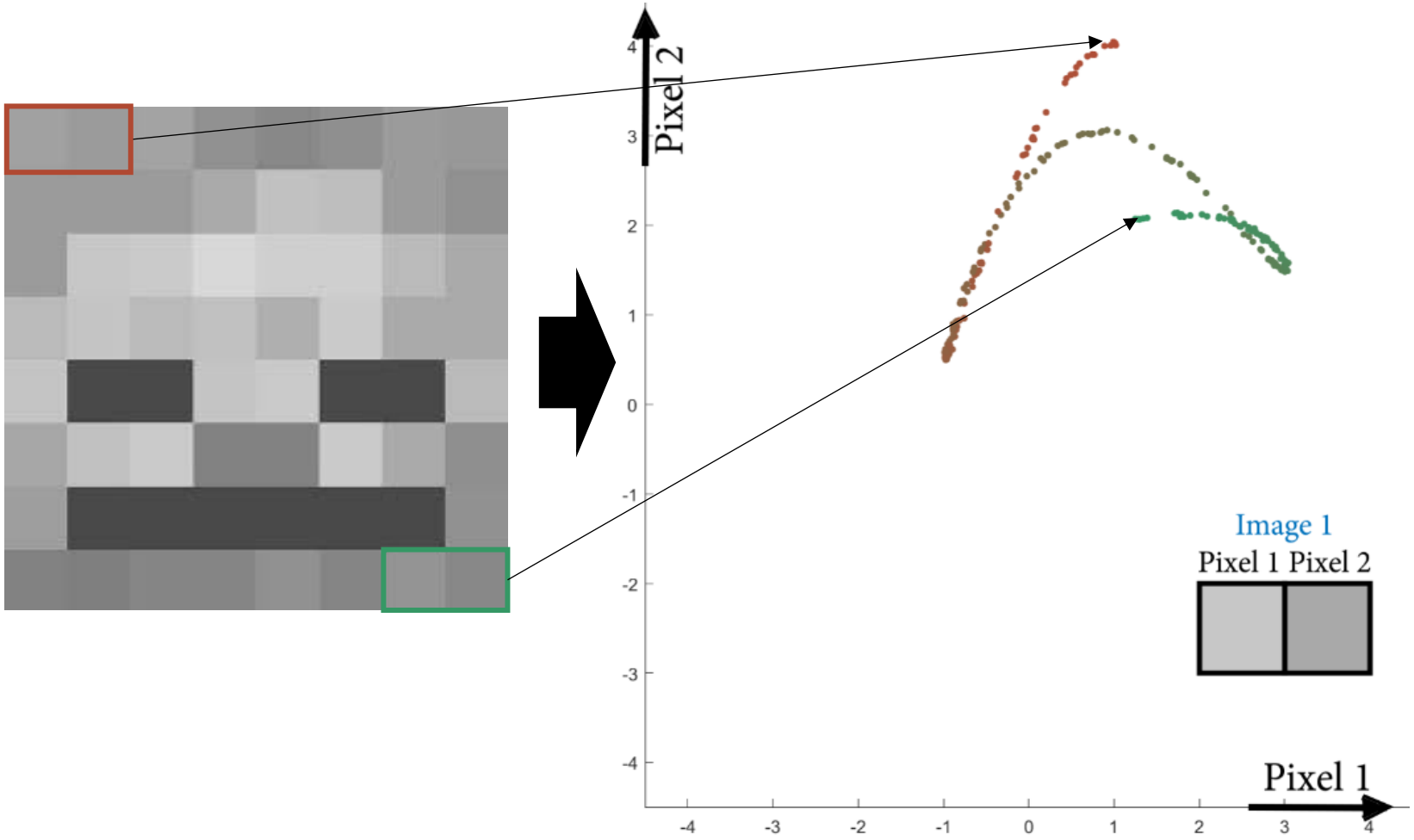


Ruido “aleatorio” puro

0.633383	0.27997	0.41456	0.671842	0.462371	0.442638	0.978315	0.629697	0.428815	0.383954	0.606949	0.431341	0.876385	0.238465	0.135506	0.142719	0.559753	0.519929	0.23130
0.380664	0.959933	0.653	0.115928	0.394821	0.019515	0.589329	0.017222	0.347459	0.984675	0.498132	0.883641	0.020806	0.719551	0.744855	0.308554	0.689079	0.326431	0.5584
0.435807	0.789487	0.733218	0.437333	0.179527	0.28529	0.59595	0.331386	0.156246	0.064081	0.137518	0.452414	0.044942	0.882833	0.019033	0.739416	0.877952	0.825258	0.59444
0.202936	0.370843	0.284145	0.882744	0.133493	0.644995	0.999096	0.919023	0.951586	0.945528	0.28035	0.917567	0.8832	0.161098	0.956	0.418125	0.113856	0.782204	0.18601
0.066574	0.927593	0.937354	0.456617	0.081768	0.766895	0.997206	0.617906	0.290246	0.115389	0.083266	0.419644	0.676305	0.192073	0.504457	0.101491	0.673661	0.909406	0.93518
0.72244	0.537	0.708692	0.697849	0.819158	0.917097	0.088864	0.31714	0.083714	0.614569	0.965145	0.440345	0.123262	0.659462	0.349817	0.895672	0.319863	0.204507	0.16285
0.422379	0.690746	0.387203	0.093788	0.64856	0.111821	0.38488	0.769695	0.014247	0.718511	0.331176	0.473251	0.331603	0.053731	0.274391	0.81464	0.969926	0.129251	0.01351
0.466008	0.643775	0.103772	0.273387	0.294557	0.620536	0.922437	0.190784	0.484889	0.542677	0.021487	0.120046	0.616444	0.881692	0.601499	0.855787	0.182124	0.866664	0.10224
0.301962	0.397797	0.961867	0.229834	0.409885	0.454353	0.996437	0.192568	0.629665	0.255687	0.150935	0.151739	0.671689	0.201022	0.259653	0.436718	0.61942	0.013653	0.5440
0.919545	0.565603	0.366608	0.163568	0.37308	0.112871	0.605056	0.898861	0.230349	0.795183	0.66931	0.465332	0.717983	0.150031	0.472154	0.166908	0.719506	0.500278	0.08111
0.879208	0.71874	0.621159	0.446365	0.330321	0.231229	0.01903	0.398505	0.352055	0.399383	0.229718	0.439555	0.259997	0.861502	0.292769	0.671323	0.59844	0.88175	0.71064
0.024089	0.162901	0.848536	0.805971	0.221097	0.678092	0.447493	0.578128	0.630915	0.446219	0.125154	0.371411	0.468723	0.253343	0.674952	0.039599	0.275928	0.07215	0.09053
0.144136	0.692886	0.102345	0.922071	0.27846	0.161075	0.036924	0.08468	0.882051	0.757441	0.848136	0.109927	0.875916	0.883092	0.141557	0.697067	0.492631	0.781786	0.91644
0.16256	0.431114	0.84911	0.59824	0.739524	0.338055	0.930245	0.945404	0.225559	0.092341	0.995564	0.569923	0.870517	0.521439	0.373577	0.830292	0.585411	0.256601	0.20496
0.367402	0.854557	0.672978	0.12829	0.901532	0.848211	0.19184	0.238983	0.634386	0.086751	0.559391	0.024547	0.494386	0.632646	0.542174	0.749229	0.45338	0.551574	0.54969
0.416931	0.923702	0.619204	0.150609	0.589722	0.686874	0.703929	0.134971	0.138123	0.369539	0.059933	0.793277	0.670178	0.123856	0.838121	0.21058	0.559885	0.533298	0.72212
0.816756	0.203664	0.671656	0.765249	0.039006	0.449085	0.109644	0.283344	0.179848	0.740615	0.694241	0.211685	0.748871	0.907466	0.621784	0.510421	0.565964	0.463038	0.31494

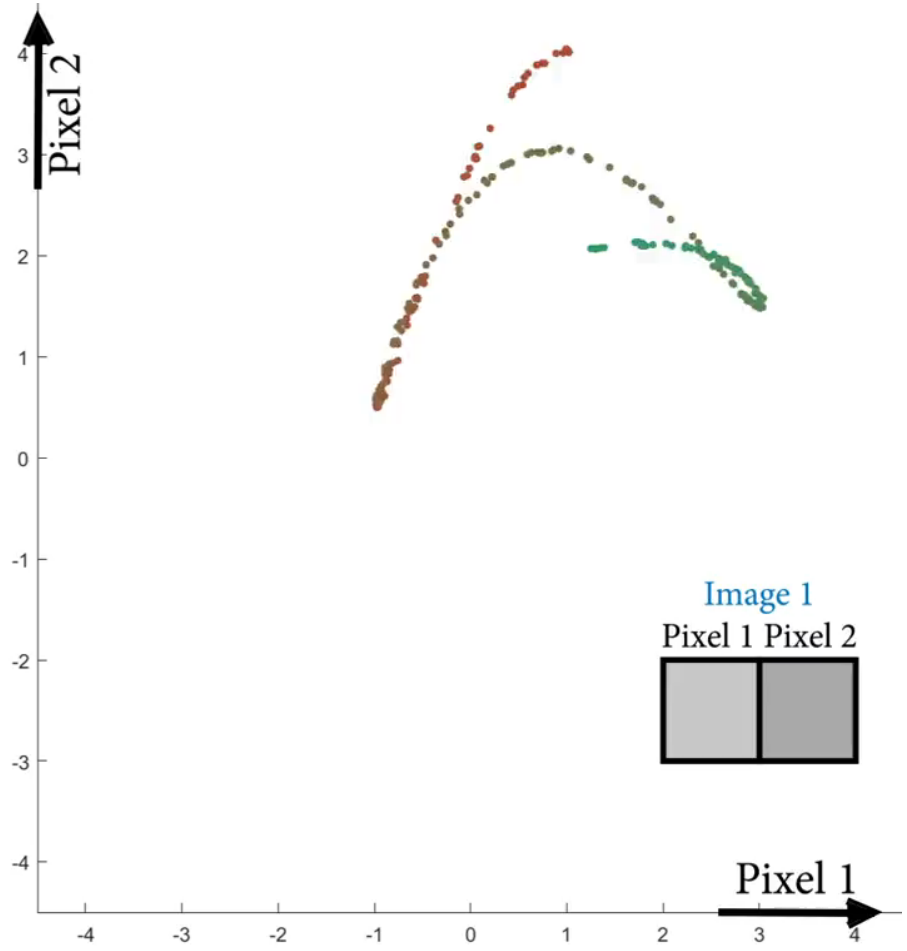
Hagamos un ejemplo con una imagen

Tomaremos un conjunto de de 2 píxeles en una imagen de escala de grises, cada uno representado como un punto 2D.



Agregando ruido “Puro”

Inicialmente, el conjunto de imágenes forma una variedad de "imágenes realistas" (aquí, ficticias). Progresivamente añado un valor aleatorio gaussiano a cada píxel, produciendo un movimiento browniano. Después de muchas iteraciones, la distribución de las imágenes todavía no es una gaussiana centrada: la variedad todavía está presente.



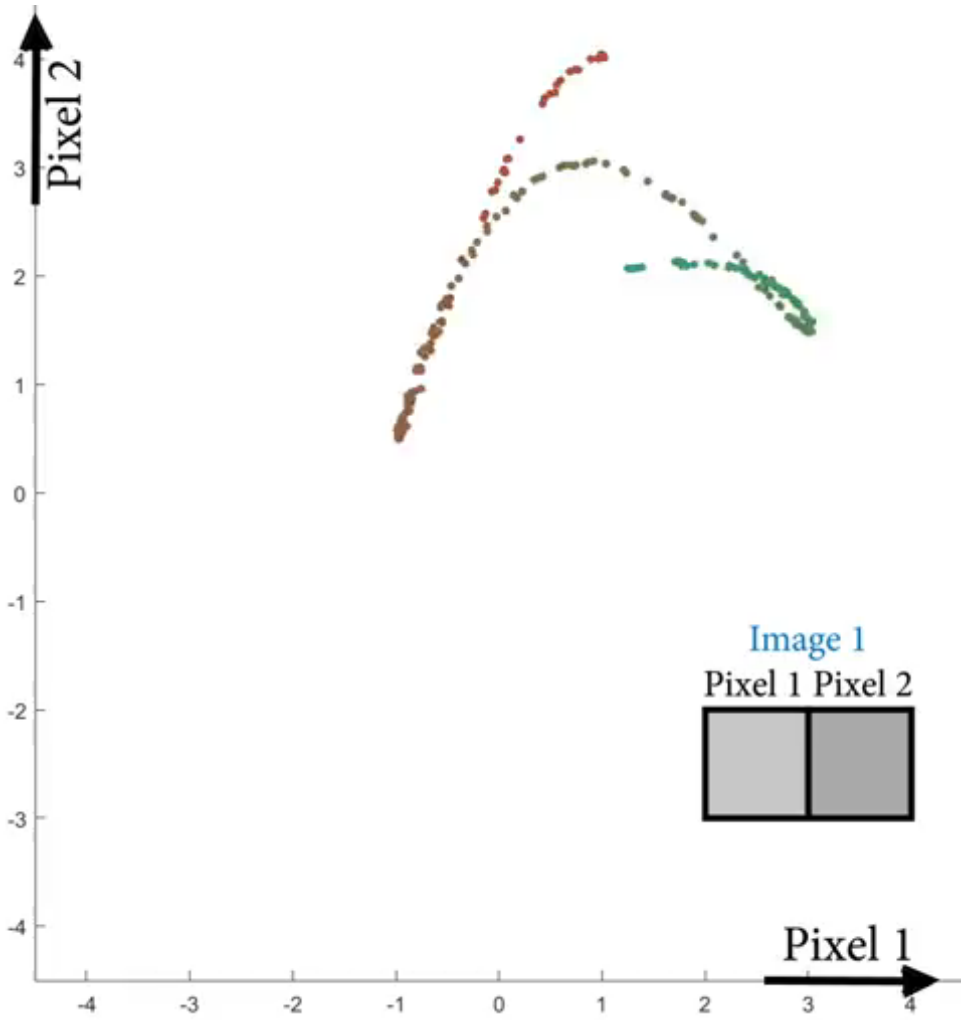
Combinando el ruido progresivamente

En cambio, para transformar la variedad de "imágenes realistas" en una distribución gaussiana, es necesario combinar progresivamente cada imagen de entrada con una gaussiana. El iterado resultante es simplemente una combinación lineal de la imagen en el paso de tiempo anterior y un valor aleatorio gaussiano.

$$I^{t+1} = \sqrt{\alpha_t}\, I^t + \sqrt{1-\alpha_t}\, \varepsilon_{t\to t+1}$$

with $\varepsilon_{t\to t+1} \sim \mathcal{N}(0,1)$

α_t some parameter (could be constant)



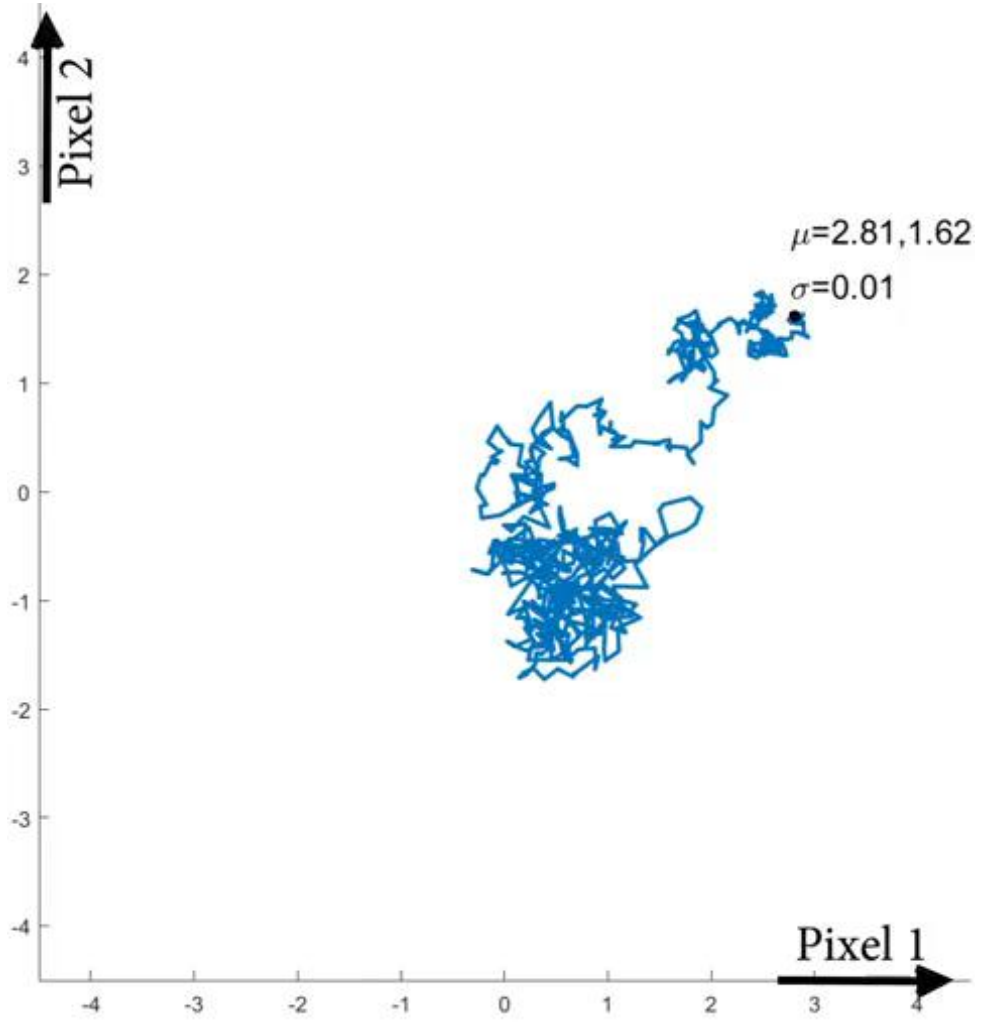
Combinando el ruido progresivamente

Para una imagen inicial I^0 sin ruido, podemos predecir su aspecto en cualquier momento t mezclándola directamente con ruido gaussiano, sin necesidad de iteraciones intermedias. Los valores \alpha_i son menores que 1 y disminuyen con el tiempo, lo que hace que la imagen original I^0 se desvanezca gradualmente.

$$I^t = \sqrt{\overline{\alpha_t}}\ I^0 + \sqrt{1-\overline{\alpha_t}}\ \varepsilon_{0\to t}$$

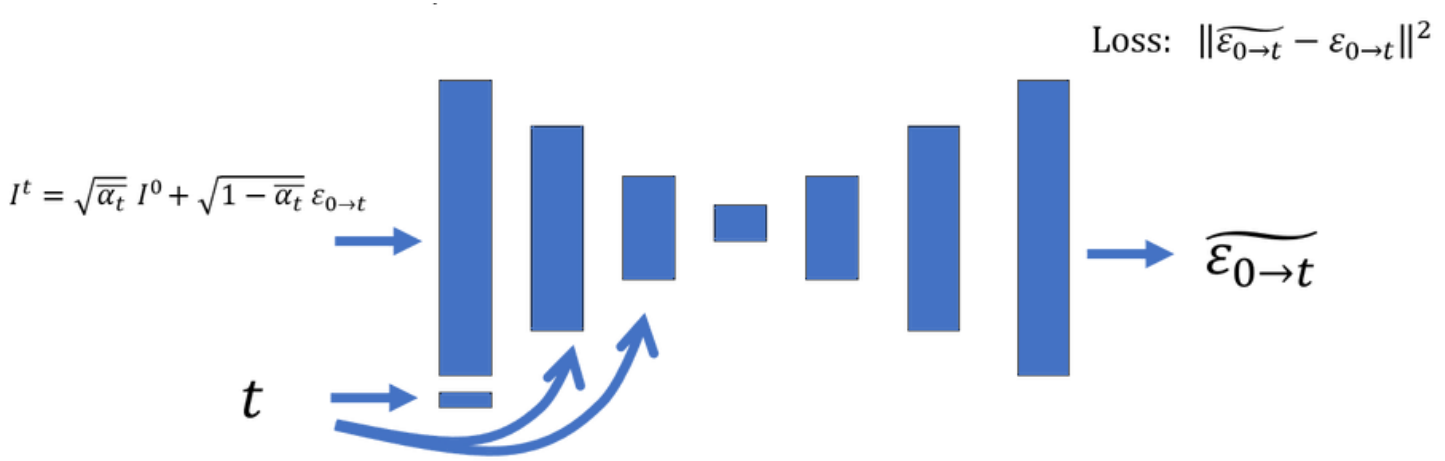
with $\varepsilon_{0\to t} \sim \mathcal{N}(0,1)$

$$\overline{\alpha_t} = \prod_{i=1}^{t} \alpha_i$$



Entrenar una red de difusión

Ahora podemos entrenar una red neuronal para eliminar ruido, por ejemplo, una U-Net. La alimentamos con un gran conjunto de datos de imágenes con cantidades aleatorias de ruido generadas por la fórmula anterior, e intentamos predecir el ruido. Minimizamos una pérdida que es simplemente el error cuadrático medio (MSE) entre el ruido de entrada y el ruido predicho.

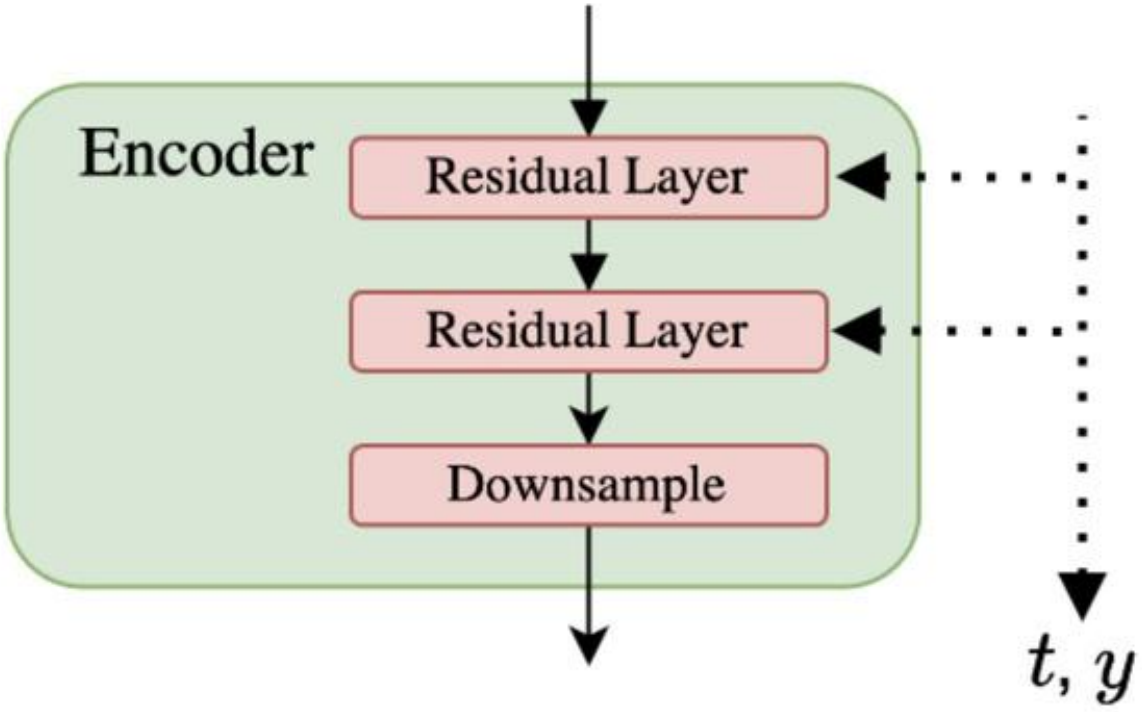


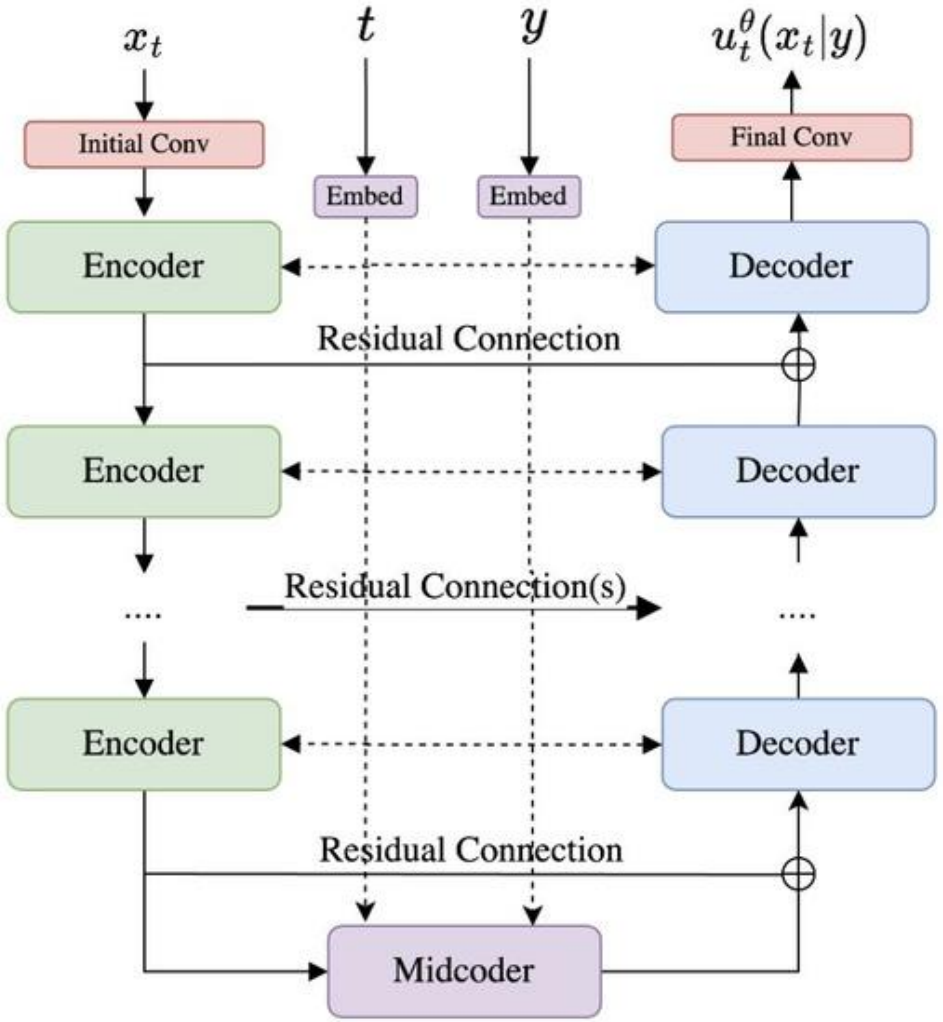
- Feed the network with many noisy images at random times $t \in [0, t_{max}]$



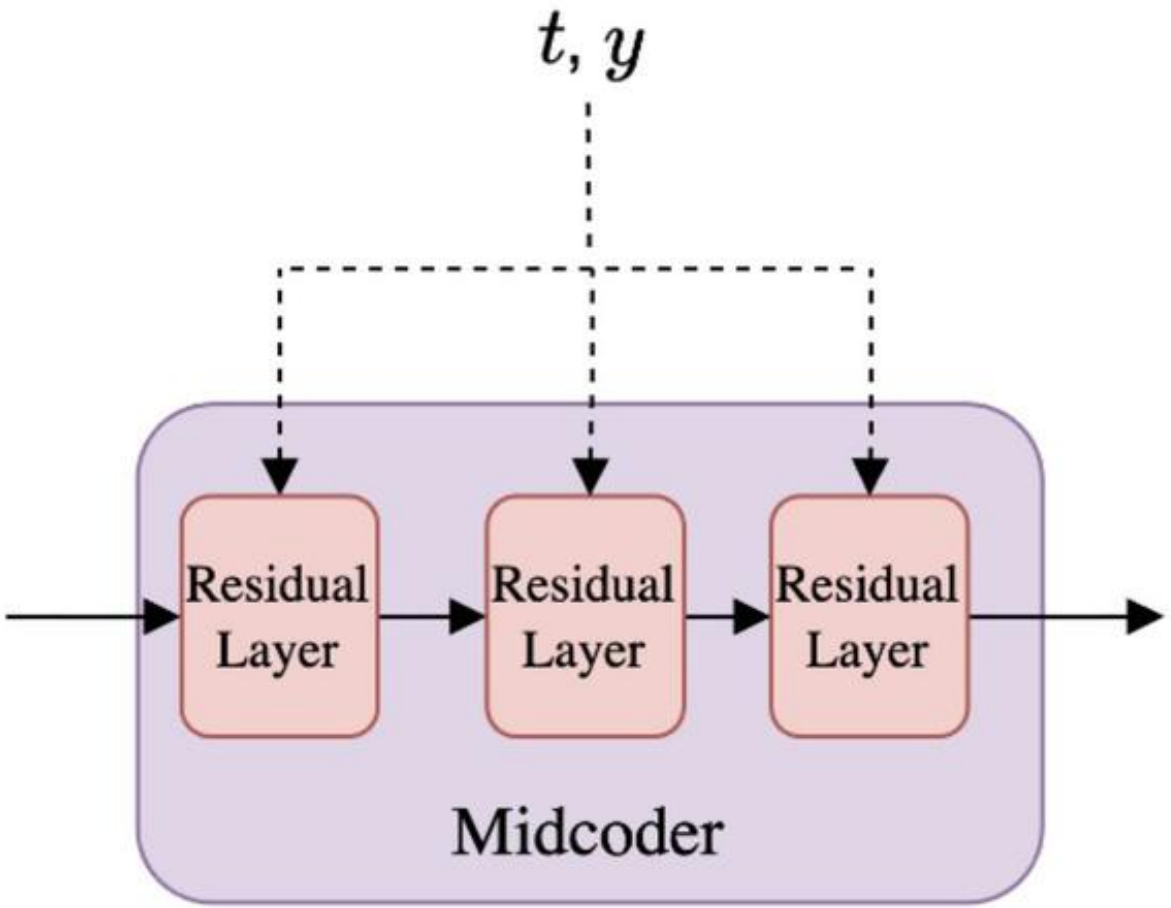
Considerar una función que demore más en destruir la señal

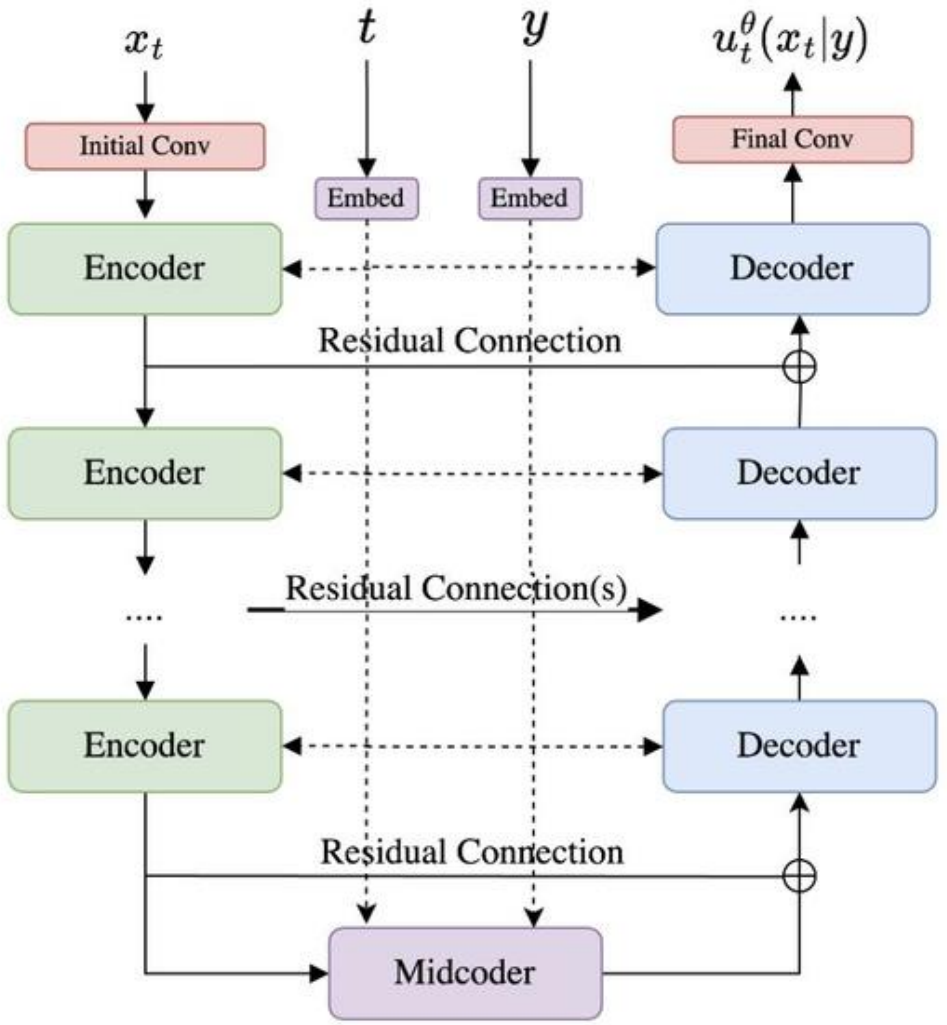
Capa codificadora



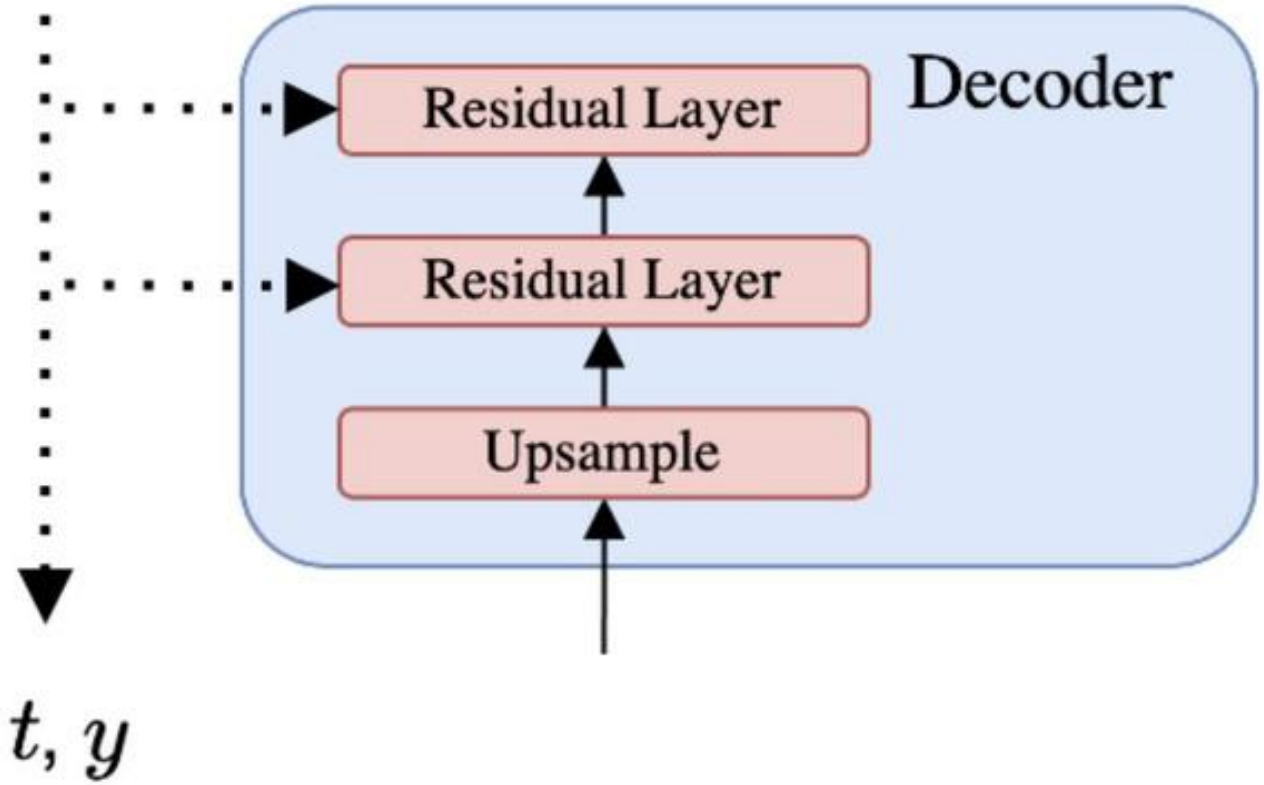


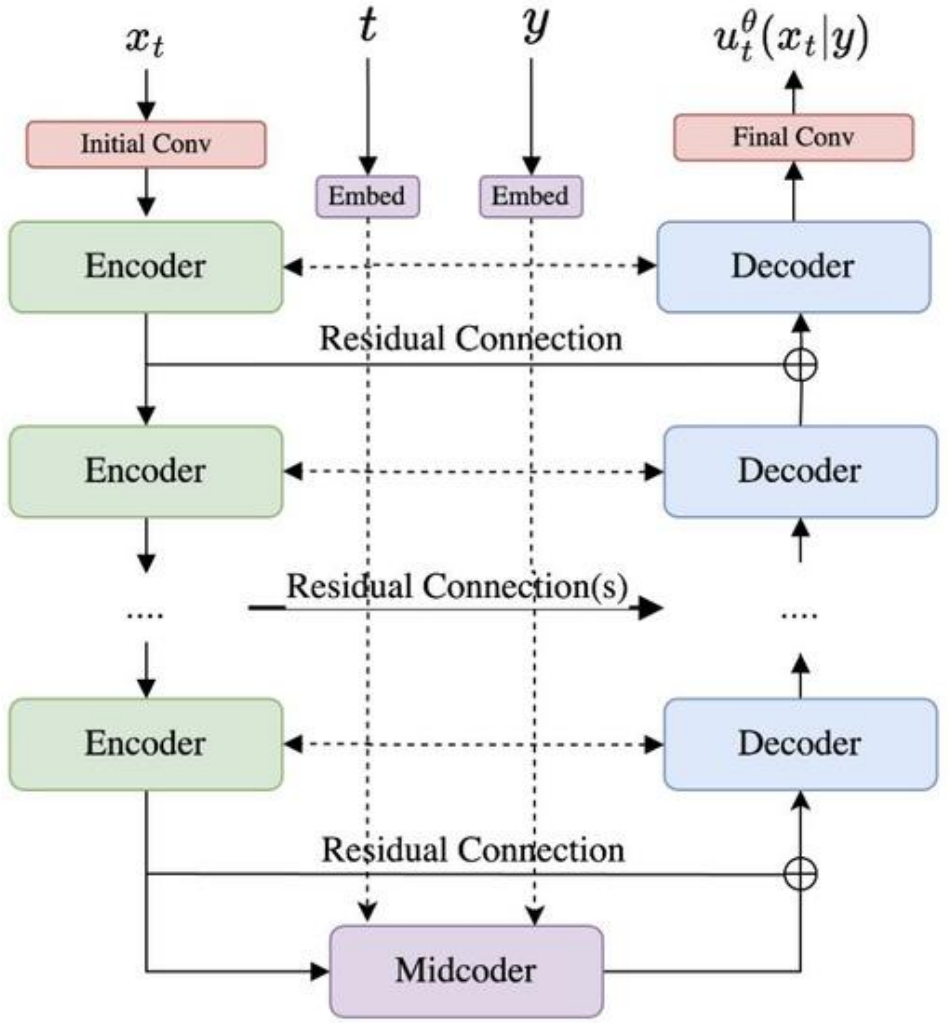
Capa de codificador intermedio



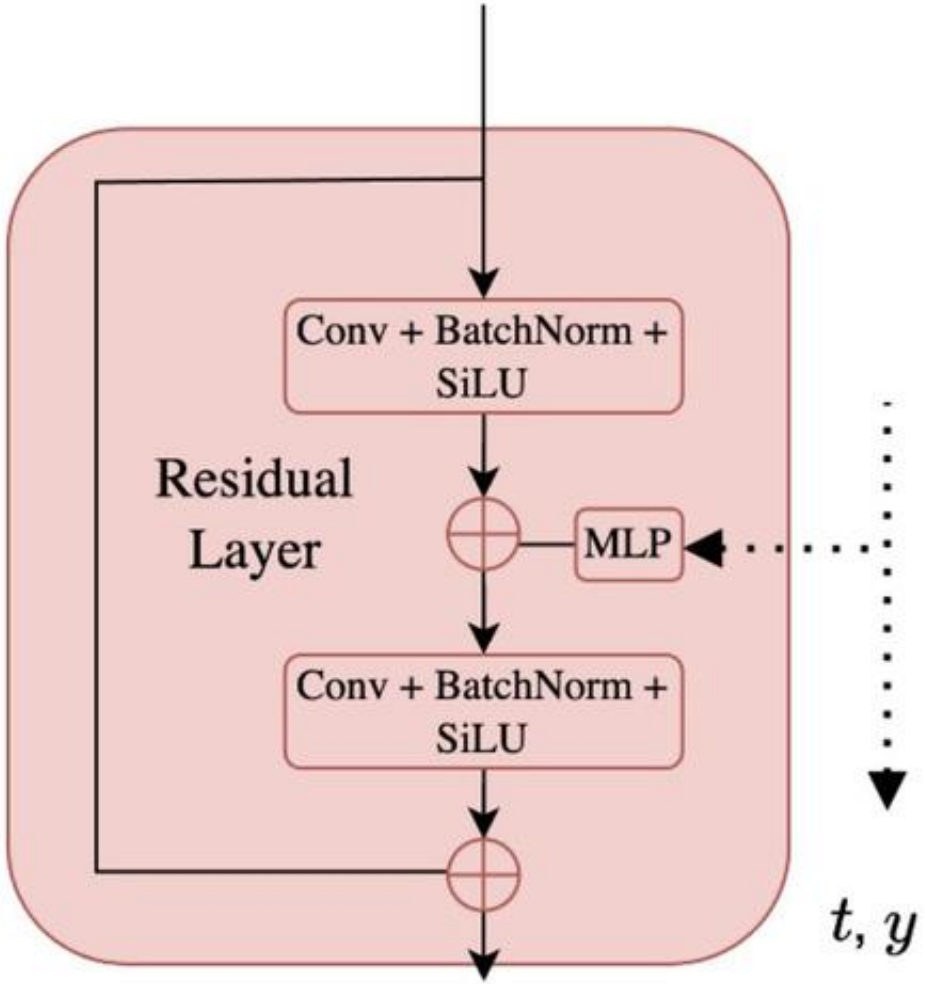


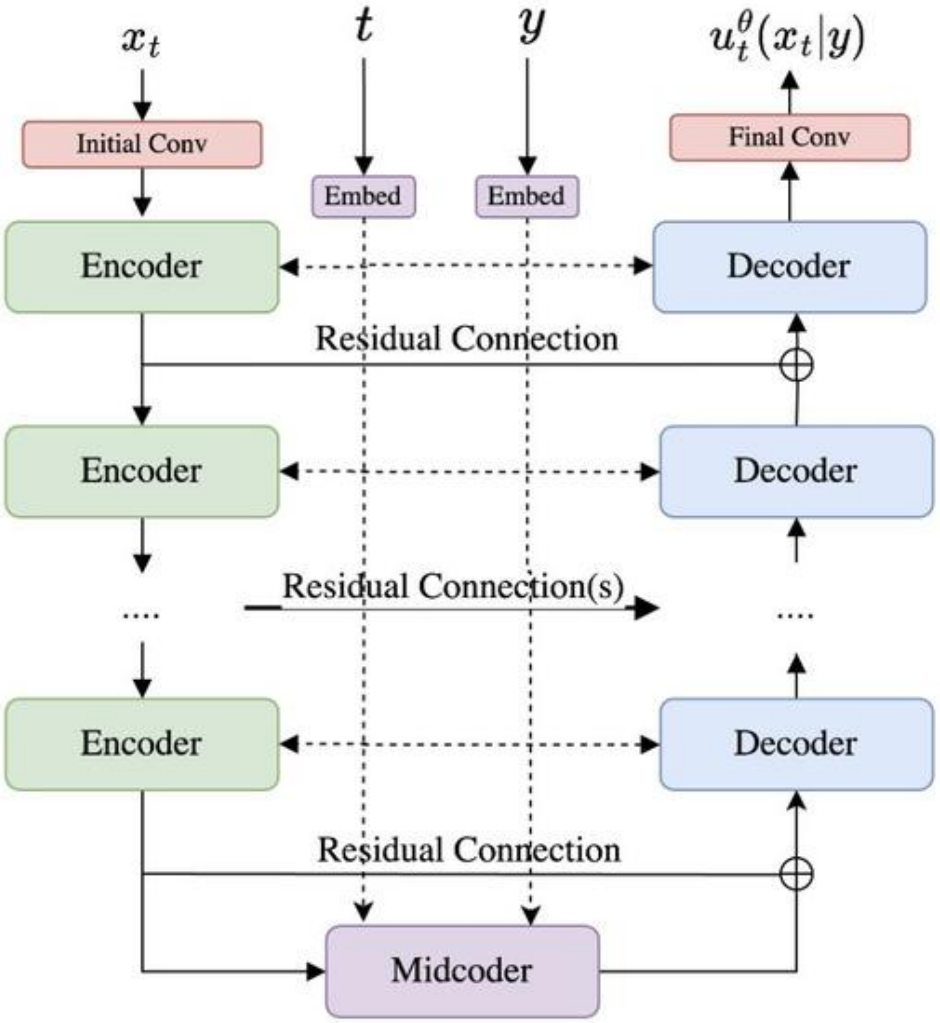
Capa decodificadora





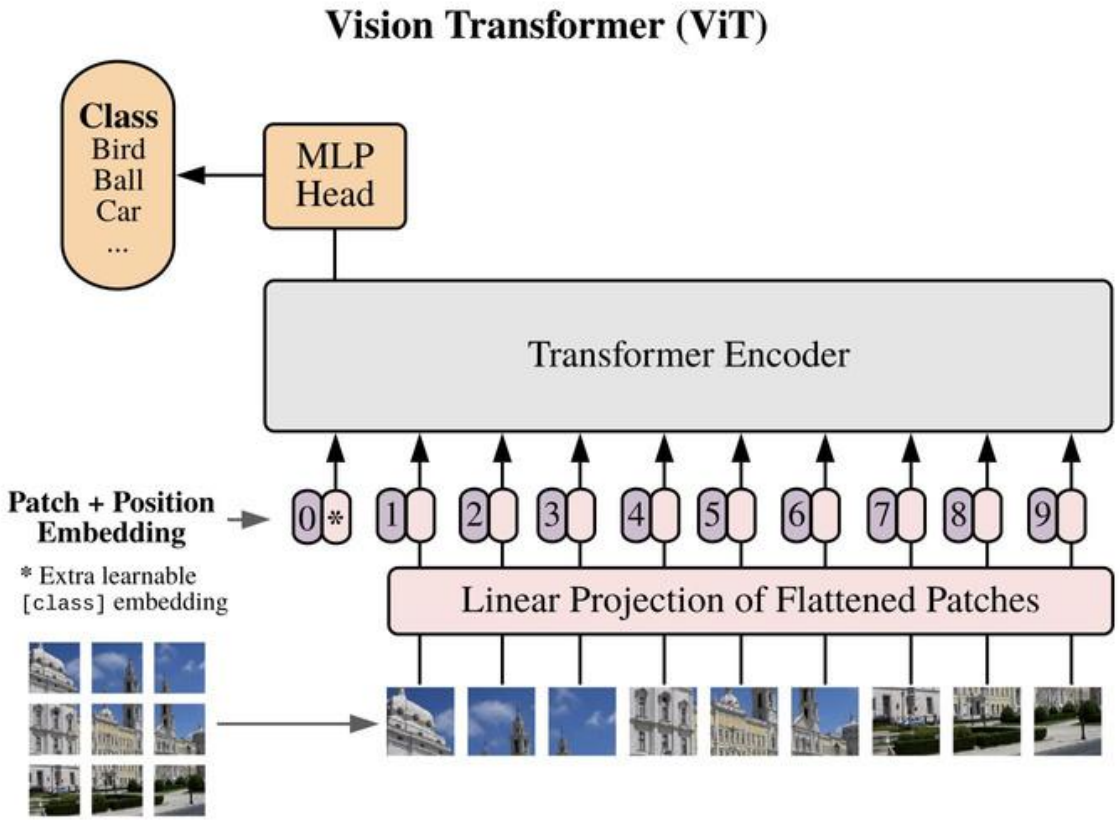
Capa residual

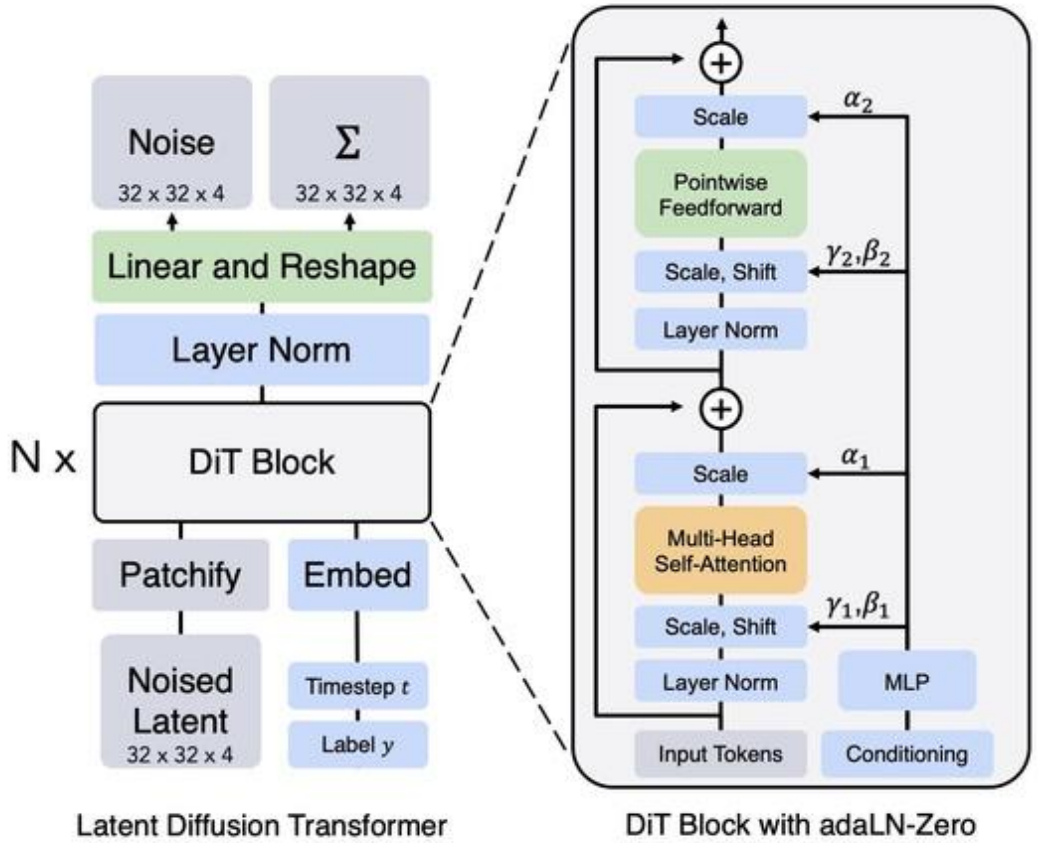




Transformador de difusión (DiT)

Idea: Dividir una imagen en fragmentos y observar entre ellos.
Basado en el transformador de visión (ViT).



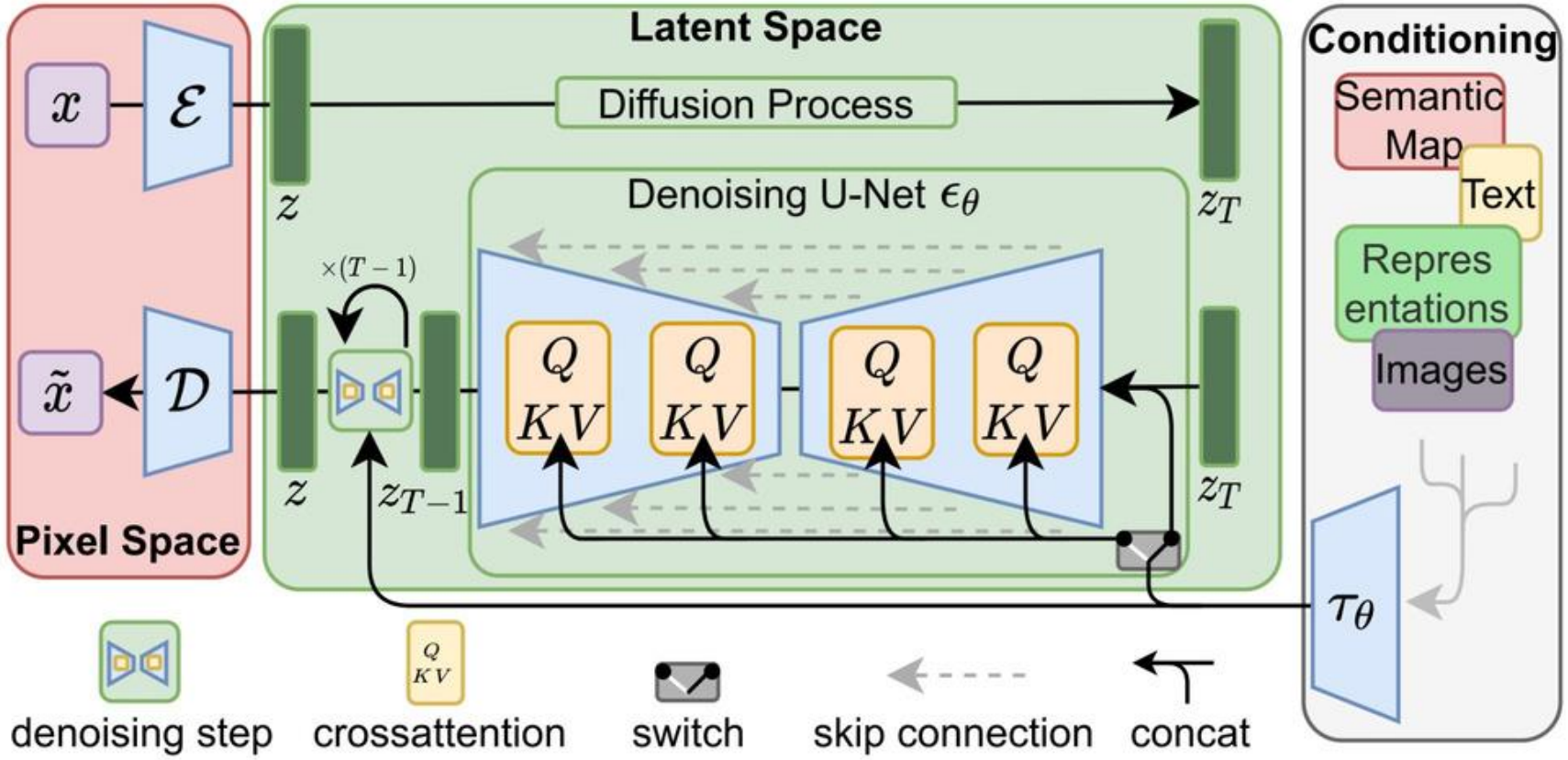


Latent Diffusion Transformer

DiT Block with adaLN-Zero

Modelado generativo en el espacio latente

Idea: Entrenar el modelo generativo en el espacio latente de un autocodificador (variacional) preentrenado.





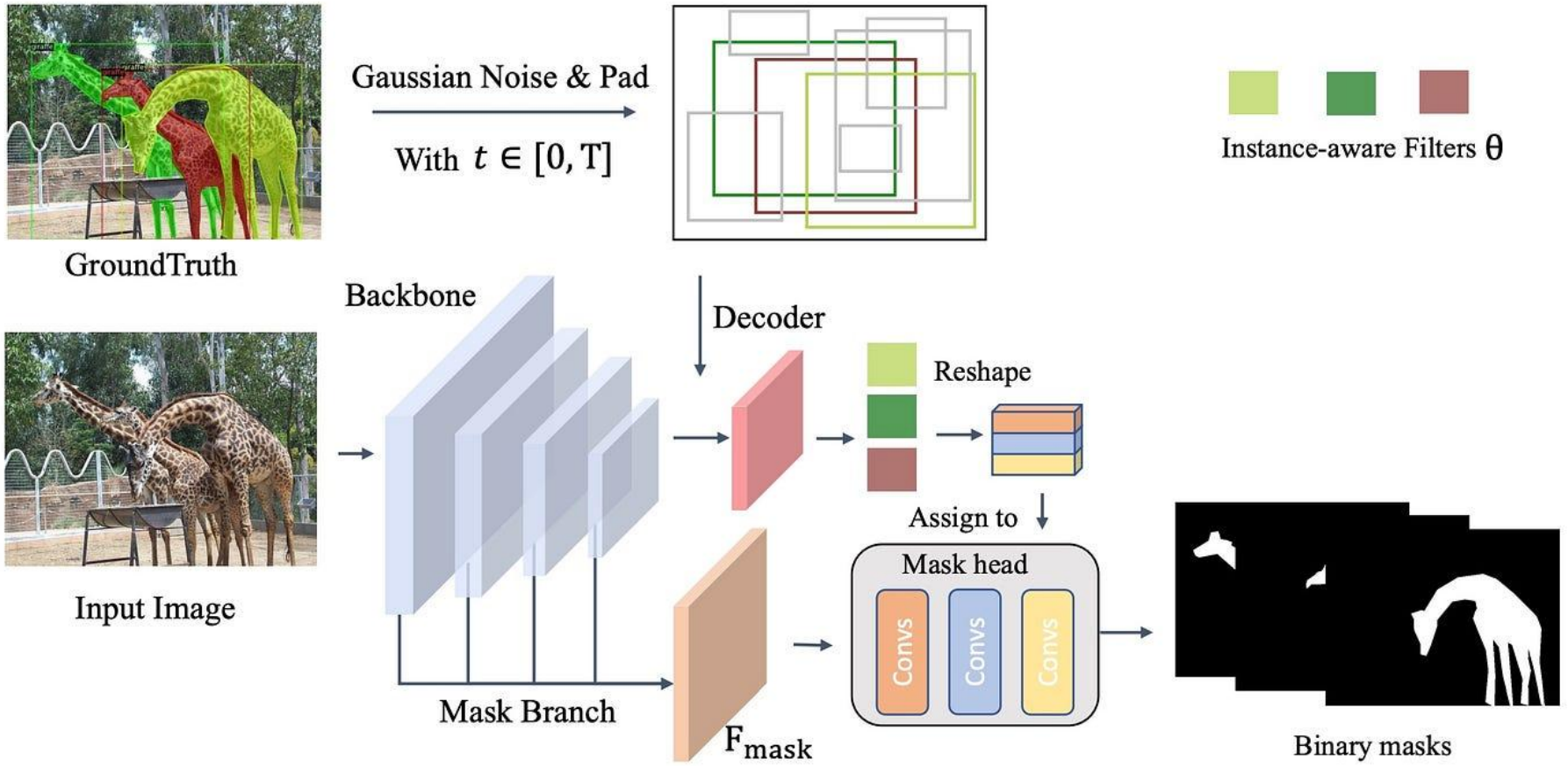
Aplicaciones Diversas de Modelos de Difusión

Inpainting con modelos de difusión



Input Image.

Segmentación

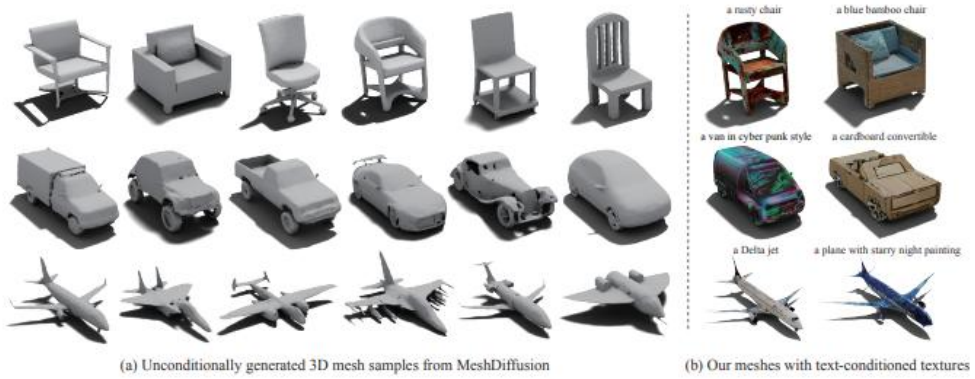


Se puede aplicar en 3D

MESHDIFFUSION: SCORE-BASED GENERATIVE 3D MESH MODELING

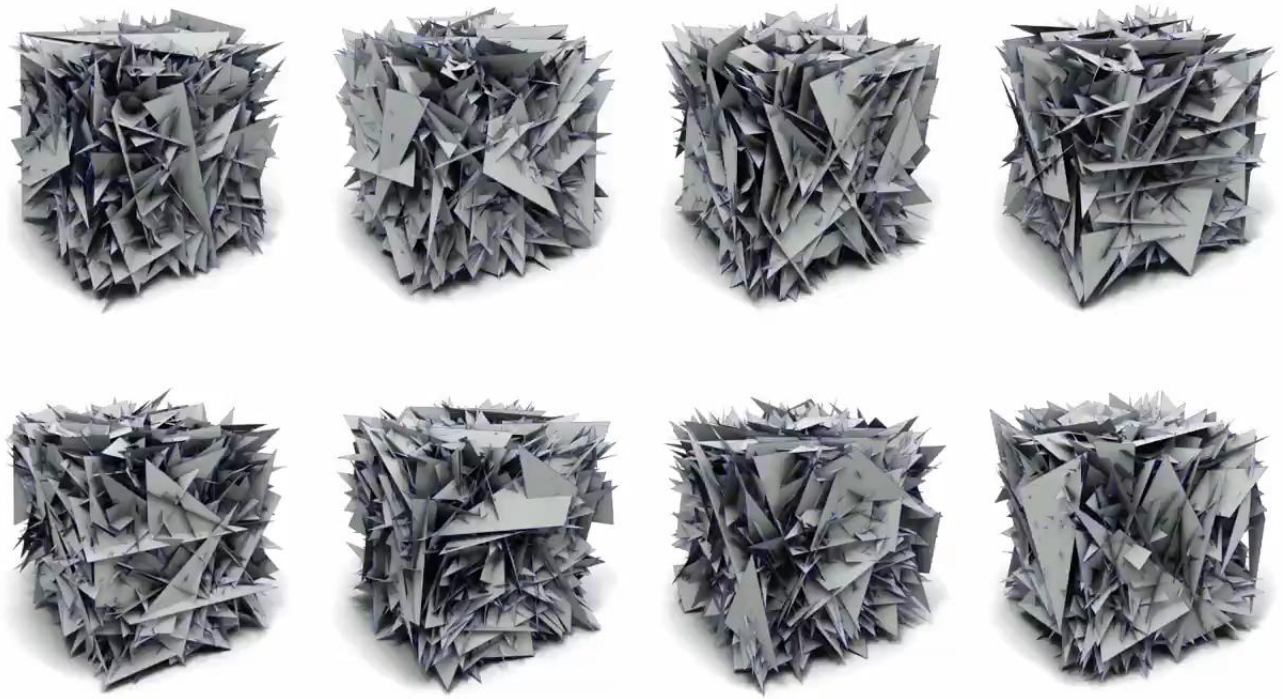
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[1]Mila, Université de Montréal [2]Max Planck Institute for Intelligent Systems - Tübingen
[3]ETH Zürich [4]McGill University [5]University of Cambridge

Project Page: meshdiffusion.github.io



(a) Unconditionally generated 3D mesh samples from MeshDiffusion (b) Our meshes with text-conditioned textures

Figure 1: (a) Unconditionally generated 3D mesh samples randomly selected from the proposed *MeshDiffusion*, a simple diffusion model trained on a direct parametrization of 3D meshes without bells and whistles. (b) 3D mesh samples generated by *MeshDiffusion* with text-conditioned textures from [39]. *MeshDiffusion* produces highly realistic and fine-grained geometric details while being easy and stable to train.





Generar código utilizable

CODEFUSION: A Pre-trained Diffusion Model for Code Generation

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José Cambronero
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Vu Le
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Cambridge, UK

Gust Verbruggen
Microsoft
Keerbergen, Belgium

Abstract

Imagine a developer who can only change their last line of code—how often would they have to start writing a function from scratch before it is correct? Auto-regressive models for code generation from natural language have a similar limitation: they do not easily allow reconsidering earlier tokens generated. We introduce CODEFUSION, a pre-trained diffusion code generation model that addresses this limitation by iteratively denoising a complete program conditioned on the encoded natural language. We evaluate CODEFUSION on the task of natural language to code generation for Bash, Python, and Microsoft Excel conditional formatting (CF) rules. Experiments show that CODEFUSION (75M parameters) performs on par with state-of-the-art auto-regressive systems (350M–175B parameters) in top-1 accuracy and outperforms them in top-3 and top-5 accuracy, due to its better balance in diversity versus quality.

approaches then select the vocabulary token with the closest embedding. In the code domain, where there are many syntactic and semantic constraints between tokens, independently projecting embeddings back to tokens can yield invalid programs.

We propose CODEFUSION, a natural language to code (NL-to-code) model that combines an encoder-decoder architecture (Raffel et al., 2020) with a diffusion process. The encoder maps the NL into a continuous representation, which is used by the diffusion model as an additional condition for denoising random Gaussian noise input. To generate syntactically correct code, we then feed the denoised embeddings to a transformer decoder, with full self-attention and cross attention with the embedded utterance, to obtain probability distributions over code tokens. Finally, we select the token with the highest probability at each index.

To pre-train CODEFUSION for code generation, we extend the continuous paragraph denoising

System description			Python (CodeBERT)		
System	**Model**	**#P**	**top-1**	**top-3**	**top-5**
T5	t5-large	770M	80.4	82.3	84.8
CodeT5	codet5-large	770M	80.5	83.1	85.0
GPT-3	text-davinci-003	175B	**82.5**	83.7	85.8
ChatGPT	gpt-3.5-turbo	20B	80.6	82.5	83.9
StarCoder	starcoder	15.5B	79.2	82.0	84.1
CodeT5+	codet5p-16b	16B	79.6	82.1	84.5
CodeGen	codegen-350m	350M	80.1	81.8	83.7
Diffusion-LM	Custom	50M	70.4	74.3	76.5
GENIE	Custom	93M	73.2	77.1	80.3
CODEFUSION	**Custom**	75M	80.7	**86.3**	**90.3**

NL:	`Copy the content of file 'file.txt' to file 'file2.txt'`
Target:	`shutil.copy('file.txt', 'file2.txt')`
$t = 1200$	`Ss ec between Useful 144 Location copyright pen they destinat Path caret corev adapter NameAnd ġiterating Ġface Ġpass ĠFH`
$t = 1000$	`(.* for copyInt filefiletxt, filefiletxt queryResult SE Screen`
$t = 800$	`destutil..copyInt'filetxt', 'filetxt') return 'filetxt'`
$t = 600$	`destutil.copyInt'file.txt', 'fi2.txt')`
$t = 400$	`shutil.copyInt'file.txt', 'file.txt')`
$t = 200$	`shutil.copy('file.txt', 'file2.txt')`
$t = 0$	`shutil.copy('file.txt', 'file2.txt')`

No es necesario que sea ruido

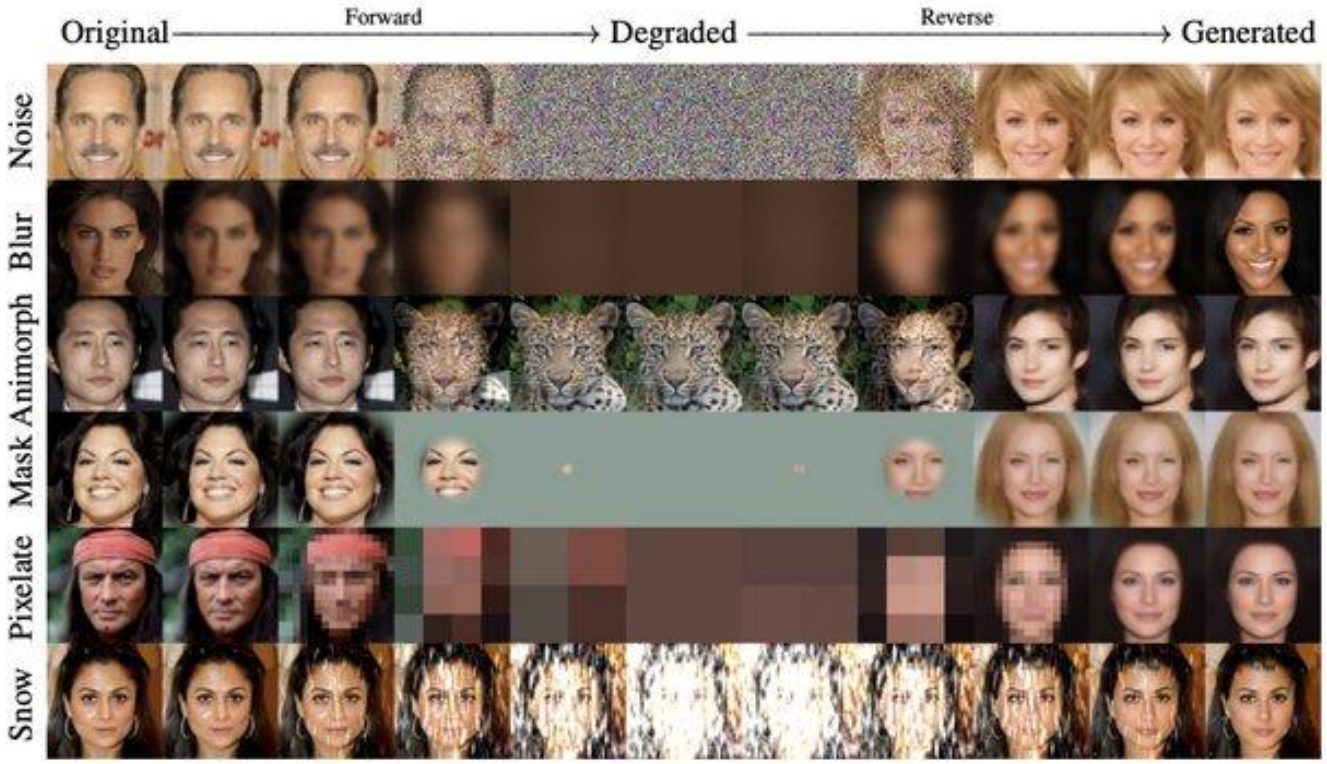


Figure 1: Demonstration of the forward and backward processes for both hot and cold diffusions. While standard diffusions are built on Gaussian noise (top row), we show that generative models can be built on arbitrary and even noiseless/cold image transforms, including the ImageNet-C *snowification* operator, and an *animorphosis* operator that adds a random animal image from AFHQ.



Ejemplos con código

Labs Basados en “How Diffusion Models Work”



https://learn.deeplearning.ai/diffusion-models/

Objetivo de la parte práctica

- **Tienes: muchas imágenes de pixel art.**

Estos son tus datos de entrenamiento.

- **Quieres: aún más pixel art.**

Esta es una red neuronal que genera más pixel art para ti.

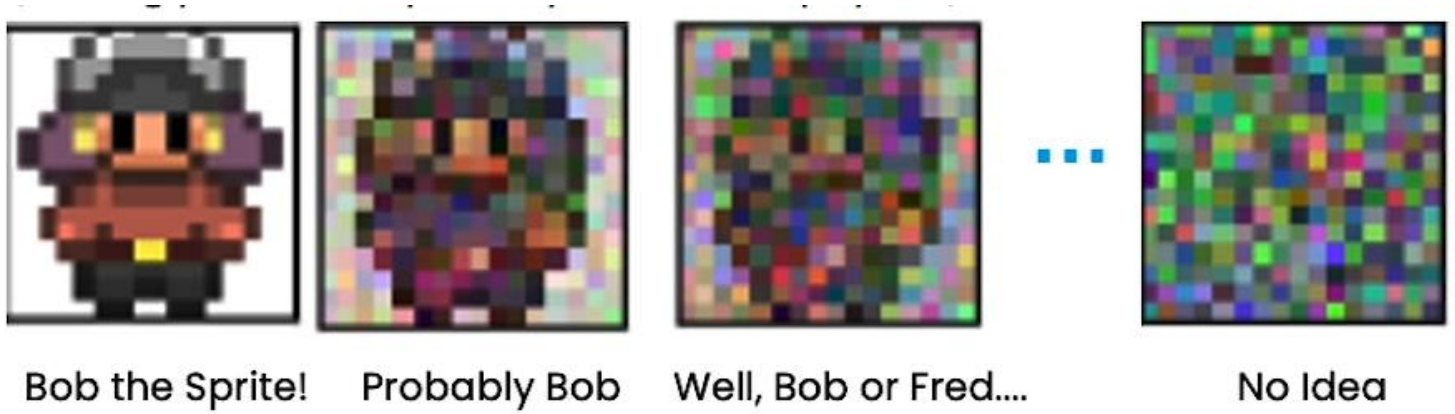


Creando imágenes útiles para una red neuronal

Quieres que una red neuronal aprenda:

- Detalles finos
- Contornos generales
- Todo lo intermedio

Una forma es añadir diferentes niveles de ruido a los datos de entrenamiento



Entrenando una red neuronal

La red neuronal aprende a tomar diferentes imágenes con ruido y convertirlas de nuevo en sprites.

Aprende a eliminar el ruido que añadiste.



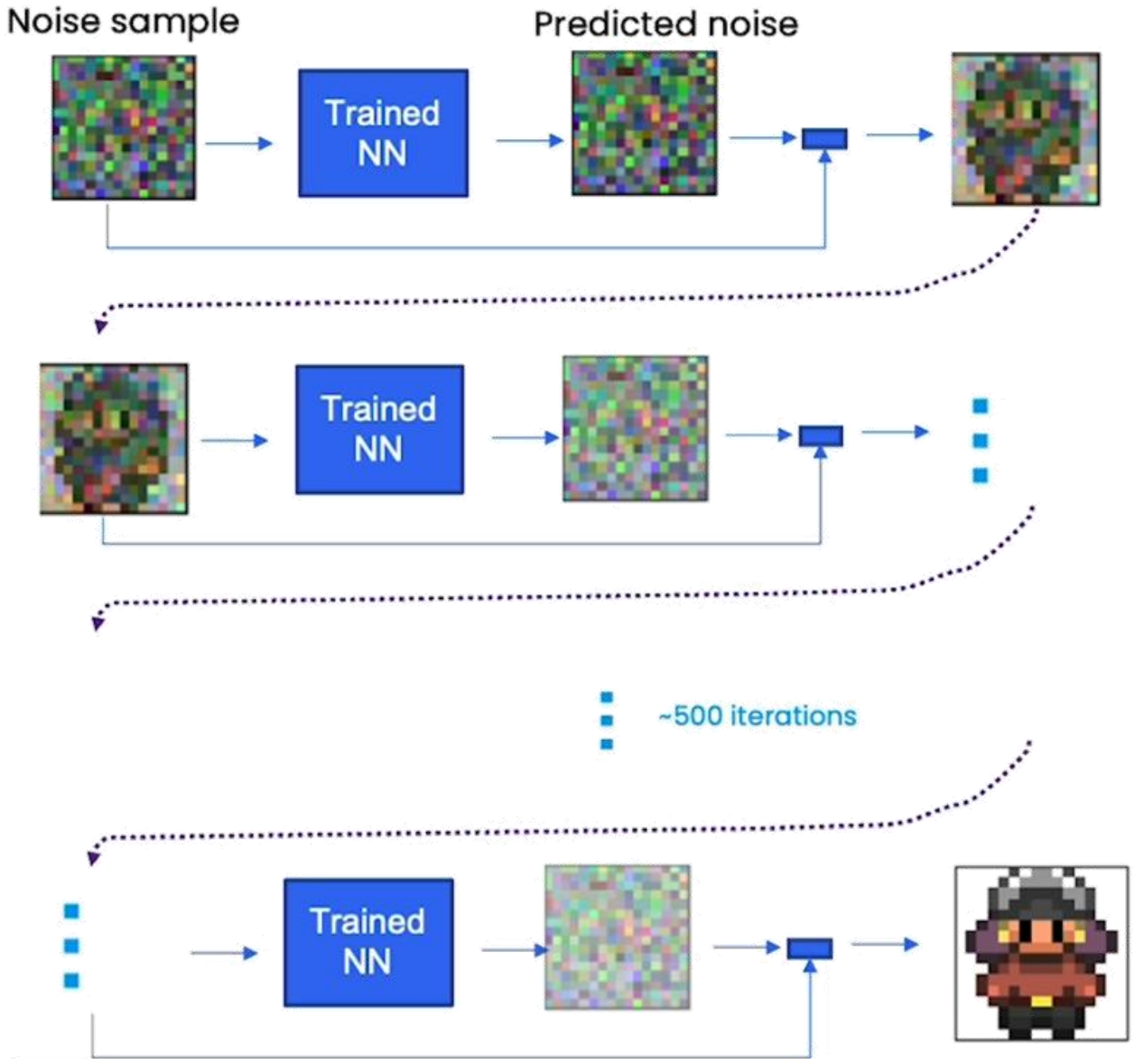
El nivel de ruido "Sin idea" es importante porque está **normalmente distribuido**.

Entonces, cuando le pides a la red neuronal un nuevo sprite:

- Puedes muestrear ruido de la distribución normal
- Obtener un sprite completamente nuevo usando la red para eliminar el ruido
- Ahora, puedes obtener aún más sprites, más allá de tus datos de entrenamiento.

Muestreo

La red neuronal intenta predecir completamente el ruido en cada paso. Realísticamente, es solo una predicción. Necesitas múltiples pasos para obtener sprites de alta calidad.





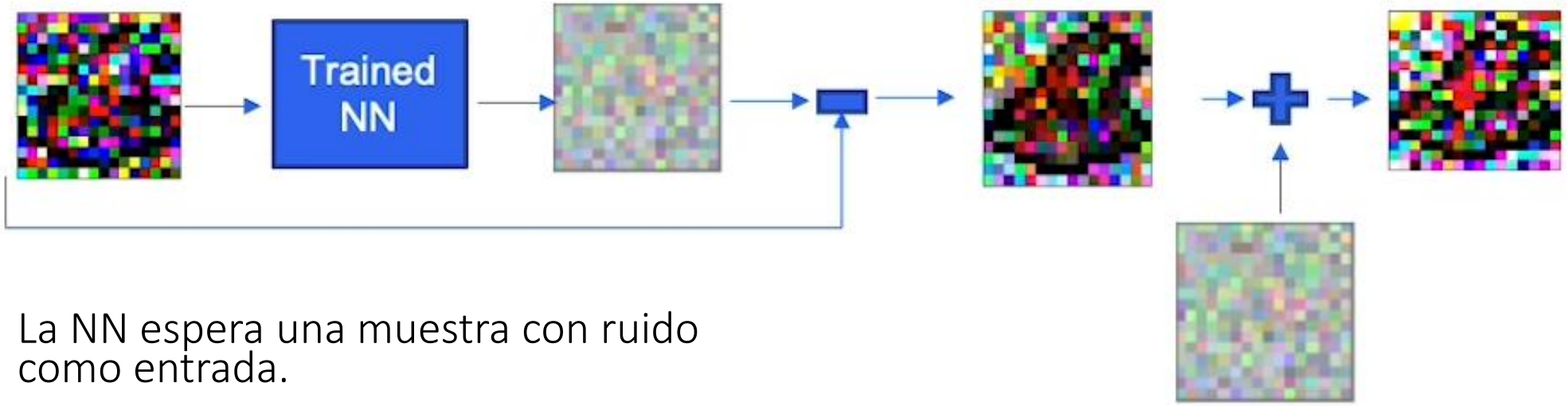
```
# sample using standard DDPM algorithm
@torch.no_grad()
def sample_ddpm(n_sample, device, save_rate=20):
    # x_T ~ N(0, 1), sample initial noise
    samples = torch.randn(n_sample, 3, height, height).to(device)
    ...
    for i in range(timesteps, 0, -1):

        # reshape time tensor
        t = torch.tensor([i / timesteps])[:, None, None, None].to(device)

        # sample some random noise to inject back in. For i = 1, don't add
back in noise
        z = torch.randn_like(samples) if i > 1 else 0

        eps = nn_model(samples, t)    # predict noise e_(x_t,t)
        samples = denoise_add_noise(samples, i, eps, z)
        ...
```

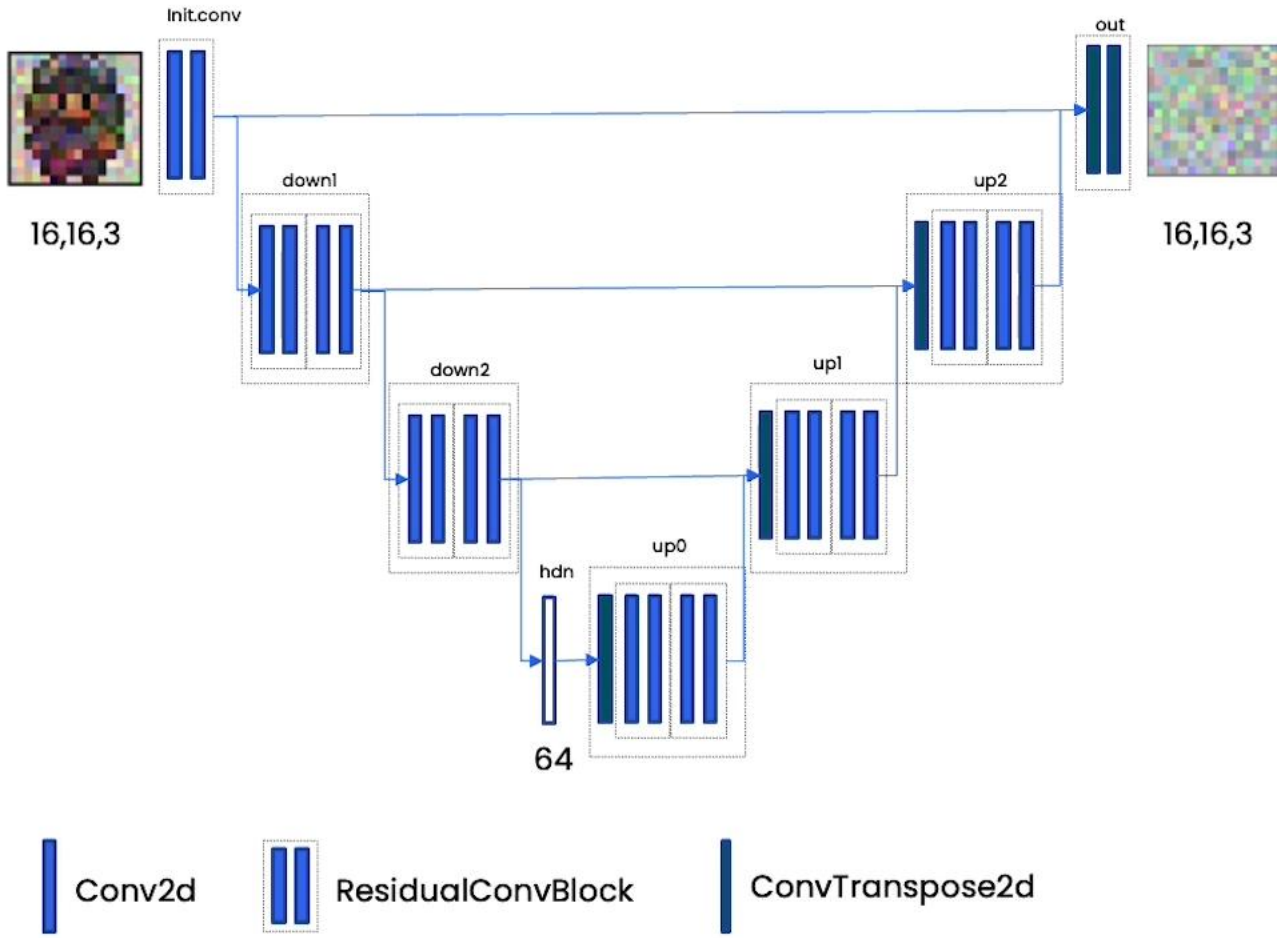
Detalles de la Iteración de Muestreo



- La NN espera una muestra con ruido como entrada.
- Puedes añadir ruido adicional antes de que se pase al siguiente paso.
- Empíricamente, esto estabiliza la NN para que no colapse a algo más cercano al promedio del conjunto de datos.



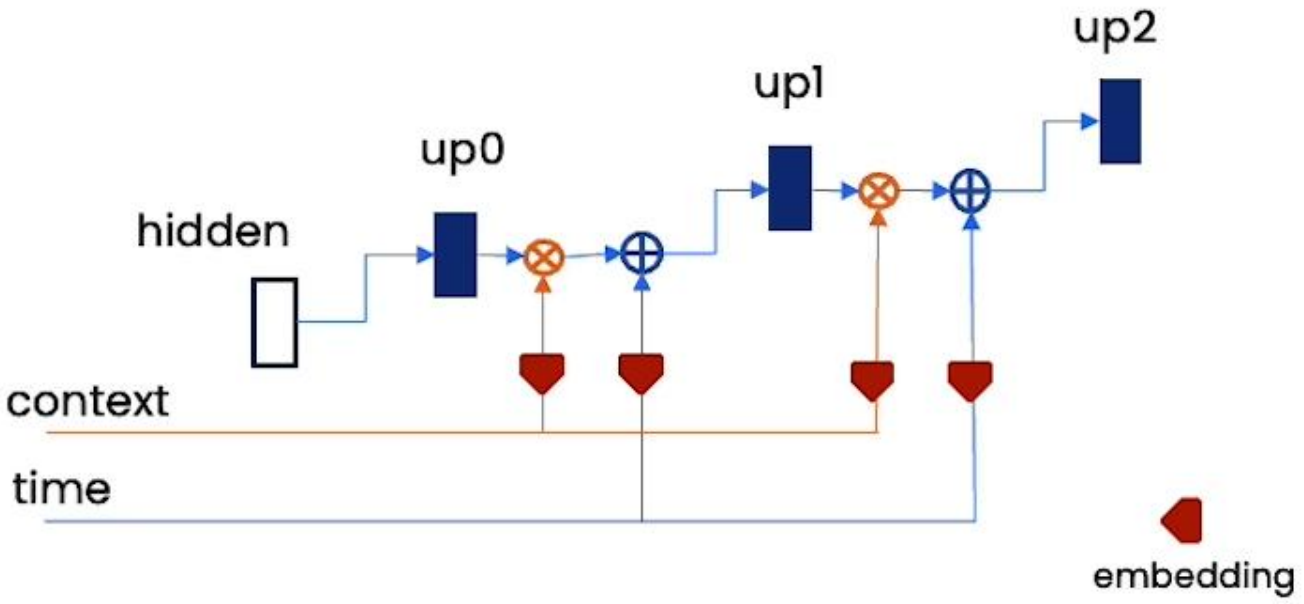
U-Net para identificar el ruido



Incorporación de Más Información

El U-Net puede incorporar más información en la forma de incrustaciones:

- Incrustación de tiempo: relacionada con el paso de tiempo y el nivel de ruido.
- Incrustación de contexto: relacionada con el control de la generación, por ejemplo, descripción de texto o factor (más adelante).

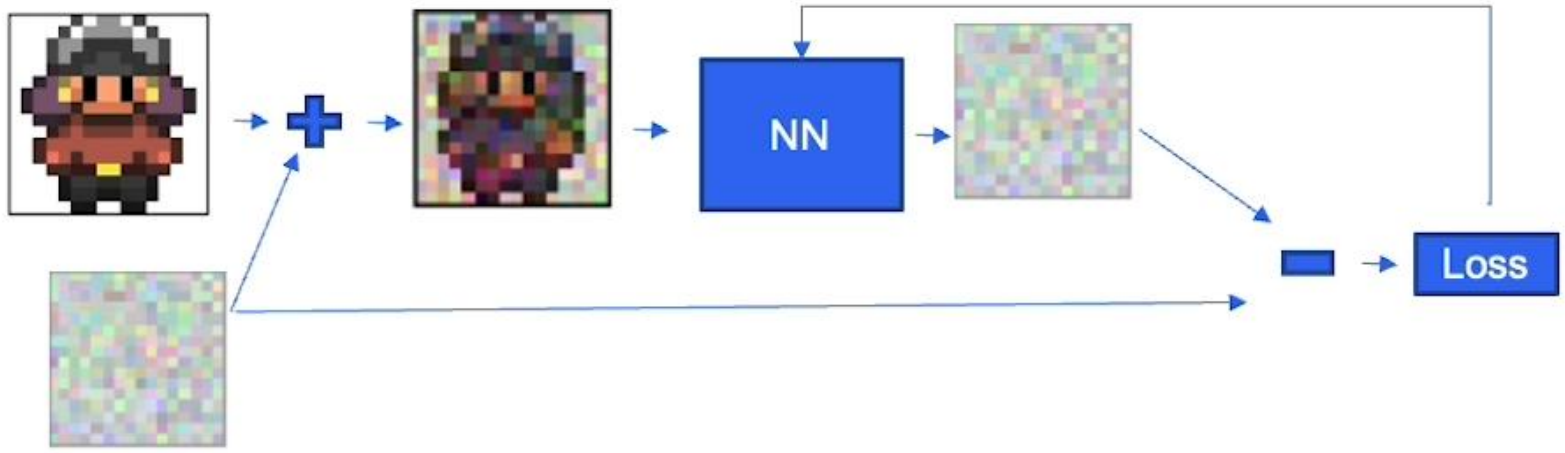



```
# embed context and timestep
cemb1 = self.contextembed1(c).view(-1, self.n_feat * 2, 1, 1)
temb1 = self.timeembed1(t).view(-1, self.n_feat * 2, 1, 1)

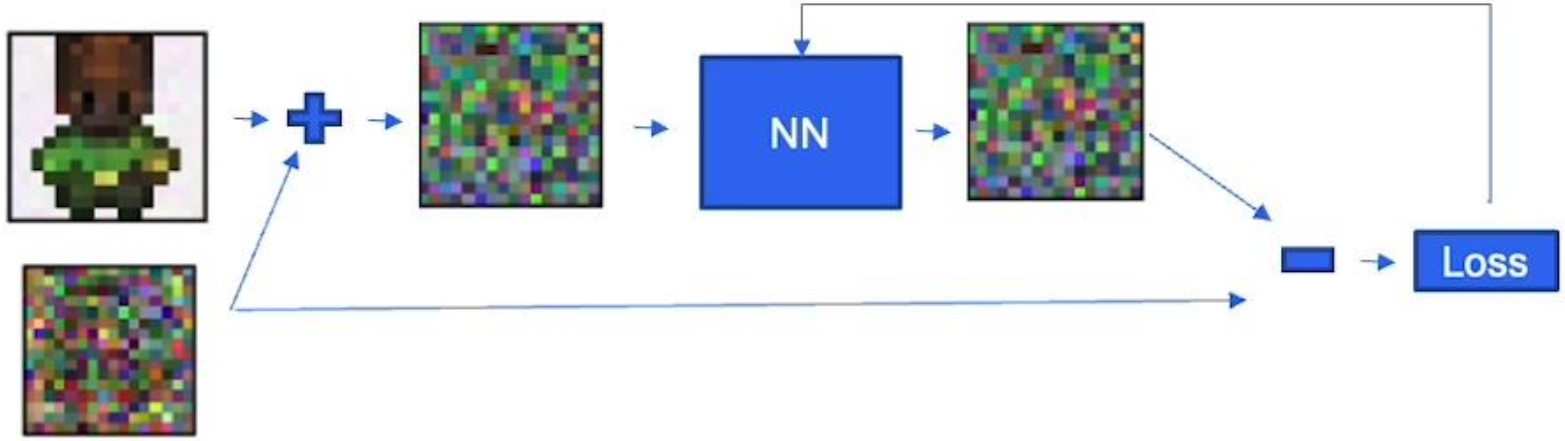
up2 = self.up1(cemb1*up1 + temb1, down2)
```

Entrenamiento

La NN aprende a predecir el ruido, lo que realmente significa aprender la distribución de lo que no es ruido.



Se toma una muestra aleatoria del paso de tiempo (nivel de ruido) por imagen para entrenar más estabilizadamente.





Real image

Input



Epoch 0



Epoch 31



El Algoritmo

- Tomar una imagen de entrenamiento.
- Tomar una muestra del paso de tiempo t. Esto determina el nivel de ruido.
- Tomar una muestra del ruido.
- Añadir ruido a la imagen.
- Introducir esto en la red neuronal. La red neuronal predice el ruido.
- Calcular la pérdida entre el ruido predicho y el ruido verdadero.
- Retropropagación y aprendizaje.

```
for ep in range(n_epoch):

    # linearly decay learning rate
    optim.param_groups[0]['lr'] = lrate*(1-ep/n_epoch)

    pbar = tqdm(dataloader, mininterval=2 )
    for x, _ in pbar:   # x: images
        optim.zero_grad()

        # perturb data
        t = torch.randint(1, timesteps + 1, (x.shape[0],)).to(device)
        noise = torch.randn_like(x)
        x_pert = perturb_input(x, t, noise)

        # use network to recover noise
        pred_noise = nn_model(x_pert, t / timesteps)

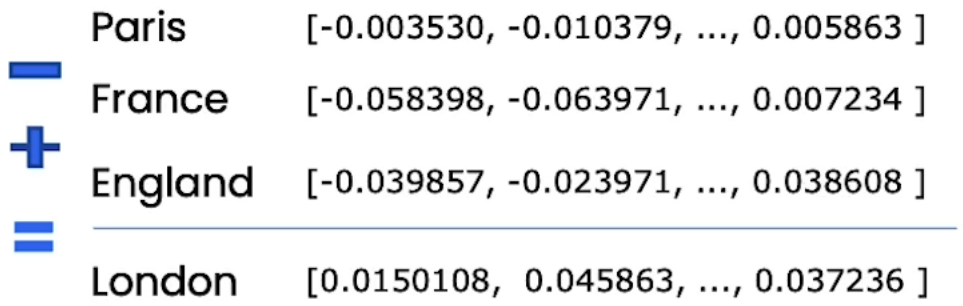
        # loss is mean squared error between the predicted and true noise
        loss = F.mse_loss(pred_noise, noise)
        loss.backward()
        optim.step()
```

Embeddings o Incrustaciones



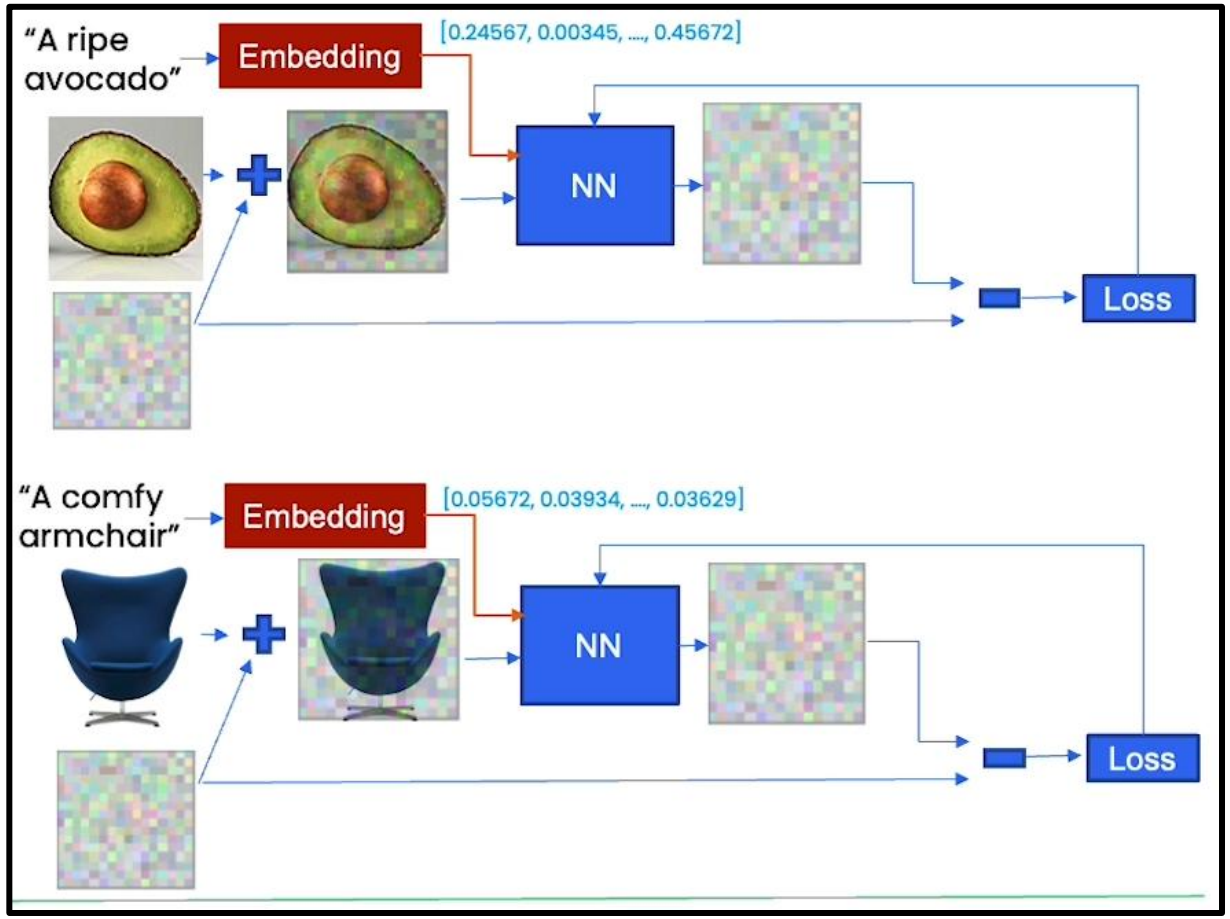
Los vectores de incrustación capturan el significado.

Textos con contenido similar tendrán vectores similares.

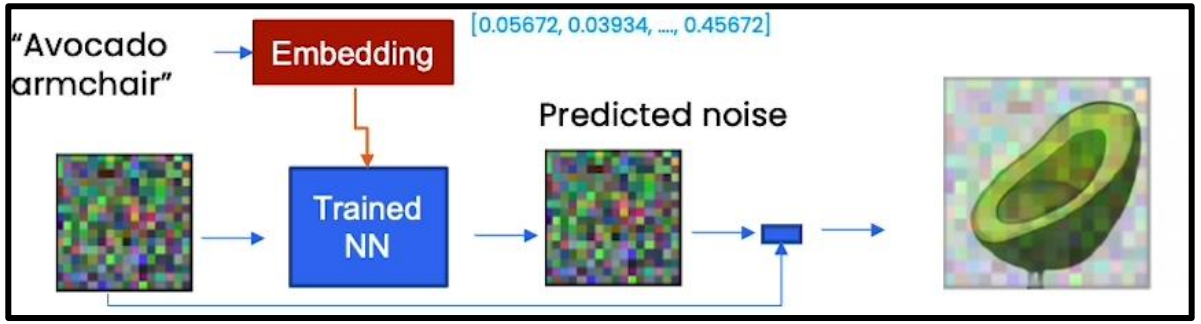


Sumando Contextos

Entrenar



Muestrear



Contexto

- El contexto es un vector que controla la generación.
- El contexto puede ser incrustaciones de texto, por ejemplo, de más de 1000 de longitud.
- El contexto también puede ser categorías, por ejemplo, de 5 de longitud.



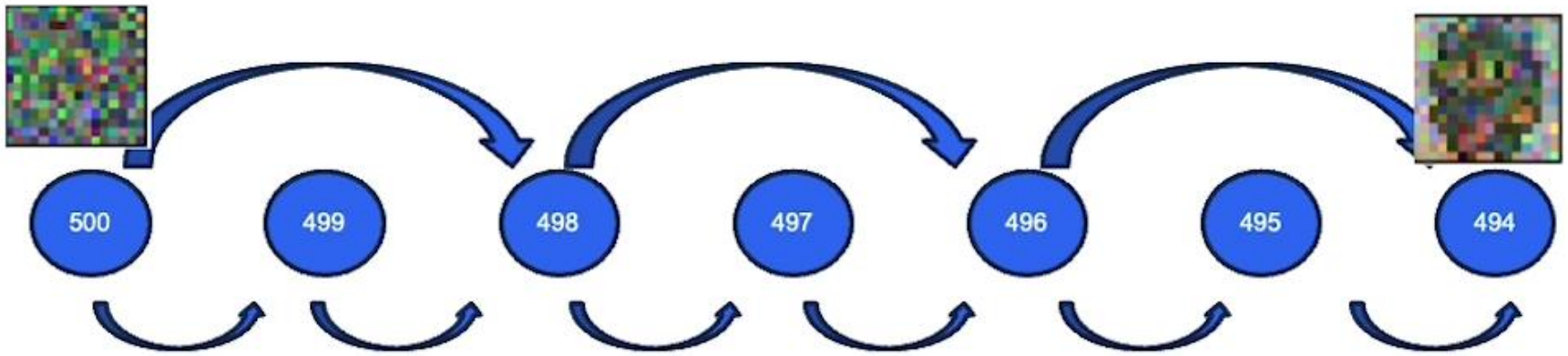
El Muestreo es Lento

- Muestreo es lento
- Quieres más imágenes rápido.

Pero el muestreo es lento porque:

- Hay muchos pasos de tiempo.
- Cada paso de tiempo depende del anterior (Markoviano).
- Muchos nuevos muestreadores abordan este problema de velocidad.

- Uno se llama DDIM: Modelos Implícitos de Difusión de Denoising (¡solo el nombre del artículo!)



DDIM es más rápido porque omite pasos de tiempo.

Recomendaciones Generales

Problemas a considerar

Training Set



Caption: Living in the light with Ann Graham Lotz

Generated Image



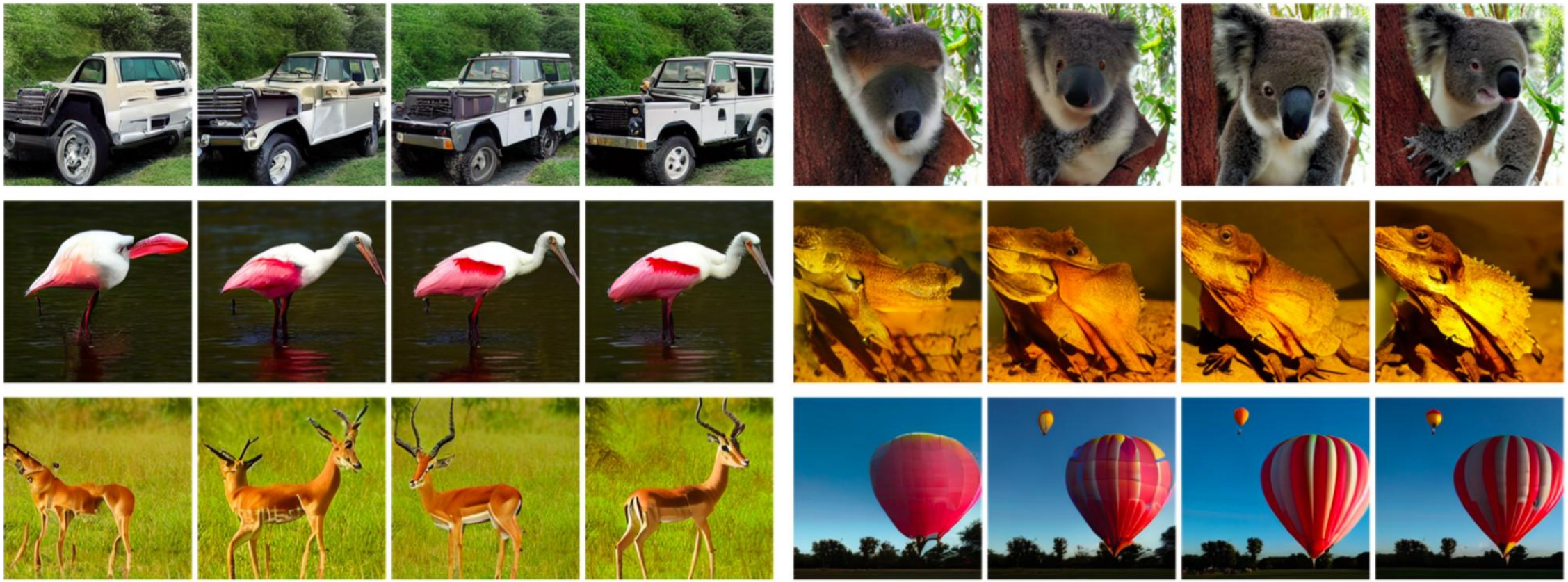
Prompt: Ann Graham Lotz

Consideraciones

- Muchas posibilidades de errores en la codificación.
- Utiliza un Modelo de Media Móvil Exponencial.
- Ten cuidado con los tiempos "cerca de los datos", usa un mínimo/máximo numérico.
- Los algoritmos de muestreo pueden cambiar mucho las cosas para un modelo fijo.
- Los modelos de difusión son una oportunidad de tener resultados importantes con modelos no tan grandes.

Consideraciones

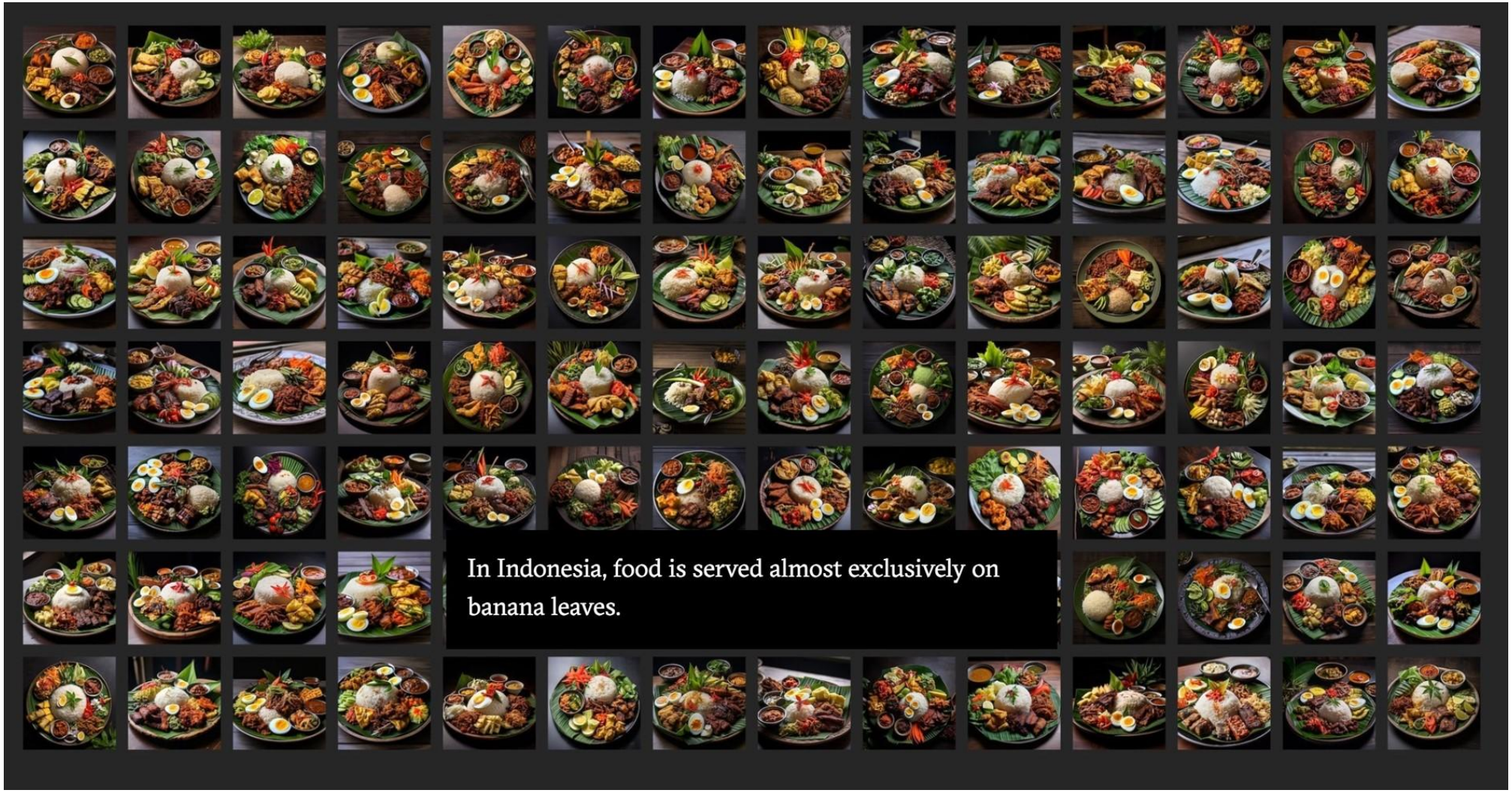
Tamaño del modelo →



Consideraciones



Consideraciones



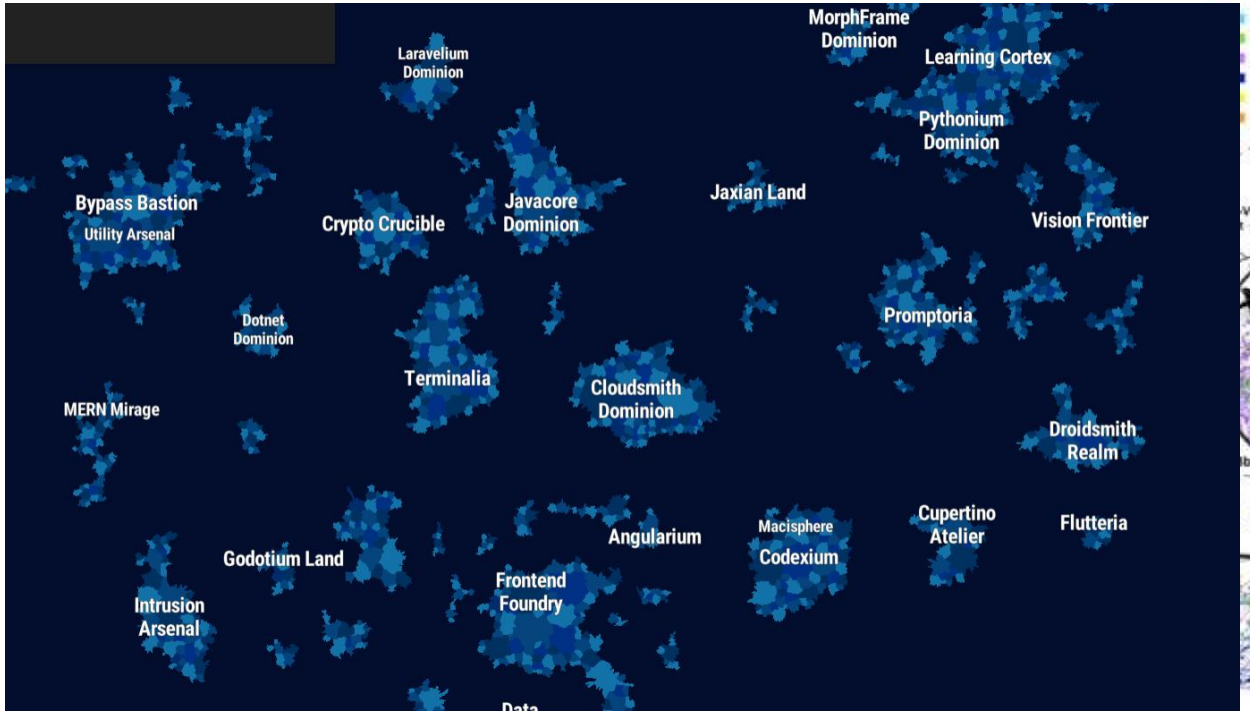
(10 min) Break



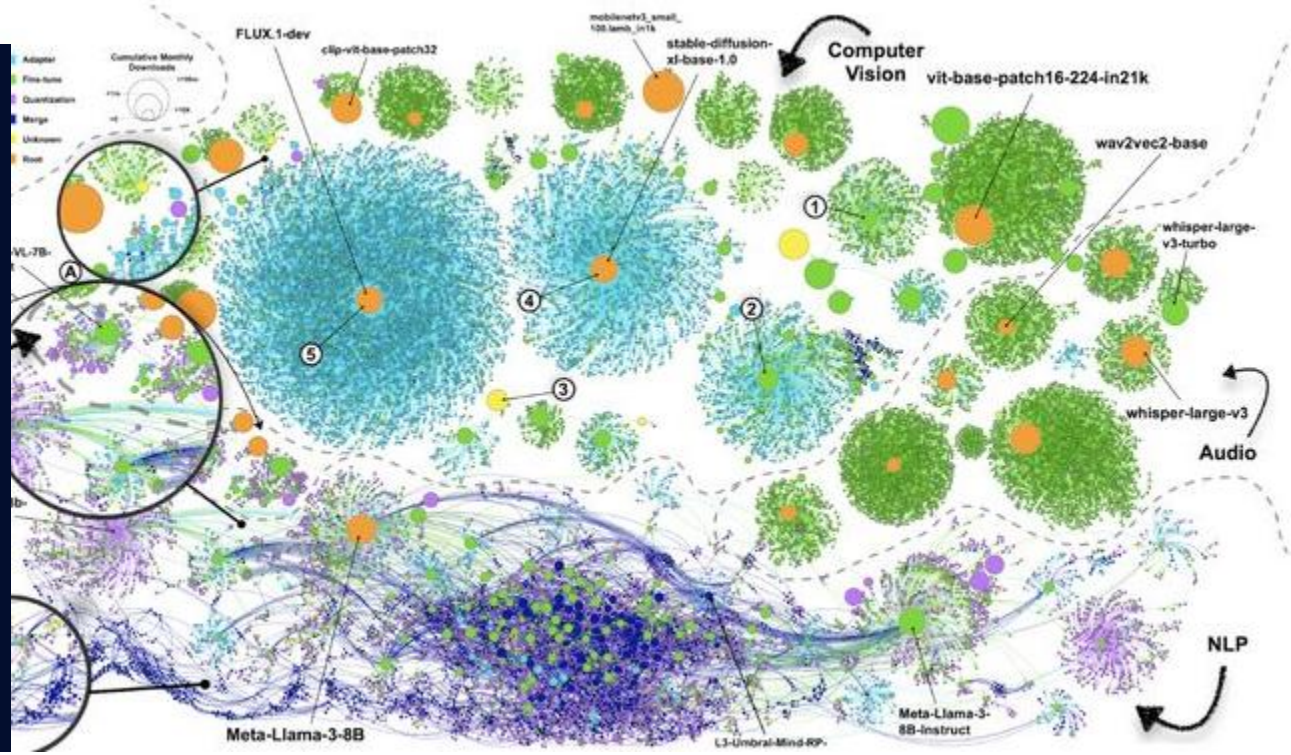
Tendencias 2026

*En base a Andrej Karpathy: Software Is Changing (Again)

Evolución del Software



Github Atlas (Software 1.0)



HuggingFace Model Atlas (Software 2.0)

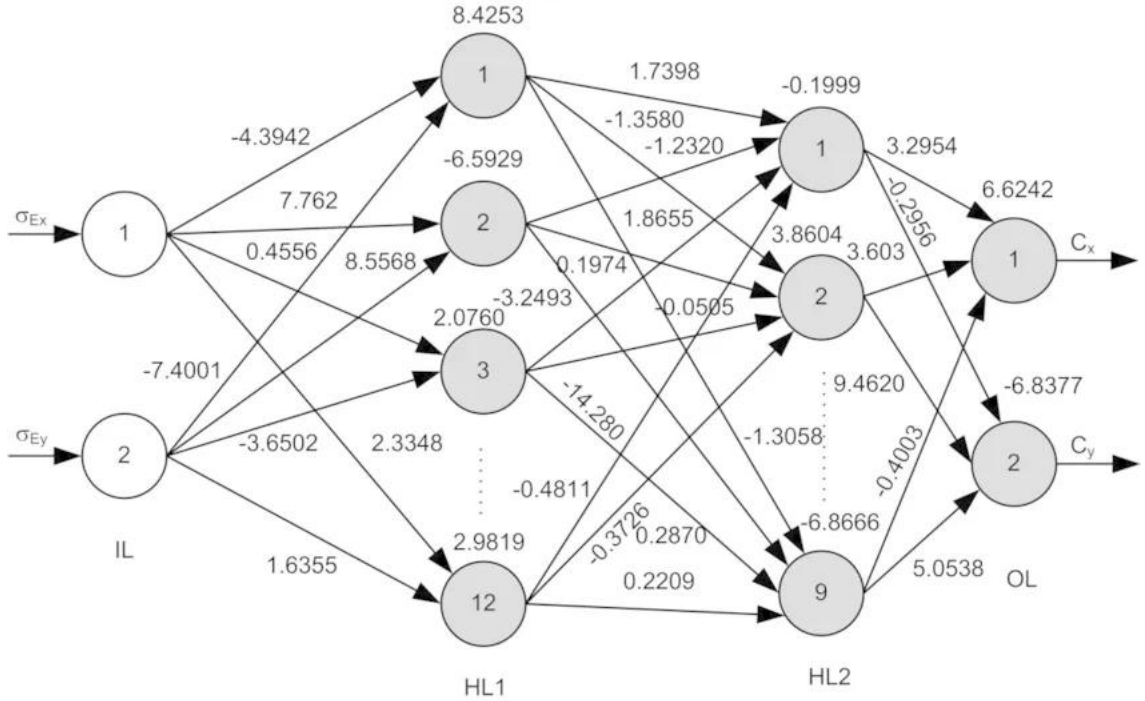
https://anvaka.github.io/map-of-github/#1.92/0.86/7.12

https://huggingface.co/spaces/Eliahu/Model-Atlas

Software 1.0 = code



Software 2.0 = weights



Finalidad de ML

Software 1.0

computer code

programs

computer



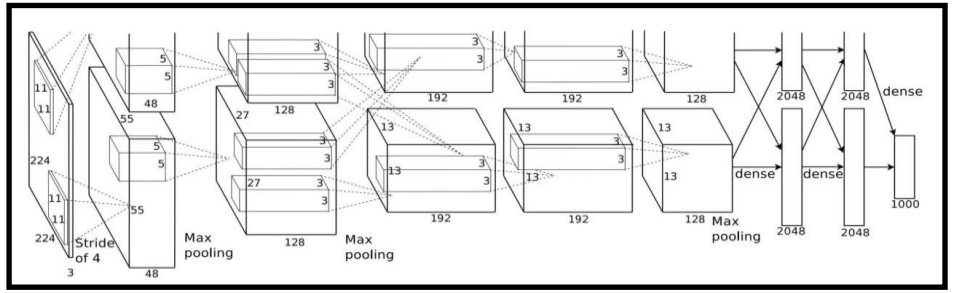
became programmable in **~1940s**

Software 2.0

weights

programs

neural net



fixed function neural net

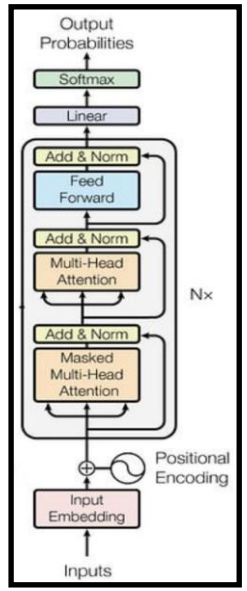
e.g. AlexNet: for image recognition (**~2012**)

Software 3.0

prompts

programs

LLM



LLM = programmable neural net!

~2019

Ejemplo: Clasificación de Sentimientos

Software 1.0

```python
def simple_sentiment(review: str) -> str:
    """Return 'positive' or 'negative' based on a tiny keyword lexicon."""
    positive = {
        "good", "great", "excellent", "amazing", "wonderful", "fantastic",
        "awesome", "loved", "love", "like", "enjoyed", "superb", "delightful"
    }
    negative = {
        "bad", "terrible", "awful", "poor", "boring", "hate", "hated",
        "dislike", "worst", "dull", "disappointing", "mediocre"
    }

    score = 0
    for word in review.lower().split():
        w = word.strip(".,!?;:")          # crude token clean-up
        if w in positive:
            score += 1
        elif w in negative:
            score -= 1

    return "positive" if score >= 0 else "negative"
```

Software 2.0

10,000 positive examples

10,000 negative examples

encoding (e.g. bag of words)

train binary classifier

parameters

Software 3.0

You are a sentiment classifier. For every review that appears between the tags
<REVIEW> … </REVIEW>, respond with **exactly one word**, either POSITIVE or NEGATIVE (all-caps, no punctuation, no extra text).

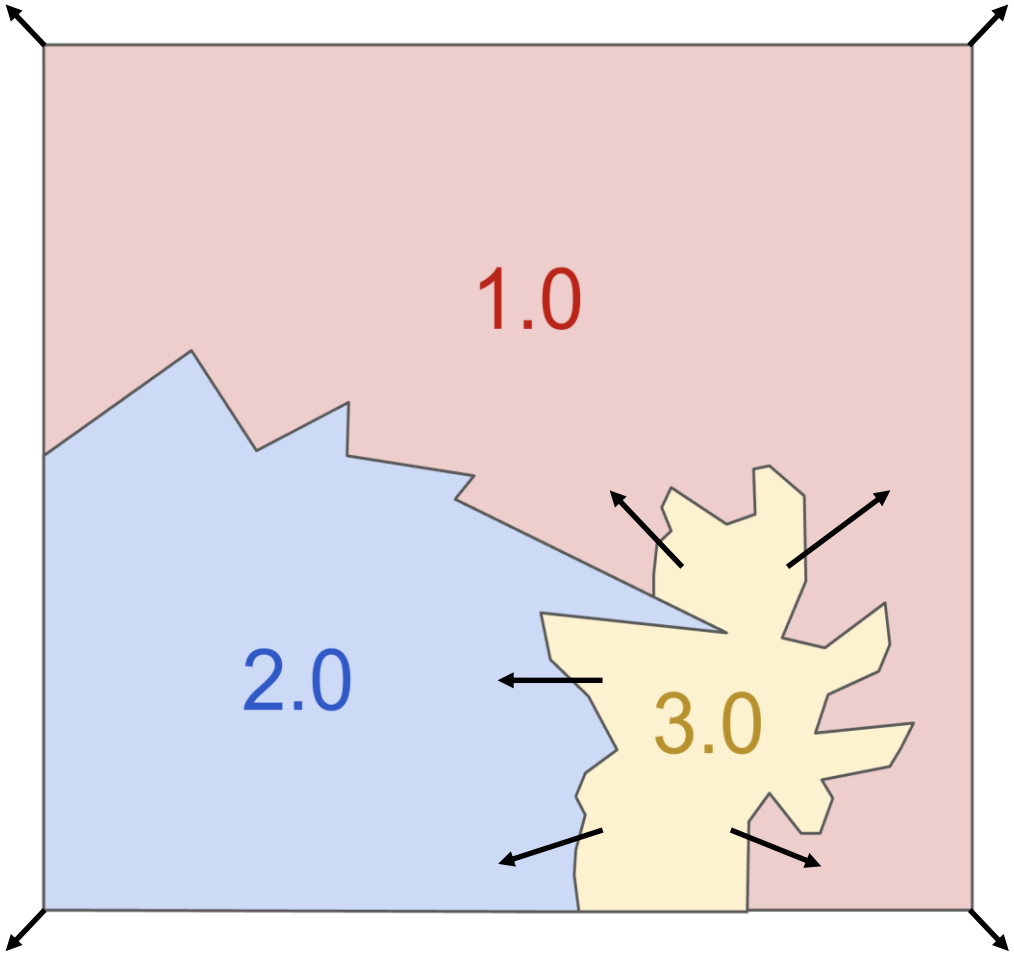
Example 1
<REVIEW>I absolutely loved this film—the characters were engaging and the ending was perfect.</REVIEW>
POSITIVE

Example 2
<REVIEW>The plot was incoherent and the acting felt forced; I regret watching it.</REVIEW>
NEGATIVE

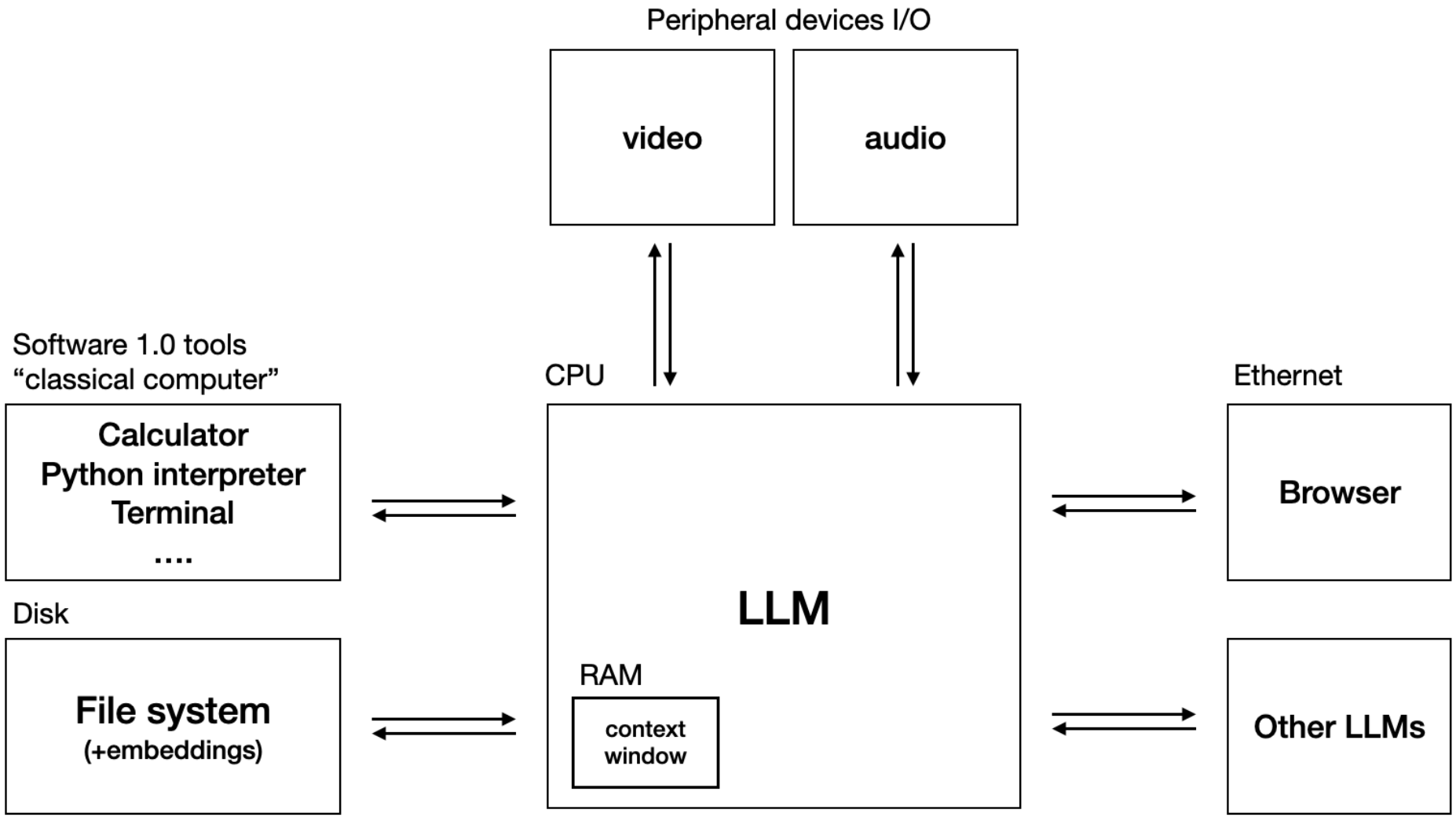
Example 3
<REVIEW>An energetic soundtrack and solid visuals almost save it, but the story drags and the jokes fall flat.</REVIEW>
NEGATIVE

Now classify the next review.

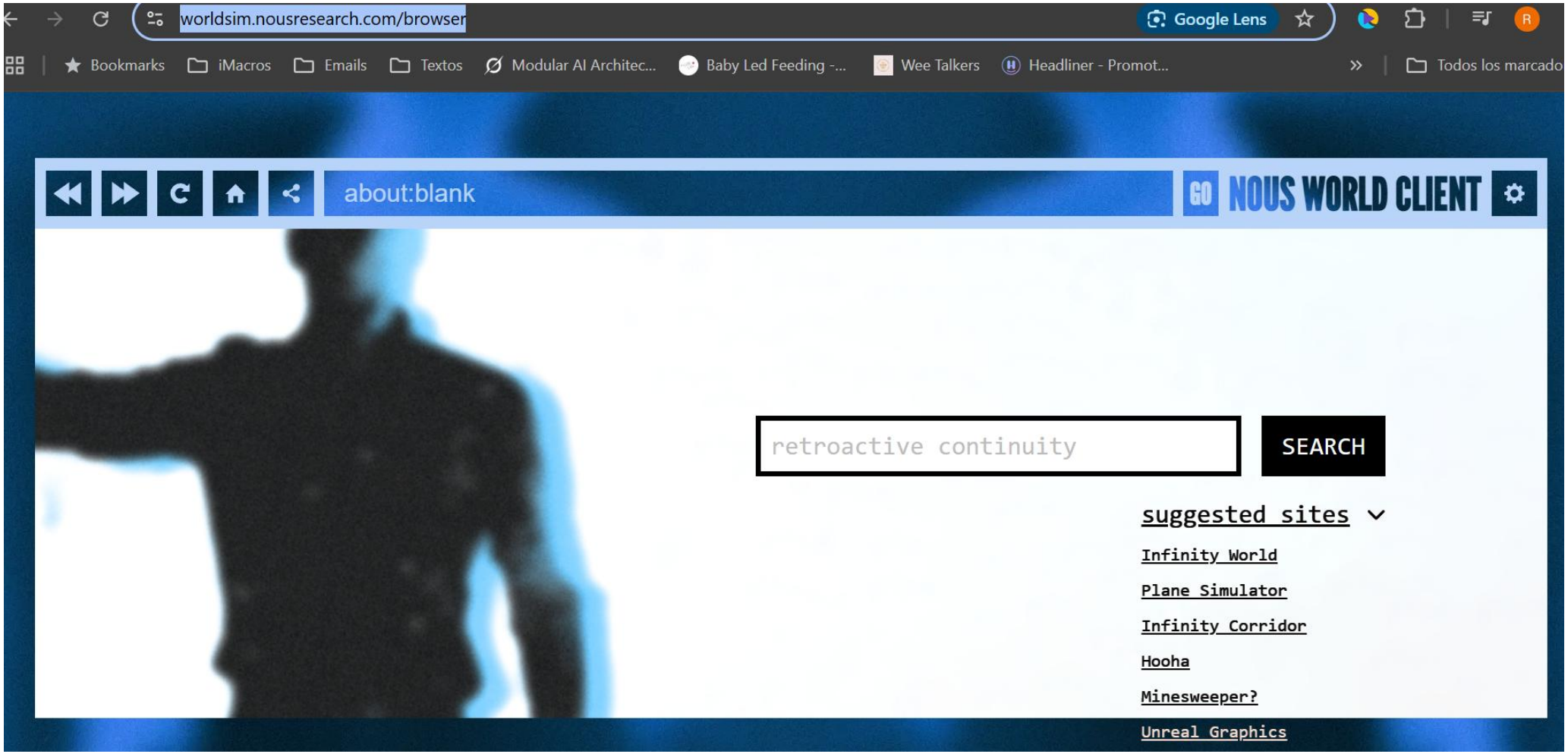
Una gran cantidad de software será desarrollado



LLM OS


Peripheral devices I/O
video
audio
Software 1.0 tools
“classical computer”
CPU
Ethernet
Calculator
Python interpreter
Terminal
....
Browser
LLM
Disk
RAM
File system
(+embeddings)
context
window
Other LLMs

Worldsim



https://worldsim.nousresearch.com/

1950s - 1970s time-sharing era

We are in the Mainframe & Time-sharing era of computing.
Centralized, expensive computers =>

- - OS runs in the cloud
- - I/O is streamed back and forth over the network
- - compute is batched over users







Early hints of Personal Computing v2

Alex Cheema - e/acc @alexocheema · Apr 5

Meta 🤝 Apple

Llama 4 + Apple Silicon is a match made in heaven.

Here's why: Like DeepSeek V3/R1, all of the new Llama 4 variants are massive sparse MoE models. They have a massive amount of parameters, but only a small number of those are active each time a token is generated.
Show more



Guillermo Flor · 3rd+
Venture Capital Investor @ GoHub Ventures
View my newsletter
4mo ·

+ Follow

GI at home

Running DeepSeek R1 across my 7 M4 Pro Mac Minis and 1 M4 Max MacBook Pro.

Total unified memory = 496GB.

Uses @exolabs
distributed inference with 4-bit quantization.

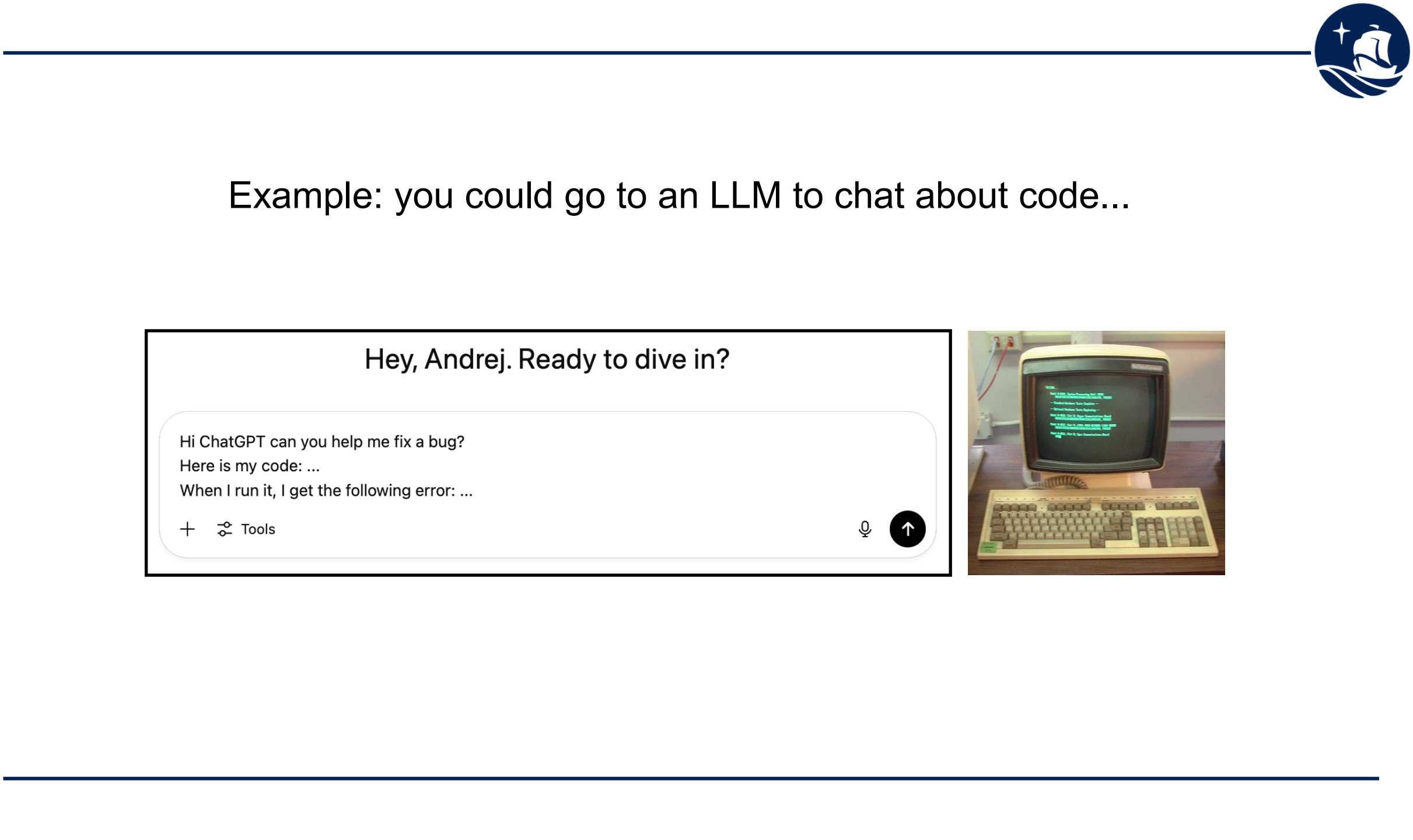
Next goal is fp8 (requires >700GB)



Partial autonomy apps



"Copilot" / "Cursor for X"

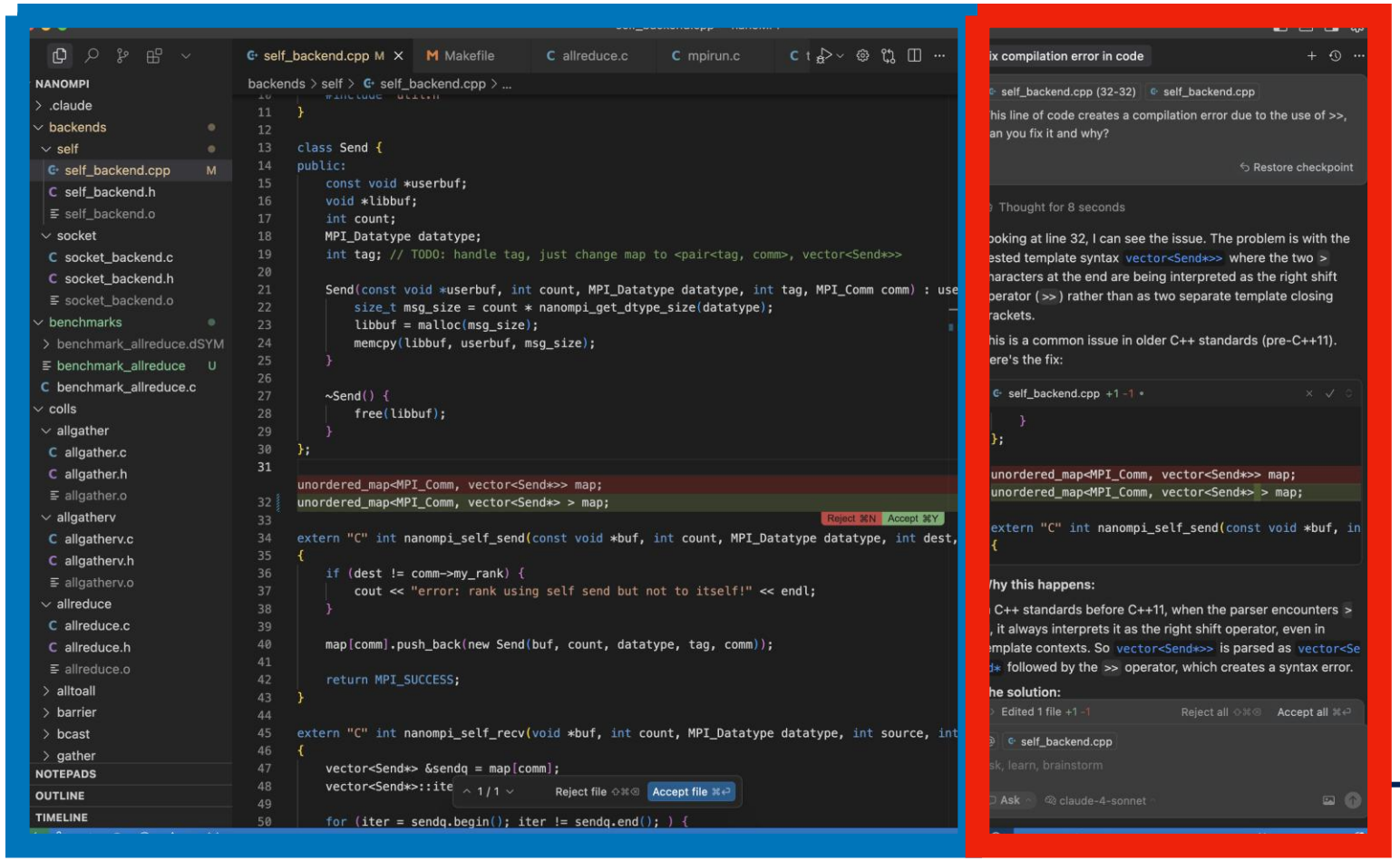


Example: you could go to an LLM to chat about code...

Example: Anatomy of Cursor

Traditional interface

LLM integration



1. Package state into a context window before calling LLM.

2. Orchestrate and call multiple models (e.g. embedding models, chat models, diff apply models, ...)

3. Application-specific GUI

4. Autonomy slider: Tab → Cmd+K → Cmd+L → Cmd+I (agent mode)

autonomy slider

Example: Anatomy of Perplexity

meta buys scale ai

Perplexity | Images | Sources · 10 | Tasks

Cointelegraph
Meta's AI shortfall prompts Zuckerberg's $15B investment i...

reuters
Meta to pay nearly $15 billion for Scale ...



ca.finance.ya...
Meta to Buy 49% of Scale AI for $14.8...



+7 sources

Meta's $14.8 Billion Investment in Scale AI: Key Details

Meta Platforms has agreed to acquire a 49% stake in Scale AI, a leading data labeling and training data company, for $14.8 billion. This deal, one of the largest in the AI sector to date, values Scale AI at $28 billion and is designed to significantly bolster Meta's artificial intelligence capabilities as it races to catch up with rivals like OpenAI, Microsoft, Google, and Amazon [2][3][5][6].

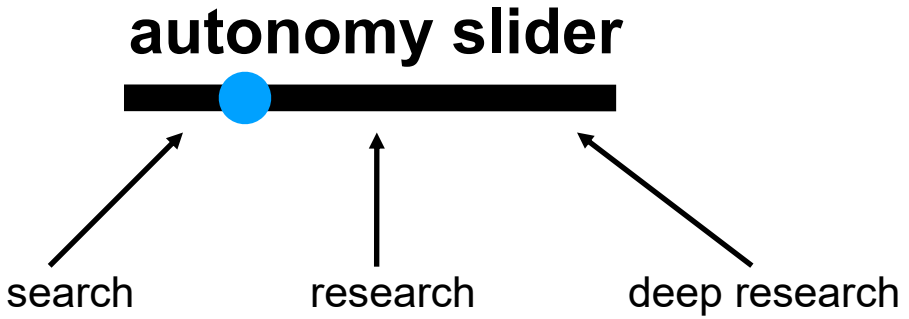
What Is Scale AI and Why Does Meta Want It?

- **Scale AI's Role:** Founded in 2016, Scale AI specializes in providing vast, high-quality labeled datasets and curated training information essential for developing advanced AI models, including those used by OpenAI's ChatGPT and other major tech companies [2][3][8].
- **Meta's Motivation:** Meta has faced criticism for lagging behind in the AI race, particularly after the underwhelming launch of its Llama 4 language model and the postponement of its flagship "Behemoth" AI model [2][6][8]. By securing a major stake in Scale AI, Meta aims to improve the quality and scale of its AI training data, a critical resource for developing state-of-the-art AI systems.

Strategic Implications

- **Superintelligence Initiative:** As part of the deal, Scale AI CEO Alexandr Wang will join Meta to lead a new "superintelligence" team, reporting directly to CEO Mark Zuckerberg. This group will focus on achieving artificial general intelligence (AGI)—AI that can perform at or above human cognitive levels [1][3][4][6].

1. Package information into a context window
2. Orchestrate multiple LLM models
3. Application-specific GUI for Input/Output UIUX



(+suggested followup questions)

What does all software look like in the partial autonomy world?



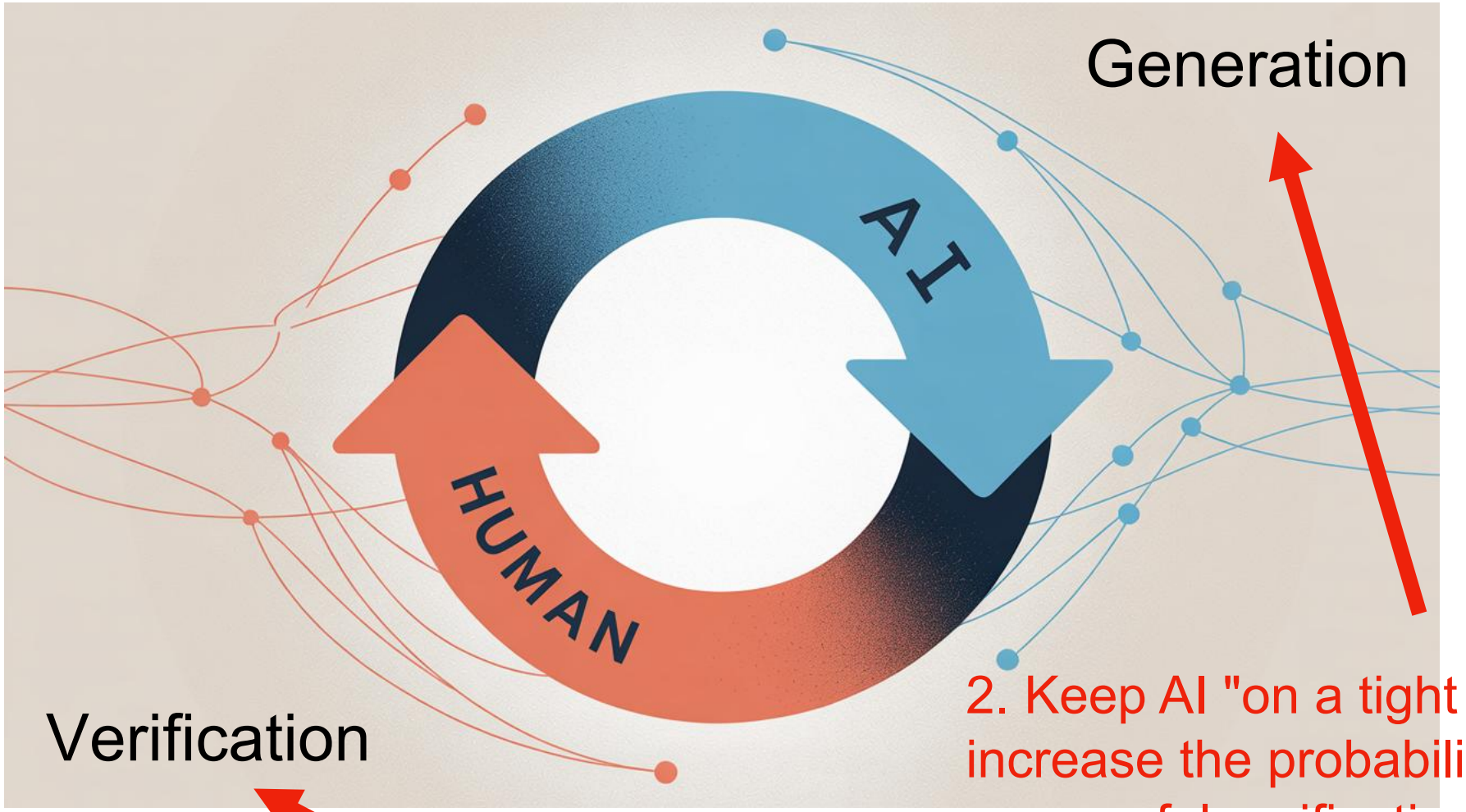
Adobe photoshop



Unreal engine

- Can an LLM "see" all the things the human can?
- Can an LLM "act" in all the ways a human can?
- How can a human supervise and stay in the loop?

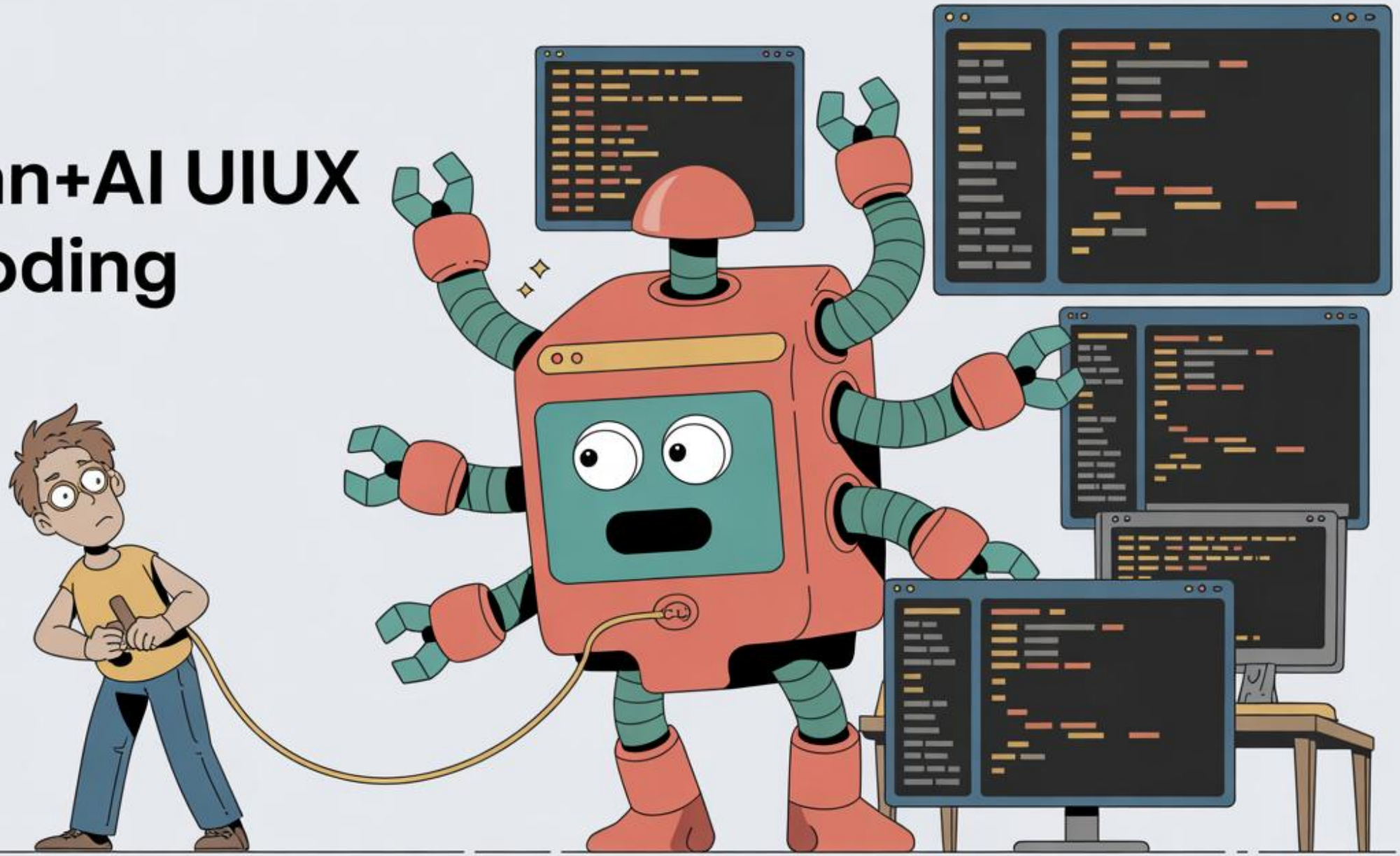
Consider the full workflow of partial autonomy UIUX



2. Keep AI "on a tight leash" to increase the probability of successful verification

1. Make this EASY, FAST to win.

Human+AI UIUX for Coding



Example: keeping agents on the leash

Here's an example. This prompt is not unreasonable but not particularly thoughtful:

```
Write a Python rate limiter that limits users to 10 requests per minute.
```

I would expect this prompt to give okay results, but also miss some edge cases, good practices and quality standards. This is how you might see someone at nilenso prompt an AI for the same task:

```
Implement a token bucket rate limiter in Python with the following requirements:

- 10 requests per minute per user (identified by `user_id` string)
- Thread-safe for concurrent access
- Automatic cleanup of expired entries
- Return tuple of (allowed: bool, retry_after_seconds: int)

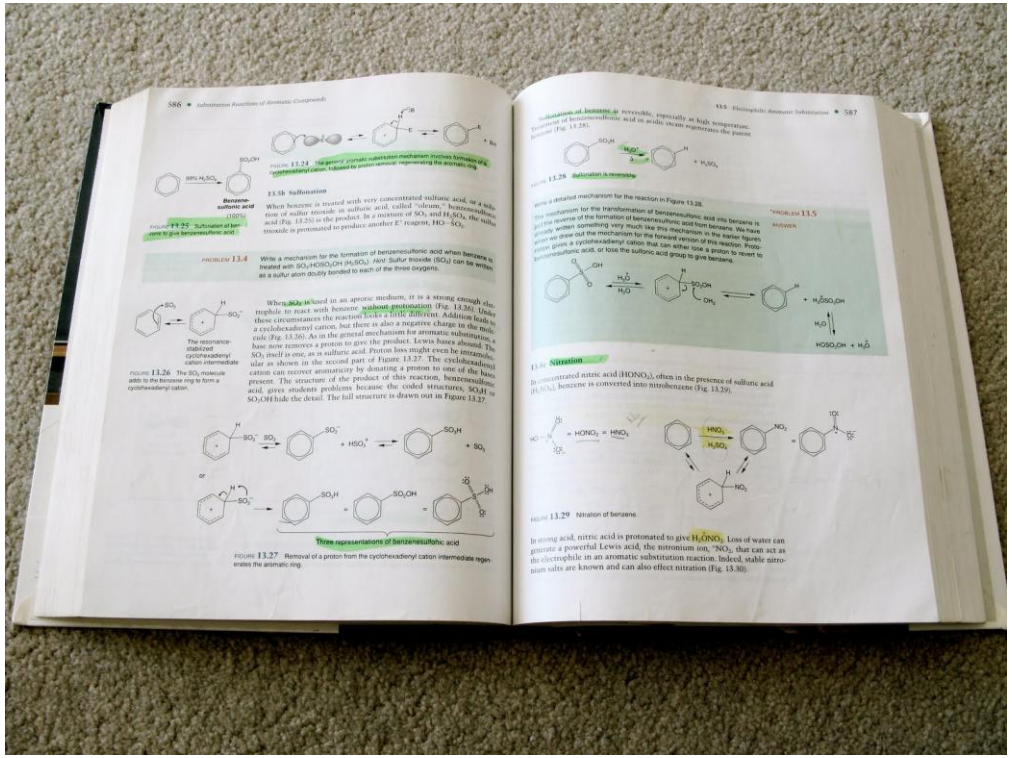
Consider:
- Should tokens refill gradually or all at once?
- What happens when the system clock changes?
- How to prevent memory leaks from inactive users?

Prefer simple, readable implementation over premature optimization. Use stdlib only (no Redis/external deps).
```



Example: keeping agents on the leash - AI + Education / LLM101n

1. App for course creation (for teacher)



2. App for course serving (for student)



Example: Tesla Autopilot



autonomy slider



- keep the lane
- keep distance from the car ahead
- take forks on highway
- stop for traffic lights and signs
- take turns at intersections

"2025 is the year of agents."

?

"2025-2035 is the decade of agents."

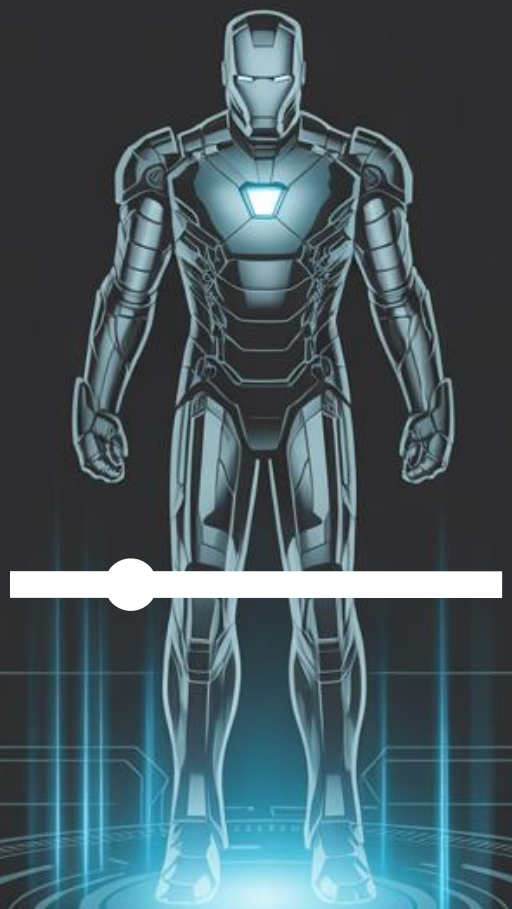
-
Andrej

THE IRON MAN SUIT

Augmentation







Agent



Building Autonomous Software

- ❌ Iron Man robots
- ❌ Flashy demos of autonomous agents
- ❌ AGI 2027

- ✅ Iron Man suits
- ✅ Partial autonomy products
- ✅ Custom GUI and UIUX
- ✅ Fast Generation - Verification loop
- ✅ Autonomy slider



Make software highly accessible 👶

(Have you heard of vibe coding by any chance?)





Andrej Karpathy
@karpathy

Promote

There's a new kind of coding I call "vibe coding", where you fully give in to the vibes, embrace exponentials, and forget that the code even exists. It's possible because the LLMs (e.g. Cursor Composer w Sonnet) are getting too good. Also I just talk to Composer with SuperWhisper so I barely even touch the keyboard. I ask for the dumbest things like "decrease the padding on the sidebar by half" because I'm too lazy to find it. I "Accept All" always, I don't read the diffs anymore. When I get error messages I just copy paste them in with no comment, usually that fixes it. The code grows beyond my usual comprehension, I'd have to really read through it for a while. Sometimes the LLMs can't fix a bug so I just work around it or ask for random changes until it goes away. It's not too bad for throwaway weekend projects, but still quite amusing. I'm building a project or webapp, but it's not really coding - I just see stuff, say stuff, run stuff, and copy paste stuff, and it mostly works.

Vibe coding

From Wikipedia, the free encyclopedia



This article **may contain an excessive number of citations**. Please help remove low-quality or irrelevant citations. *(June 2025)* *(Learn how and when to remove this message)*

Vibe coding (or **vibecoding**) is an approach to producing software by using artificial intelligence (AI), where a person describes a problem in a few natural language sentences as a prompt to a large language model (LLM) tuned for coding. The LLM generates software based on the description, shifting the programmer's role from manual coding to guiding, testing, and refining the AI-generated source code.[1][2][3]

Advocates of vibe coding say that it allows even amateur programmers to produce software without the extensive training and skills required for software engineering.[4] The term was introduced by Andrej Karpathy in February 2025[5][2][4][1] and listed in the Merriam-Webster Dictionary the following month as a "slang & trending" noun.[6]

Thomas Wolf
@Thom_Wolf

we've seen nothing yet! hosted a 9-13 yo vibe-coding event w. @robertkeus this w-e (h/t @antonosika @LovableBuild)

takeaway? AI is unleashing a generation of wildly creative builders beyond anything I'd have imagined

and they grow up *knowing* they can build anything!



Vibe coding

MenuGen

https://www.menugen.app/



Spago

WOLFGANG PUCK

SPAGO BREAKFAST SIGNATURES

Kaya French Toast 26
Pandan-Coconut Jam, Soy Caramel, Egg Jam

Iberico Pork Dumplings "Agnolotti" 20
Soy, Chili Oil, Black Vinegar, Coriander, Garlic

Spago's Laksa Noodle 26
Little Neck Clams, Tiger Prawns, Calamari, Onion Sambal, Calamansi, Laksa Leaf

"Curry & Waffle" 20
Tamarind Fish Curry, Japanese Seabream, Prata Waffles, Curry Leaves

BREAKFAST CLASSICS

Assorted Fresh Fruits 18
Sarawak Pineapple, Watermelon, Chitose Farm Strawberry, Thai Mango, Passion Fruit Syrup

French Omelet 28
Sauteed Wild Mushrooms, Gruyère, Thyme, Rocket Salad

Open-Faced Monte Cristo Sandwich 24
Canadian Style Smoked Bacon, Sunny Side Up Egg, Gruyère, Soy Caramel

Avocado Toast 26
Salmon Cream Cheese, Watercress, Poached Egg, Hollandaise, Dill, Chives

Spago Full Breakfast 34
2 Eggs, Canadian Style Smoked Bacon, Spiced Pork Sausage, Pommes Anna, Fontina

Shakshuka 24
Braised Tripe, 1 Egg, Spiced Tomato Ragu, Grilled Baguette, Mint, Parsley, Chili

BOWLS AND SALADS

"Yogurt and Berries" 26
Fage Greek Yogurt, Banana, Blueberry, Raspberry, Almond

Homemade Muesli 20
Steele Cut Oats, Apple, Apricot, Yogurt, Homemade Granola

Healthy Breakfast Bowl 26
Egg White Scramble, Sauteed Spinach, French Bean, Wild Mushrooms, Parmesan Crumble
(Add Grilled Chicken +9)

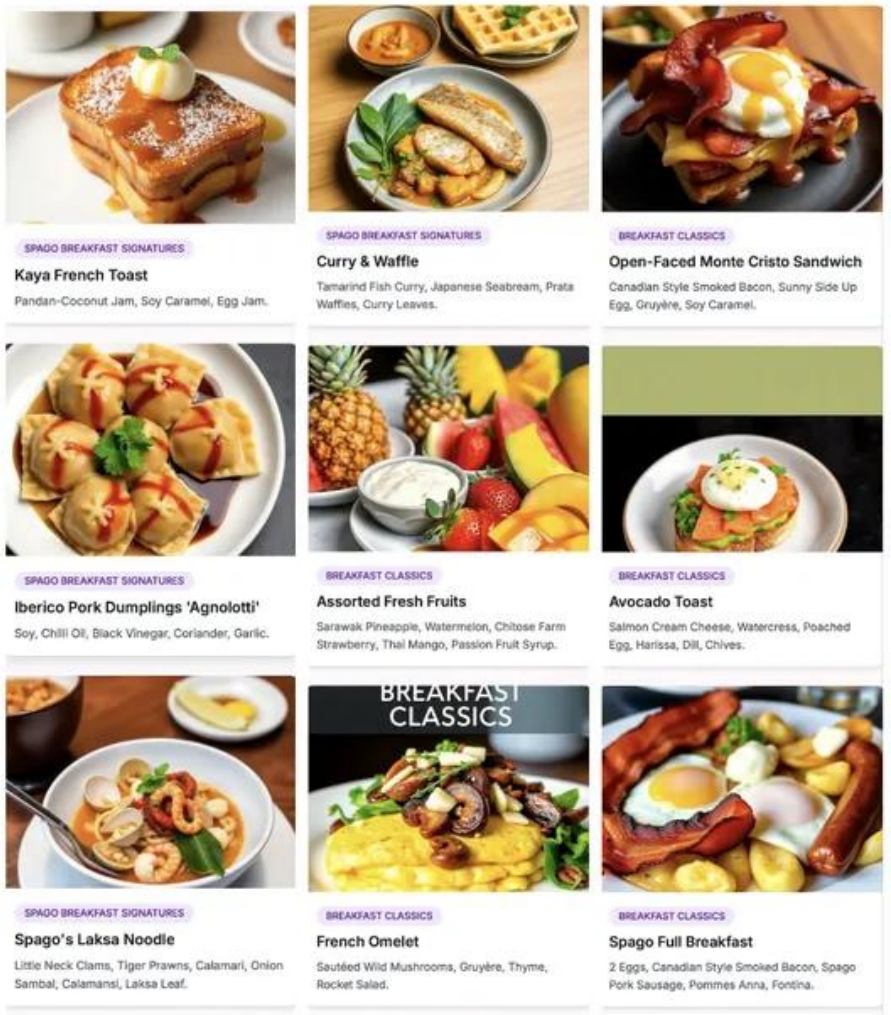
Kaviari Smoked Salmon 28
Dill Crème Fraîche, Cucumbers, Amela Tomatoes, Capers, Rye Sourdough

COFFEE SELECTION
Espresso (single) 8
Americano 8
Cappuccino 10
Latte 10
Mocha 10

TEA SELECTION
Earl Grey 14
English Breakfast 12
Alphonso (Mango Black Tea) 12
Sencha 14
Chamomile 12

FRESH JUICES & SMOOTHIES
Orange 8
Watermelon 8
Stimulate – Carrot, Pineapple, Celery, Ginger & Lime 9
ABC – Apple, Beetroot & Carrot 9
Strawberry Banana 14

menugen!



SPAGO BREAKFAST SIGNATURES
Kaya French Toast
Pandan-Coconut Jam, Soy Caramel, Egg Jam.

SPAGO BREAKFAST SIGNATURES
Curry & Waffle
Tamarind Fish Curry, Japanese Seabream, Prata Waffles, Curry Leaves.

BREAKFAST CLASSICS
Open-Faced Monte Cristo Sandwich
Canadian Style Smoked Bacon, Sunny Side Up Egg, Gruyère, Soy Caramel.

SPAGO BREAKFAST SIGNATURES
Iberico Pork Dumplings 'Agnolotti'
Soy, Chilli Oil, Black Vinegar, Coriander, Garlic.

BREAKFAST CLASSICS
Assorted Fresh Fruits
Sarawak Pineapple, Watermelon, Chitose Farm Strawberry, Thai Mango, Passion Fruit Syrup.

BREAKFAST CLASSICS
Avocado Toast
Salmon Cream Cheese, Watercress, Poached Egg, Harissa, Dill, Chives.

SPAGO BREAKFAST SIGNATURES
Spago's Laksa Noodle
Little Neck Clams, Tiger Prawns, Calamari, Onion Sambal, Calamansi, Laksa Leaf.

BREAKFAST CLASSICS
French Omelet
Sautéed Wild Mushrooms, Gruyère, Thyme, Rocket Salad.

BREAKFAST CLASSICS
Spago Full Breakfast
2 Eggs, Canadian Style Smoked Bacon, Spago Pork Sausage, Pommes Anna, Fontina.



The code was the easiest part! :O

Most of the work was in the browser clicking things.

😐 LLM API keys

😐 Flux (image generation) API keys

✅ Running locally (ez)

😕 Vercel deployments

😕 Domain names

😖 Authentication

😕 Payments



https://karpathy.bearblog.dev/vibe-coding-menugen/

Example: adding Google login

Add Google as a social connection

Before you start

- A Clerk application is required.
- A Google Developer account is required.

Enabling OAuth with Google allows your users to sign up and sign in to your Clerk application with their Google account.

> **Warning**
> Google OAuth 2.0 **does not** allow apps to use WebViews for authentication. See the dedicated Google blog post for more information. If your app requires users to sign in via in-app browsers, follow the setup instructions in the Google Help guide.

Configure for your development instance

For *development instances*, Clerk uses preconfigured shared OAuth credentials and redirect URIs—no other configuration is needed.

1. In the Clerk Dashboard, navigate to the **SSO connections** page.
2. Select **Add connection** and select **For all users**.
3. In the **Choose provider** dropdown, select **Google**.
4. Select **Add connection**.

Configure for your production instance

For *production instances*, you must provide custom credentials.

To make the setup process easier, it's recommended to keep two browser tabs open: one for the Clerk Dashboard and one for your Google Cloud Console.

1 Enable Google as a social connection

1. In the Clerk Dashboard, navigate to the **SSO connections** page.
2. Select **Add connection** and select **For all users**.
3. In the **Choose provider** dropdown, select **Google**.
4. Ensure that both **Enable for sign-up and sign-in** and **Use custom credentials** are toggled on.
5. Save the **Authorized Redirect URI** somewhere secure. Keep this modal and page open.

2 Create a Google Developer project

1. Navigate to the Google Cloud Console.
2. Select a project or create a new one. You'll be redirected to your project's **Dashboard** page.
3. In the top-left, select the menu icon (≡) and select **APIs & Services**. Then, select **Credentials**.
4. Next to **Credentials**, select **Create Credentials**. Then, select **OAuth client ID.** You might need to configure your OAuth consent screen. Otherwise, you'll be redirected to the **Create OAuth client ID** page.
5. Select the appropriate application type for your project. In most cases, it's **Web application**.
6. In the **Authorized JavaScript origins** setting, select **Add URI** and add your domain (e.g., `https://your-domain.com` and `https://www.your-domain.com` if you have a `www` version). For local development, add `http://localhost:PORT` (replace `PORT` with the port number of your local development server).
7. In the **Authorized Redirect URIs** setting, paste the **Authorized Redirect URI** value you saved from the Clerk Dashboard.
8. Select **Create**. A modal will open with your **Client ID** and **Client Secret**. Save these values somewhere secure.

3 Set the Client ID and Client Secret in the Clerk Dashboard

1. Navigate back to the Clerk Dashboard where the modal should still be open. Paste the **Client ID** and **Client Secret** values that you saved into the respective fields.
2. Select **Add connection**.

> **Note**
> If the modal or page is no longer open, navigate to the **SSO connections** page in the Clerk Dashboard. Select the connection. Under **Use custom credentials**, paste the values into their respective fields.

4 Test your connection

The simplest way to test your connection is to visit your Clerk app's Account Portal, which is available for all Clerk apps out-of-the-box.

1. In the Clerk Dashboard, navigate to the **Account Portal** page.
2. Next to the **Sign-in** URL, select **Visit**. The URL should resemble:
   - **For development** – `https://your-domain.accounts.dev/sign-in`
   - **For production** – `https://accounts.your-domain.com/sign-in`
3. Sign in with your connection's credentials.

> **Warning**
> Google sign-in **does not** allow users to sign in via in-app browsers.



Build for agents 🤖

There is new category of consumer/manipulator of digital information:

1. Humans (GUIs)
2. Computers (APIs)
3. **NEW**: Agents <- computers... but human-like

robots.txt →

The /llms.txt file

A proposal to standardise on using an `/llms.txt` file to provide information to help LLMs use a website at inference time.

AUTHOR
Jeremy Howard

PUBLISHED
September 3, 2024

```
# FastHTML

> FastHTML is a python library which brings together Starlette, Uvicorn, HTMX,
and fastcore's `FT` "FastTags" into a library for creating server-rendered
hypermedia applications.

Important notes:

- Although parts of its API are inspired by FastAPI, it is *not* compatible with
FastAPI syntax and is not targeted at creating API services
- FastHTML is compatible with JS-native web components and any vanilla JS
library, but not with React, Vue, or Svelte.

## Docs

- [FastHTML quick start]
(https://answerdotai.github.io/fasthtml/tutorials/quickstart_for_web_devs.html.md)
A brief overview of many FastHTML features
- [HTMX reference](https://raw.githubusercontent.com/path/reference.md): Brief
description of all HTMX attributes, CSS classes, headers, events, extensions, js
lib methods, and config options

## Examples

- [Todo list application](https://raw.githubusercontent.com/path/adv_app.py):
Detailed walk-thru of a complete CRUD app in FastHTML showing idiomatic use of
FastHTML and HTMX patterns.

## Optional

- [Starlette full documentation]
(https://gist.githubusercontent.com/path/starlette-sml.md): A subset of the
Starlette documentation useful for FastHTML development.
```

Docs for people

Copy page

Vercel Documentation

Start with an idea

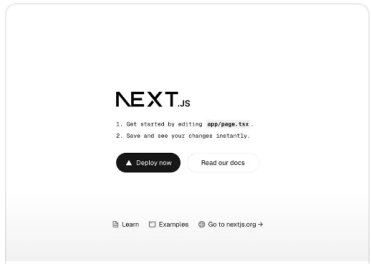
Vercel builds tools to help you create products faster.

Like v0, which is your web development assistant. Paste a screenshot or write a few sentences and v0 will generate a starting point for your next app, including the code for how it looks *and* how it works. v0 then connects to Vercel, takes your code, and creates a URL you can share.

Get started in minutes

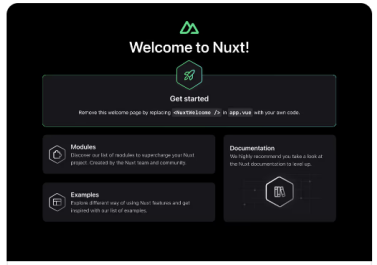
Deploy a Template

View All Templates



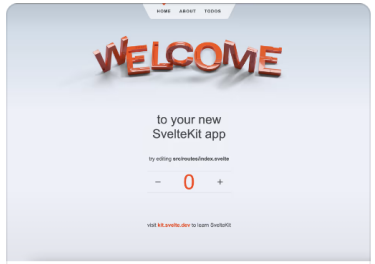
Next.js Boilerplate

Get started with Next.js and React in seconds.



Nuxt.js 3 Boilerplate

A Nuxt.js 3 app, bootstrapped with create-nuxt-app.



SvelteKit Boilerplate

A SvelteKit app including nested routes, layouts, and page endpoints.

Configure for your production instance

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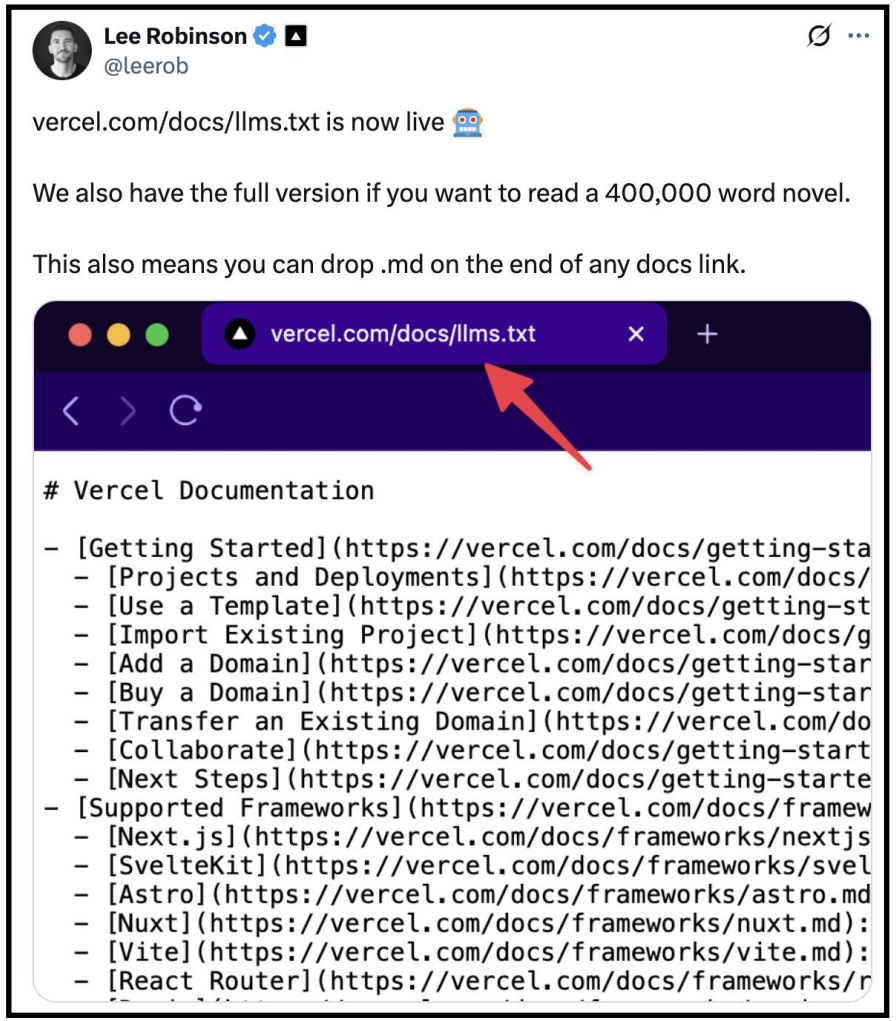
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Docs for people LLMs



Home / Get started

Build on Stripe with LLMs

Copy page

Use LLMs in your Stripe integration workflow.

You can use large language models (LLMs) to assist in the building of Stripe integrations. We provide a set of tools and best practices if you use LLMs during development.

Plain text docs

You can access all of our documentation as plain text markdown files by adding `.md` to the end of any url. For example, you can find the plain text version of this page itself at https://docs.stripe.com/building-with-llms.md.

This helps AI tools and agents consume our content and allows you to copy and paste the entire contents of a doc into an LLM. This format is preferable to scraping or copying from our HTML and JavaScript-rendered pages because:

- Plain text contains fewer formatting tokens.
- Content that isn't rendered in the default view (for example, it's hidden in a tab) of a given page is rendered in the plain text version.
- LLMs can parse and understand markdown hierarchy.

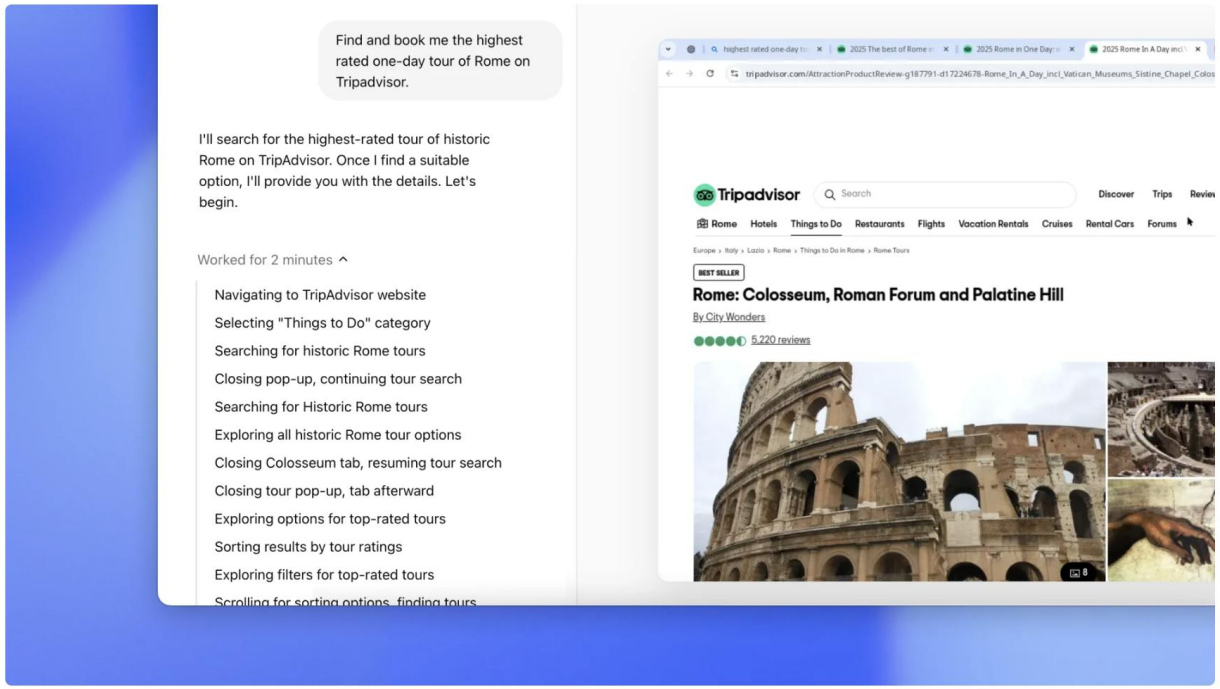
We also host an /llms.txt file which instructs AI tools and agents how to retrieve the plain text versions of our pages. The `/llms.txt` file is an emerging standard for making websites and content more accessible to LLMs.

January 23, 2025 Product

Introducing Operator

A research preview of an agent that can use its own browser to perform tasks for you. Available to Pro users in the U.S.

Go to Operator ↗





Más sobre Software 3.0



https://www.youtube.com/watch?v=LCEmiRjPEtQ